

Such methods, including QSPR, are useful in the first instance to indicate the qualitative/quantitative adsorption coefficient of a substance. In some instances an estimated value may be sufficient for this endpoint. In all such cases the estimated method must be proven to be valid for the type of substance considered (see the general guidance for use and applicability of QSPR), and if possible a sensitivity analysis should be conducted with values generated from different models. Using a range of values in the CSA will help to highlight if the adsorption coefficient is an important factor for environmental behaviour of the substance. In general an estimated value will be sufficient if it is indicated that the adsorption coefficient will not affect the CSA, i.e. no risk is identified for the sediment/soil compartments. Estimated values are essential for substances for which experimental measurement is not feasible i.e. for *difficult substances*. Estimated values are also useful for comparing screening tests [e.g. HPLC method (OECD 121; EC C19)]. A number of reviews of K_{oc} prediction have been published recently (Lyman 1990, Reinhard & Drefahl 1999, Doucette 2000, Delle Site 2001, Doucette 2003, Dearden 2004). That of Doucette (2000) contains a number of worked examples of the estimation of log K_{oc} values. Additional information on the K_{oc} can be found in Gerstl (1990), Briggs (1981) and Nendza (1998).

Grouping of substances and read-across approach

Read-across and/or QSPR prediction for K_{oc} are important predictive tools and should be the first method used to predict K_{oc} if reliable measured data do not exist and the model is valid for the substance. However if these options do not give meaningful and valid information or if K_{oc} is an important factor in the CSA (i.e. risks are indicated for sediment/soil compartments based on a predicted value and log $K_{ow} > 3$), then an experimental value should be measured.

Testing is technically not possible

Testing should always be considered, if none of the waiving possibilities applies. In general, partition coefficients that are measured with a suitable standard method are preferred (and they are usually essential for surfactants and ionic substances that dissociate at environmentally relevant pH).

Further adaptation possibilities

Not foreseen. The K_{oc} is not directly relevant for environmental classification or the PBT assessment. However, it is a key property for exposure assessment so the information requirement should not be waived.

R.7.1.15.5 Impurities;uncertainties

Impurities can have an impact on the measurement of adsorption/desorption. Expert judgement should be used when considering whether impurities may affect the determination of the adsorption/desorption. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

R.7.1.15.6 Endpoint specific information in the registration dossier/ in IUCLID

HPLC method (OECD TG 121, EU C.19)

Materials and methods

- description of the HPLC equipment and operating conditions (column, mobile phase, means of detection, temperature);
- dead time and method used for its determination;

- reference substances (identity, purity, K_{oc}, retention times) with results of at least 6 measurements with at least one of them above and one below the expected value for the test substance;
- quantities of test and reference substances introduced in the column.

Results and discussion

- average retention data and estimated d log K_{oc} value for test compound;
- all values of log K_{oc} derived from individual measurements.

Batch equilibrium method (OECD TG 106, EU C.18)

Materials and methods

- details on soil types (nature and sampling site(s), organic C, clay content and soil texture, and pH, if relevant Cation Exchange Capacity);
- information on the test substance (nominal and analytical test concentrations, stability and adsorption on the surface of the test vessel, solubilising agent if relevant (and justification for its use), radiochemical purity if relevant);
- details on test conditions (e.g. soil/solution ratio, number of replicates and controls, sterility, test temperature, and pH of the aqueous phase before and after contact with the soil);
- details on sampling (e.g. frequency, method);
- details on the analytical methods used for determination of the substance (detection limit, recovery %).

Results and discussion

- soil dry mass, total volume of aqueous phase, concentration of test substance in solution and/or soil after agitation and centrifugation, equilibration time, K_{oc}, if appropriate mass balance;
- explanations of corrections made in the calculations, if relevant (e.g. blank run).

Leaching in soil columns (OECD TG 312)

Materials and methods

- details on soil types (nature and sampling site(s), organic C, clay content and soil texture, Cation Exchange Capacity, bulk density (for disturbed soil), water holding capacity and pH);
- information on the test substance (amount of test substance and, if appropriate, reference substance applied, solubilising agent if relevant (and justification for its use), radiochemical purity if relevant);
- details on test conditions (number of replicates and controls, test temperature, amount, frequency and duration of application of artificial rain);
- details on the analytical methods used for determination of the substance (detection limit, recovery %);
- reference substance used.

Results and discussion

- K_{oc}, tables of results expressed as concentrations and as % of applied dose for soil segments and leachates;
- mass balance, if appropriate;
- leachate volumes;
- leaching distances and, where appropriate, relative mobility factors.

Adsorption control within an inherent biodegradability test (OECD TG 302B)

Materials and methods

- details on inoculum;
- information on the test substance (toxicity to bacteria, test concentration);
- details on test conditions (blank controls used, inoculum and test compound ratio (as DOC));
- details on sampling (frequency);
- details on the analytical methods used for determination of the DOC or COD;
- reference substance.

Results and discussion

- estimate of the extent of adsorption to STP sludge made from the elimination level in this Zahn-Wellens inherent biodegradation test, based on the 3-hour value if possible;
- values beyond 24 hours should not normally be used but where data is not available for adsorption up to 24 hours, data from time scales beyond this can only be used if adsorption is the only removal mechanism, with an upper limit of 7 days;
- if relevant results of testing of inhibition of biodegradation.

Simulation test/field measurement (OECD TG 22)

Materials and methods

- details on soil types (nature and sampling site(s); if relevant: organic C, clay content and soil texture, Cation Exchange Capacity and pH);
- details on lysimeter;
- information on the test substance (nominal and analytical test concentrations, solubilising agent if relevant (and justification for its use), radiochemical purity if relevant);
- details on test climate conditions (e.g. air temperature, solar radiation, humidity, potential evaporation or rate of artificial rainfall), soil temperature and soil moisture and duration of the study;
- details on sampling (frequency, method);
- details on the analytical methods used for determination of the test substance (detection limit, recovery %).

Results and discussion

- concentration of test substance in soil layers; Koc, if appropriate mass balance and concentrations and as % of applied dose for soil segments and leachates;
- explanations of corrections made in the calculations, if relevant (e.g. blank run).

Distribution modelling

Materials and methods

- model name and version;
- date of the model development;
- model type description e.g. steady-state, dynamic, fugacity, Gaussian, Level I-IV, etc.;
- environmental compartments which the model covers;
- information on model segmentation and environmental properties;
- input parameters (minimum information required for assessing the partitioning and degradation behaviour):
 - vapour pressure;

- water solubility;
- molecular weight;
- octanol-water partition coefficient;
- information on ready biodegradability;
- for inorganic substances: it is recommended to have information on the partition coefficients and possible abiotic transformation products;
- temperature effect.

Results and discussion

- key exposure routes and distribution of the substance among them.

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on adsorption/desorption can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
5.4.1	VIII 9.3.1	Adsorption / desorption	E.5.5.2	4.1.4
5.4.2	/	Henry's Law constant	E.5.5.3	4.1.4
5.4.3	X 9.3.4	Distribution modelling	E.5.5.4	4.1.4
5.4.4	X 9.3.4	Other distribution data	E.5.5.5	4.1.4

R.7.1.15.7 Further information/references

Briggs G.G. (1981) Theoretical and experimental relationships between soil adsorption, octanol-water partition coefficients, water solubilities, bioconcentration factors and the parachor. *J. Agric. Food Chem.* 29, 1050-1059.

Dearden J.C. (2004) QSAR modelling of soil sorption. In Cronin M.T.D. and Livingstone D.J. (Eds.), *Predicting Chemical Toxicity and Fate*, CRC Press, Boca Raton, FL, pp. 357-371.

Delle Site, A., (2001) Factors affecting sorption of organic compounds in natural sorbent/water systems and sorption coefficients for selected pollutants. *J. Phys. Chem. Ref. Data* 30, 187-439.

Doucette W.J. (2000) Soil and sediment sorption coefficients. In Boethling R.S. and Mackay D. (Eds.), *Handbook of Property Estimation Methods for Chemicals: Environmental and Health Sciences*. Lewis, Boca Raton, FL, pp. 141-188.

Doucette W.J. (2003) Quantitative structure-activity relationships for predicting soil/sediment sorption coefficients for organic chemicals. *Environ. Toxicol. Chem.* 22, 1771-1788

ECETOC (1998). Technical Report No. 74: QSARs in the Assessment of the Environmental Fate and Effects of Chemicals. European Centre for Ecotoxicology and Toxicology of Chemicals. Brussels.

EU C.18 Adsorption – desorption using a batch equilibrium method.

EU C.19 Estimation of the adsorption co-efficient (K_{oc}) on soil and on sewage sludge using high performance liquid chromatography (HPLC).

Gerstl Z. Estimation of organic chemical sorption by soils. *J. Contaminant Hydrology* (1990) 6, 357-375.

ISO (2004). *Water quality: adsorption of substances on activated sludge – batch test using specific analytical methods*. International Standard ISO 18749. First edition February 2004.

Lyman W.J. Adsorption coefficient for soils and sediments. In Lyman W.J., Reehl W.F. and Rosenblatt D.H. (Eds.), *Handbook of Chemical Property Estimation Methods*, American Chemical Society, Washington DC, 1990, pp. 4.1-4.33.

Mueller, M. and Kordell, W. (1996). Comparison of screening methods for the estimation of adsorption coefficients on soil. *Chemosphere* 32(12), 2493-2504.

Nendza M. *Structure-Activity Relationships in Environmental Sciences*. Chapman & Hall, London, 1998.

OECD (2000a). Adsorption – desorption using a batch equilibrium method. Organisation for Economic Co-operation and Development (OECD) Guideline for the testing of chemicals 106.

OECD (2000b) Guidance Document No. 22: Performance of Outdoor Monolith Lysimeter Studies. Organisation for Economic Co-operation and Development (OECD), Paris.

OECD (2001). Estimation of the adsorption co-efficient (K_{oc}) on soil and on sewage sludge using high performance liquid chromatography (HPLC). Organisation for Economic Co-operation and Development (OECD) Guideline for the testing of chemicals 121.

OECD (2002) Guidelines for Testing of Chemicals (Draft): Leaching in Soil Columns. Organisation for Economic Co-operation and Development (OECD), Paris.

OPPTS (1996). Sediment and soil adsorption/desorption isotherm. United States Environmental Protection Agency, Office of Prevention, Pesticides and Toxic Substances Fate, Transport and Transformation Test Guideline 835.1220. Draft of April 1996.

Poole S.K. and Poole C.F. (1999) Chromatographic models for the sorption of neutral organic compounds by soil from air and water. *J. Chromatogr. A* 845, 381-400.

Reinhard M. and Drefahl A. (1999). *Handbook for Estimating Physico-chemical Properties of Organic Compounds*. Wiley, New York.

SETAC (1993). Guidance Document on Sediment Toxicity Tests and Bioassays for Freshwater and Marine Environments. In *Workshop on Sediment Toxicity Assessment at Renesse, Netherlands on 8-10 November 1993*. Hill I, Mathiessen P, Heimbach F (eds). Society of Environmental Toxicology and Chemistry – Europe, Brussels.

R.7.1.16 Stability in organic solvents and identity of relevant degradation products

R.7.1.16.1 Type of property

The stability in organic solvents is required for substances manufactured or imported in quantities of ≥ 100 t/a only if their stability in organic solvent is considered critical (REACH Annex IX, section 7.15).

There are rare occasions when it is important to have information on the stability of a compound in an organic solvent, to ensure confidence in the test results. However, for many substances, the stability in organic solvents will not be critical and testing need not be conducted.

Examples of when stability in organic solvents could be important are:

- for certain solubility measurements (e.g. octanol–water partition coefficient);
- to check on the stability of reagent solutions, fortification standards or calibration standards;
- when a test substance is dosed as a solution in an organic solvent (e.g. ecotoxicity studies);
- when a test substance is extracted from an environmental sample, plant or animal tissue or diet matrix (arising from a variety of physicochemical property, ecotoxicity and animal toxicity studies) into an organic solvent and stored pending analytical measurement.

R.7.1.16.2 Definition

A study of the stability of a test compound in an organic solvent is normally undertaken for a specific time period to confirm whether the test compound is stable under these conditions for the duration of the storage of the organic solvent or extract containing the test substance. Often several time periods are selected to check whether there is any particular downward trend in stability over time.

The stability of the test substance at a particular time period during the study is normally expressed as a percentage of the concentration of the test substance in the solvent extract, at that time period compared with the initial starting concentration of the test substance at $t = 0$, namely:

$$\frac{C_t}{C_0} \times 100 \%$$

where C_t is the concentration of test substance in solvent extract at $t = t_1, t_2, t_3, \dots, t_n$; and C_0 is the concentration of test substance in solvent extract at $t = 0$.

R.7.1.16.3 Test method(s)

A number of physical, chemical and biological processes can result in a decline in the actual concentration of a test substance in an organic solvent over time. Information on the stability of a test substance in a solvent is desirable, particularly when samples are to be stored. However, there does not appear to be any generally accepted methodology for performing

such stability studies. Factors affecting the rate of degradation include rates of hydrolysis, of photolysis and of oxidation.

Typically, one or more concentrations of the test substance in the solvent are made up and analysed immediately after preparation (i.e. $t = 0$). They are then stored in appropriate vessels under the required test conditions (e.g. temperature, absence of light) and analysed, along with a freshly prepared solution of the test substance at the original test concentration(s), at regular intervals during the period of interest.

At each time of analysis, a sample is withdrawn from storage and mixed thoroughly before taking any aliquot for analysis. The analysis is carried out using the recommended method to determine whether any significant loss of the test substance has occurred during storage. It is important to analyse freshly made standards of the test substance in the organic solvent at the same time as analysing stored samples, so that any losses that may occur of the test substance during sampling, sample treatment and analysis are taken into consideration.

It is important to be able to have a check on the temperature to ensure that the temperature regime has been maintained throughout the period of the stability study.

Unlabelled reference material of suitable known purity may be used where a reliable method of analysis is available. Where an analytical method is still under development or is unlikely to be sufficiently sensitive, radio-labelled compounds should be used if available. Use of radio-labelled compounds can shorten the analysis time and help facilitate identification of any degradation products, should the test substance not be stable in the organic solvent.

Recovery or spiking experiments should normally be run. The number of spiking levels or the range of concentrations tested within a project should be left to the judgement of the analyst.

Further information should be obtained by checks on the stability of standards of the test substance in organic solvents as part of routine analytical protocols, to confirm whether the test substance is unstable under normal storage conditions.

Further tests may be necessary to identify storage conditions which minimise any degradation of the test substance not only in organic solvents, but also during the conducting of other tests, such as water solubility, surface tension and in the preparation of test media for ecotoxicity studies (OECD, 2000). Identification of the degradation product(s) will allow an assessment of whether they are likely to be more toxic than the parent material in subsequent ecotoxicity studies.

R.7.1.16.4 Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex IX to REACH

Column 2 of REACH Annex IX provides the following specific rules for adaptation of the standard information requirement for stability in organic solvents and identity of relevant degradation products:

'The study does not need to be conducted if the substance is inorganic.'

Adaptation possibilities according to Annex XI to REACH

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

Stability data of substances in organic solvents are not normally reported in standard published sources of physicochemical data. Relevant sources of basic information regarding stability and storage conditions of substances are the Hazardous Substances Data Base (HSDB) and Sax's 'Dangerous Properties of Industrial Materials'.

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

(Q)SAR

At present (Q)SAR is generally not applicable for determination of stability in organic solvent and degradation products. Application of (Q)SAR is not possible.

Grouping of substances and read-across approach

At present grouping and read across are not applicable.

Testing is technically not possible

Testing should always be considered, if none of the waiving possibilities applies.

Further adaptation possibilities

Not foreseen.

R.7.1.16.5 Impurities; uncertainties

Impurities can have an impact on the measurement of stability in organic solvent and degradation products. Expert judgement should be used when considering whether impurities may affect the determination of the stability in organic solvent and degradation products. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

R.7.1.16.6 Endpoint specific information in the registration dossier / in IUCLID

This endpoint needs to be fulfilled on a case by case basis. As several different methods can be used to document this intrinsic property, we recommend the same strategy for drafting robust study summaries as described for the other endpoints. The general aspects described in section 2 of the Practical guide 3: How to report robust study summaries should also be applied for this endpoint. All endpoint specific characteristics should be described in such a way that the robust study summary allows an independent assessment of the endpoints reliability and completeness. The objectives, methods, results and conclusions of the full study report should be reported in a transparent manner as described for all other endpoints in this guidance.

Any deviation from the guideline method used or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on stability in organic solvents can be found in:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.17	IX 7.15	Stability in organic solvents and identity of relevant degradation products	E.4.18	3.15

R.7.1.16.7 Further information / references

OECD Series on Testing and Assessment Number 23 Guidance Document on Aquatic Toxicity Testing of Difficult Substances and Mixtures, ENV/JM/MONO(2000)6 ([http://www.oecd.org/officialdocuments/displaydocumentpdf?cote=env/jm/mono\(2000\)6&doclanguage=en](http://www.oecd.org/officialdocuments/displaydocumentpdf?cote=env/jm/mono(2000)6&doclanguage=en)).

R.7.1.17 Dissociation constant

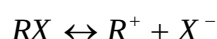
R.7.1.17.1 Type of property

Information on the dissociation constant is **supplementary data** for hazard assessment (OECD TG 112, 1981). The dissociation of a substance in water is of importance in assessing its impact upon the environment and may also influence the ADME of a substance and consequently its effects on human health. It governs the form of the substance which in turn determines its behaviour and transport. It may affect the adsorption of the substance on soils and sediments and absorption into biological cells.

The dissociation constant may also be an important factor in deciding which method or conditions should be used to determine the octanol-water partition coefficient (K_{ow}) and soil adsorption partition coefficient (K_{oc}). Slight changes in pH can considerably affect the form in which the substance is present in solution, especially if the pK_a value is within the environmentally-relevant pH range²¹. The dissociated and non-dissociated species may have significantly different water solubilities and partition coefficients. Therefore, significantly different bioavailability and toxicity may result. It is important to note that the dissolution of salts from their crystal lattice into individual ions is not intended to be covered by the endpoint dissociation constant. Therefore this section refers only to acid dissociation (pK_a).

R.7.1.17.2 Definition

Dissociation is the reversible splitting of a substance into two or more chemical species, which may be ionic (OECD TG 112, 1981). The process can be represented as:



The dissociation constant (K) for this process is expressed as the ratio of concentrations of the species on either side of the equation in water at equilibrium:

$$K = \frac{[R^+][X^-]}{[RX]}$$

Where the cation R^+ is hydrogen, the substance can be considered an acid, and so this constant becomes an acid dissociation constant (K_a).

$$K_a = \frac{[H^+][X^-]}{[HX]}$$

A substance can have more than one acidic (or basic²²) group, and the dissociation constant can be derived for each dissociation step in a similar way.

²¹ Fresh surface waters have pH values in the range 4-9, whereas marine environments have a stable pH of about 8. pH normally varies between 5.5 and 7.5 for agricultural soils and sewage treatment plant tanks.

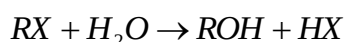
²² Base strength is expressed as the acidity of the conjugate acid. The term pK_b was once used to express basicity so that the same scale could be used alongside acidity – care should be taken when citing older sources to check which term has been used. For consistency, dissociation of bases should preferably be expressed using the pK_a of the conjugate acid.

The K_a is related to pH as follows (where p is $-\log_{10}$):

$$pK_a = pH - \log_{10} \left(\frac{[X^-]}{[HX]} \right)$$

In practice for a simple substance having one dissociating group, the pK_a is equivalent to the pH at which the ionised and non-ionised forms are present in equal concentration (i.e. the substance has undergone 50% dissociation).

It is important to differentiate between dissociation and hydrolysis as hydrolysis is a separate standard information requirement according to Annex VIII of the REACH regulation. Hydrolysis is defined as reaction of a substance RX with water, with the net exchange of the group X with OH at the reaction centre (OECD TG 111, 2004).



R.7.1.17.3 Test method(s)

OECD test guideline 112 (Dissociation constants in water, adopted May 1981) describes three laboratory methods to determine the pK_a of a substance. The three methods are appropriate for particular types of substances as described in the test guideline²³.

R.7.1.17.4 Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex IX to REACH

Column 2 of REACH Annex IX provides the following specific rules for adaptation of the standard information requirement for dissociation constant:

'A study does not need to be conducted if:

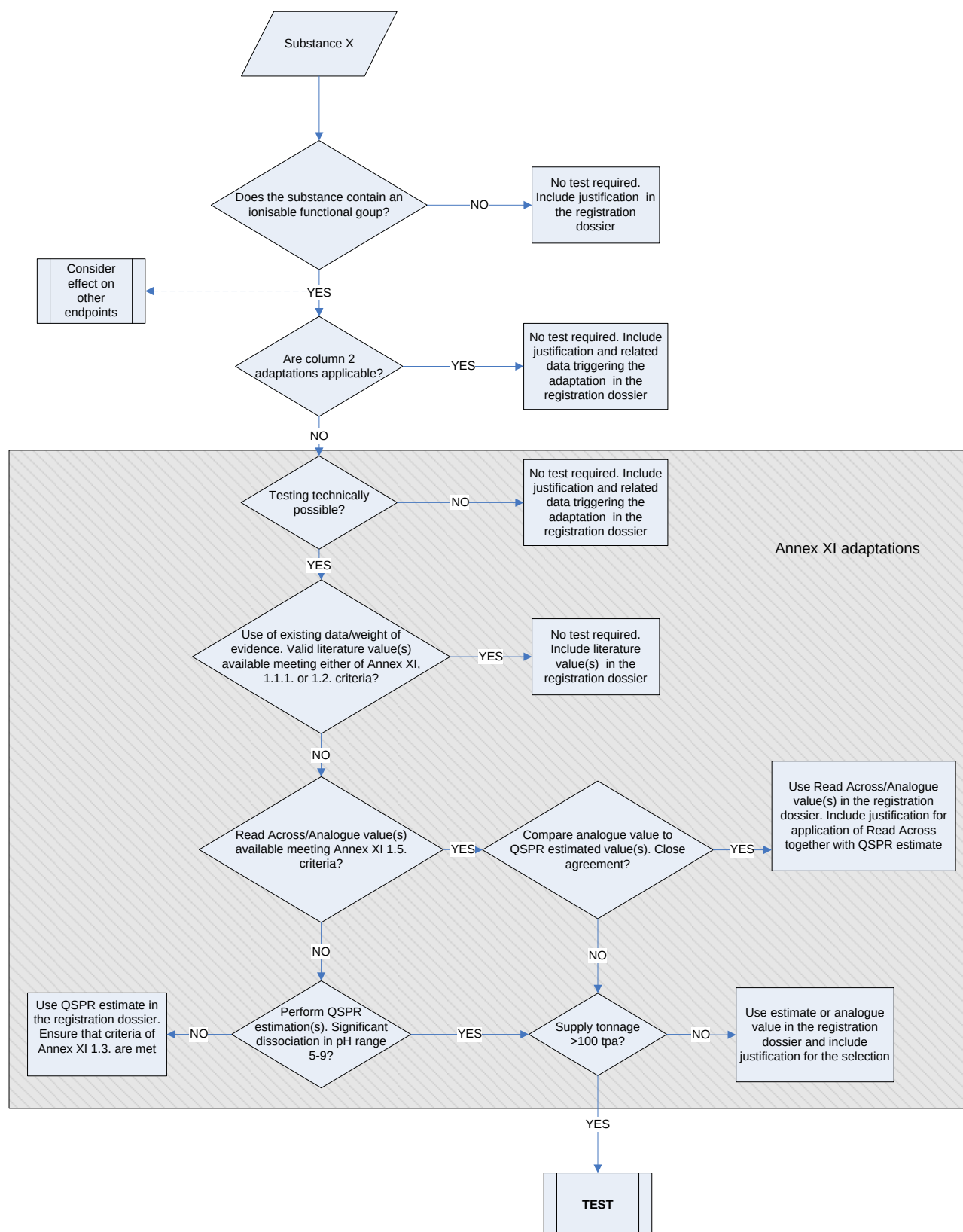
- the substance is hydrolytically unstable (half-life less than 12 hours) or is readily oxidisable in water; or*
- it is scientifically not possible to perform the test (e.g. because the analytical method is not sensitive enough).'*

In all cases where the above specific rules for adaptation are used to waive testing, evidence demonstrating the existence of that property of the substance which triggers the adaptation rule should be provided in the IUCLID dossier, e.g. if the test is not performed because the substance is hydrolytically unstable (half life < 12 hours) then the dossier must contain valid data on the hydrolysis clearly indicating a half life < 12 hours.

It is important to note that OECD TG 112 allows the use of a small amount of a water-miscible solvent to aid dissolution of sparingly soluble substances. Therefore low solubility will only prevent performance of the test in the context of the column 2 rules above for substances which remain highly insoluble and undetectable by analytical techniques in the presence of water miscible solvents.

²³ The test method is available at the following link: http://www.oecd-ilibrary.org/environment/test-no-112-dissociation-constants-in-water_9789264069725-en

Figure R.7.1-7 Integrated testing strategy for dissociation constant



Adaptation possibilities according to Annex XI to REACH

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

Many literature sources for dissociation constant exist; some reference textbooks and on-line sources are listed in Section [R.7.1.17.7](#). These should be searched for published, valid data. As mentioned in section 1.1.1 of Annex XI to REACH a number of conditions need to be met before any such data can be used. Namely:

- *'adequacy for the purpose of classification and/or risk assessment;*
- *sufficient documentation is provided to assess the adequacy of the study; and*
- *the data are valid for the endpoint being investigated and the study is performed using an acceptable level of quality assurance.'*

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used provided that data from a number of distinct sources indicate a similar value for the dissociation constant which is supported by one or more relevant QSPR predictions. The criteria in section 1.2 of Annex XI to REACH must then be met.

(Q)SAR

Estimated pK_a data can be generated by valid QSPR methods. In general, pK_a values that are measured with a suitable method are preferred to QSPR predictions. If an estimated pK_a value suggests that the substance will dissociate significantly at environmentally relevant pH, a test may be required to confirm the result.

There have been a few attempts to model pK_a values of diverse sets of substances. Klopman and Fercu (1994) used their MCASE methodology to model the pK_a values of a set of 2464 organic acids, and obtained good predictions; a test set of about 600 organic acids yielded a standard error of 0.5 pK_a unit. Klamt *et al.* (2003) employed their COSMO-RS methodology to predict pK_a values of 64 organic and inorganic acids, with a standard error of 0.49 pK_a unit. A comparison of commercially available software for the prediction of pK_a was done by Dearden *et al.* (2007).

Grouping of substances and read-across approach

For most ionisable substances supplied at greater than 100 t/y that are predicted to dissociate at environmentally relevant pHs, a test will typically be required for dissociation constant. Similar substances (analogues) for which measured pK_a data according to a reliable method are available may be considered for read-across. Such values should be reinforced by estimated methods for pK_a (e.g. the result of a QSPR prediction; see section above). In some instances it may be acceptable to read-across dissociation constant from an analogue. However if there is significant variation between the analogue read-across and the predicted pK_a then a test should be conducted.

Testing is technically not possible

Testing should always be considered if none of the waiving possibilities applies. Instances where testing is technically not possible as a consequence of the properties of the substance are expected to be limited to highly reactive or unstable substances, and substances which in contact with water emit flammable gases.

Further adaptation possibilities

As stated in Annex IX of REACH, when for certain endpoints, it is proposed to not provide information for other reasons than those mentioned in column 2 of that Annex or in Annex XI of REACH, this fact and the reasons must also be clearly stated. Such an approach may then be used.

No dissociating groups

If the substance cannot dissociate due to a lack of relevant functional groups, the dissociation constant is irrelevant and testing information does not need to be provided. However, ionisable groups might not always be obvious (e.g. in sulphonyl urea herbicides, which contain the function $-S(=O)_2NH.C(=O)NH-$, the acid group is $S(=O)_2NH$).



If a substance is much more soluble in water than expected, this may be an indication that dissociation has occurred.

UVCBs

For complex mixtures (e.g. UVCBs) containing ionisable components the assessment of pK_a is clearly complicated. Estimation of the representative constituent's pK_a values, if appropriate, should be considered.

R.7.1.17.5 Impurities; uncertainties

Impurities can have an impact on the measurement of dissociation constant. Expert judgement should be used when considering whether impurities may affect the determination of the dissociation constant. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

The presence of multiple dissociation/equilibrium reactions can complicate determination of the dissociation constant(s). In cases where multiple dissociation reactions can take place due to the presence of numerous dissociating groups and/or the presence of tautomerisation and/or zwitterionic forms, care should be taken in the interpretation of experimental results. QSPR predictions for such substances should also be carefully analysed as the models may not account for concurrent equilibria/dissociations. Additionally QSPR predictions may not account for intramolecular hydrogen bonding effects which can have a pronounced effect on the observed dissociation constant. In some cases, formation of intramolecular hydrogen bonding depends on the cis/trans isomerism of the substance, as is the case for the isomers fumaric and maleic acid. Care should be taken when using QSPR predictions for such molecules, as cis/trans isomerism is typically not taken into account.

The extent of ionisation may vary according to pH, ionic strength and/or the level of common ions in the test medium (common ion effect), and relatively small changes may significantly alter the equilibrium between dissociated and non-dissociated species.

R.7.1.17.6 Endpoint specific information in the registration dossier / in IUCLID

Knowledge of an ionisable substance's pK_a is important for all such substances. For substances supplied at levels below 100 tonnes per annum dissociation constant is not a testing requirement. Ideally however, a literature value, analogue value and/or QSPR prediction can be obtained and provided for such substances, especially if dissociation is relevant for interpreting the results of other physicochemical or fate and (eco)toxicological tests and for chemical safety assessment. For ionisable substances supplied at tonnages greater than 100 t/y, dissociation constant is a standard information requirement.

For substances which contain multiple ionisable functionalities, all measured macro pK_a values should be reported and preferably assigned to specific micro-reactions.

With regard to study summaries of experimental data, the IUCLID dossier should contain all relevant information regarding the endpoint and as a minimum the items listed below:

Materials and methods

- type of method;
- test guideline followed.

Test Materials

- test material identity.

Results and discussion

- concentration of the substance;
- test results as pK_a -value(s);
- temperature of the test medium ($^{\circ}C$);
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on dissociation constant can be found in:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Guide 3	Practical
4.21	IX 7.16	Dissociation constant	E.4.22	3.16	

R.7.1.17.7 References on dissociation constant

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R.7.1.18 Viscosity

R.7.1.18.1 Type of property

Viscosity is a property:

- needed for substance characterization;
- needed for the classification of aspiration hazard of liquids;
- which gives an indication of the penetration of the substance within soil.

R.7.1.18.2 Definition

Viscosity: viscosity is the (inner) resistance of a substance (gas, liquid) to a shift caused by laminar flow.

Dynamic viscosity (= dynamic viscosity coefficient) η :

Quantifies the property 'viscosity' by the quotient shear stress τ / shear rate $\dot{\gamma}$ ($\eta = \tau / \dot{\gamma}$)

Kinematic viscosity (= kinematic viscosity coefficient) ν :

is given by the quotient dynamic viscosity to density ($\nu = \eta / \rho$).

R.7.1.18.3 Test method(s)

Five different types of test methods are standardized for liquid substances:

- capillary viscometer;
- flow cup;
- rotational viscometer;
- rolling ball viscometer;
- drawn-shear viscometer.

There exist a lot of standardized determination methods with sometimes very specialised application ranges with respect to products, especially mixtures. For substances (within the scope of the REACH Regulation) the following standardised determination methods are recommended:

- Capillary viscometer: EN ISO 3104, EN ISO 3105, DIN 51562, BS 188, NF 60-100, ASTM D445, ASTM D4486;
- Flowcup: EN ISO 2431;
- Rotational viscometer: EN ISO 3219, DIN 53019;
- Rolling ball viscometer: DIN 53015.

For newtonian liquids (liquids for which the viscosity is independent of the shear stress and shear rate) any determination method may be used within the scope and applicability specifications. For non-newtonian liquids (liquids for which the viscosity depends on the shear rate) only the use of rotational viscometers is possible. Because the viscosity is remarkably

temperature dependent each determination must be accompanied by the temperature at which the measurement was made. It is recommended to use the mean of two test runs. It is also recommended to determine the viscosity at at least two different temperatures. The classification criteria for aspiration hazard refer to kinematic viscosity at 40°C.

If explosives, pyrophorics or self-reactives are to be characterized, determination of the viscosity may not be practicable. For pyrophorics and self-reactives testing under inert gas should be considered. In any case the determination method has to be chosen carefully.

The use of the most recent update of the standard is advised; they are accessible *via* numerous websites, see [R.7.1.1.3](#).

R.7.1.18.4 Adaptation of the standart testing regime

Within the REACH Regulation requirements testing of viscosity is only of interest for liquid substances.

Adaptation possibilities according to column 2 of Annex IX to REACH

Column 2 of REACH Annex IX does not provide any specific rules for adaptation from column 1.

Adaptation possibilities according to Annex XI to REACH

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

(Q)SAR

For the determination of the viscosity, (Q)SAR approaches are discouraged for the purpose of classification / risk assessment, except when the mean absolute error of the (Q)SAR is less than 5 %.

Grouping of substances and read-across approach

For the determination of the viscosity read across is not possible.

Testing is technically not possible

Testing should always be considered if none of the waiving possibilities applies. But the testing is technically not possible:

- if the substance is a solid;

- if liquid explosives, pyrophorics or self-reactives are to be characterized, determination of the viscosity may not be practicable (see above section Test method(s)).

Further adaptation possibilities

- the viscosity does not have to be determined experimentally if conclusive and consistent literature data are available;
- data for viscosity generated with the same tests and classification principles as specified in the CLP Regulation generated in conjunction with transport classification can satisfy the REACH requirements on a case-by-case basis.

As stated in Annex IX of REACH, when for certain endpoints, it is proposed to not provide information for other reasons than those mentioned in column 2 of that Annex or in Annex XI of REACH, this fact and the reasons must also be clearly stated. Such an approach may then be used.

R.7.1.18.5 Impurities; uncertainties

The influence of impurities is negligible if their concentration is below 1 %. The influence of higher concentrations may be significant. There exists no generalised tendency of the influence on the viscosity. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

R.7.1.18.6 Endpoint specific information in the registration dossier / in IUCLID

Materials and methods

- type of method;
- test guideline followed.

Results and discussion

- viscosity value and unit according to the used test method;
- preferred units are m Pa·s (for dynamic viscosity) and mm²/s (for static viscosity) but other units are also accepted;
- each measured value should be accompanied with temperature (in °C). Usually two values are needed. Preferably one value is measured at approximately 20°C and another at an approximately 20°C higher temperature. Two determinations of viscosity should be measured for each temperature;
- for non-Newtonian liquids, the results obtained are preferably in the form of flow curves, which should be interpreted;
- individual and mean values should be provided at each temperature (from OECD Guideline 114 'Viscosity of liquids').

Any deviation from the guideline method used or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on viscosity can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Guide 3	Practical
4.22	IX 7.17	Viscosity	E.4.23	3.17	

R.7.1.19 Shape

Please check *Appendix R7-1 Recommendations for nanomaterials applicable to: Chapter R7a Endpoint specific guidance* of the [Guidance on IR&CSA](#), section 2.2.3.3 Recommendations for shape.

R.7.1.20 Surface area

Please check *Appendix R7-1 Recommendations for nanomaterials applicable to: Chapter R7a Endpoint specific guidance* of the [Guidance on IR&CSA](#), section 2.2.3.4 Recommendations for surface area.

R.7.1.21 Further information to be submitted for classification and labelling in hazard classes of the substance in accordance with article 10 (a) (iv) REACH

The criteria listed in the table below ([Table R.7.1-15](#)) should be provided for general registration purposes according to Article 10 (a) (iv) and section 4 of Annex VI to REACH. The assignment of hazard classes to relevant subchapters in [0](#) to [R.7.1.21.3](#) should therefore only be understood as a means to structure this document in accordance with Annexes VII to XI to the REACH Regulation.

Table R.7.1-15 Information to be submitted for general registration purposes according to Article 10 (a) (iv) REACH, CLP hazards classes and corresponding tests methods according to the Test Method Regulation and CLP²⁴

CLP Regulation (EC) No. 1272/2008 (the no. in brackets is the respective chapter no. in Annex I to CLP)	Corresponding test method according to the Test Method Regulation, Regulation (EC) No. 440/2008	Chapter in revised R.7(a) guidance	Information requirement according to REACH Regulation (EC) No. 1907/2006	Corresponding test method according to CLP Regulation
Flammable aerosols (2.3) ²⁵	n.a.	R.7.1.21.1	See Article 10 (a) (iv) REACH requirements	Test methods according to 75/324/EC amended by 2008/47/EC (harmonised with UN-MTC Section 31)
Gases under pressure (2.5)	n.a.	R.7.1.21.2	See Article 10 (a) (iv) REACH requirements	n.a.
Corrosive to metals (2.16)	n.a.	R.7.1.21.3	See Article 10 (a) (iv) REACH requirements	UN Test C.1 (UN-MTC Section 37.4)

²⁴ Please note that REACH information requirements regarding classification and labelling in accordance with Article 10(a)(iv) of the REACH Regulation are not limited to the items listed in this table. This table stresses that, while the REACH Regulation does not require the generation of information regarding the following hazard classes (Article 10(a)(vi) of the REACH Regulation, see [Table R.7.1-1](#)), any information available on these hazard classes must be included in a REACH registration dossier for a substance pursuant to Article 10(a)(iv) of the REACH Regulation.

²⁵ The 4th ATP to the CLP Regulation amends the criteria in the CLP Annex I, Section 2.3 Flammable aerosols by changing the scope and title to Section 2.3 Aerosols.

R.7.1.21.1 Flammable aerosols

For further guidance on these please check the [Guidance on the Application of the CLP criteria](#), chapter 2.3²⁶.

R.7.1.21.2 Gases under pressure

For further guidance please check the [Guidance on the Application of the CLP criteria](#) chapter 2.5.

R.7.1.21.3 Corrosive to metals

For further guidance please check the [Guidance on the Application of the CLP criteria](#) chapter 2.16.

²⁶ The 4th ATP to the CLP Regulation amends the criteria in the CLP Annex I, Section 2.3 Flammable aerosols by changing the scope and title to Section 2.3 Aerosols. Consequently the [Guidance on the Application of the CLP criteria](#), Part 2: Physical hazards has been restructured to take account of the 4th ATP, which applies to substances from 1 December 2014 and to mixtures from 1 June 2015. Once the 4th ATP is applied a Guidance corrigendum will be made to delete the outdated sub-chapter 2.3.1 Flammable aerosols in the [Guidance on the Application of the CLP criteria](#).

Appendix R.7.1-1 to Section R.7.1

Appendix R.7.1–1 Henry's law constant and evaporation rate

The Henry's law constant (HLC) is one of the most important factors in determining the environmental fate of chemicals. Henry's law states that the mass of gas dissolved by a given volume of solvent is proportional to the pressure of the gas with which it is in equilibrium. HLC is the ratio of the equilibrium concentration of the chemical in the gas phase (C_G) and that in the liquid phase (C_L):

$$\text{HLC} = \frac{C_G}{C_L}$$

Therefore, HLC quantifies the partitioning of substances between the aqueous phase and the gas phase such as rivers, lakes and seas with respect to the atmosphere (gas phase). Indeed, this constant is a fundamental input for fugacity models that estimate the multimedia partitioning of chemicals (Mackay, 1991). As HLC is a ratio of two concentrations, it is without unit if both concentrations are expressed in the same unit. Some prefer to express the gas concentration in pascals and the liquid concentration in mol/m³, thus giving the unit Pa·m³/mol for the HLC.

For many chemicals, volatilisation can be an extremely important removal process, with half lives as low as several hours. HLCs can give qualitative indications of the importance of volatilisation. For substances with HLC values less than 0.01 Pa·m³/mol, the substance is less volatile than water and as water evaporates the concentration of the substance in the aqueous phase will increase; for substances with HLC values around 100 Pa·m³/mol, volatilisation will be rapid.

However, the degree of volatilisation of substances from the aquatic environment is highly dependent on the environmental parameters for the specific water bodies in question, such as the depth and the gas exchange coefficient (influenced e.g. by wind speed and water flow rate). The HLC cannot be used for evaluation of the removal of a substance from the water phase without considering these factors. As the n-octanol/water partition coefficient (K_{ow}) is used to predict bioaccumulation potential in air-breathing organisms, this aspect is especially important in a PBT context.

For example, where a substance has both a low vapour pressure and low water solubility, HLC can be relatively large if calculated using the ratio of vapour pressure and water solubility, which might imply that volatilisation is an important fate process. In practice, adsorption to dissolved organic carbon is likely to be much more relevant, and volatilisation will be lower than the HLC value suggests.

Experimental determination of Henry's law constant

The experimental approaches can be classified into two major groups: dynamic equilibration approach (often referred to as the *gas purge* approach) and the static equilibration approach. The following table ([Table R.7.1–16](#)) briefly summarises the reviewing work done by Staudinger and Roberts (1996).

Table R.7.1–16 Experimental approaches for the determination of HLC

Approach	Average Relative Standard Deviations (RSDs)/Notes
Dynamic approach	
<i>Batch air stripping (bubble column)</i> Henry's law constant (HLC) values are determined by measuring the rate of loss of the substance of interest from water by isothermally stripping with a gas (typically air) in a suitable bubble column apparatus.	Average RSDs determined from different literature sources ranged from 2.8 to 21
<i>Concurrent flow (wetted wall column)</i> Values are determined based on the use of a wetted wall (desorption) column. The wetted wall column equilibrates an organic solute between a thin film of water and a concurrent flow of gas. Substance-laden water is introduced into the wetted wall column where it comes in contact with a substance-free gas stream flowing concurrently. HLC: The knowledge of flow rates and compound masses present in the separated phase streams enables the direct calculation of HLC.	Average RSDs determined from different literature sources ranged from 19 to 52 Preliminary work must be performed to ensure that phase equilibrium is reached.
Static approach	
<i>Single equilibration</i> A known mass of a substance is introduced into an air-tight vessel with a known volume of water and air. When the equilibrium is attained the substance concentration is determined in one or both phases.	Average RSDs determined from different literature sources ranged from 2.8 to 30
<i>Multiple Equilibration</i> A liquid sample containing a known quantity of solute is allowed to equilibrate with a known volume of solute-free air. The air is expelled and a new equilibration with the same amount of solute-free air is started. This process can be repeated until the number of equilibrations exhausts the mass of solute remaining in the system.	RSDs ranged from 0.7 to 3.5 This method is applicable for substances with $0.1 \leq \text{HLC} \leq 2$ The experimental error is reduced with a larger number of equilibrations.
<i>EPICS Technique</i> HLC is determined by measuring the gas headspace concentration ratios from pairs of sealed bottles. Relative rather than absolute air-phase concentrations are required.	Average RSDs determined from different literature sources ranged from 2.9 to 19
<i>Variable Headspace</i> The method is based upon the measurement of the relative equilibrium air-phase concentration (gas chromatography peak areas) from aliquots of the same solution in multiple containers having different headspace-to-liquid volume ratios.	Average RSDs determined from different literature sources ranged from 0.5 to 7.9

A data-analysis of reviewed experimental studies for HLC can be found in Staudinger and Roberts (1996). HLC values can also be found in one or more of the following references: Sander (1999), CRC Handbook of Chemistry and Physics (2000), the NIST Chemistry WebBook

(1998), and 'The Handbook of Environmental Data on Organic Chemicals' (Verschuere K, 2001).

Main factors affecting Henry's Law Constant values

Staudinger and Roberts (1996) thoroughly explain all the factors affecting HLC values and report equations that quantify the effect of temperature and pH. According to their work, in a majority of cases temperature is the main parameter affecting HLC values for natural waters with moderate contamination (1 mg/ml or less). Other conditions that have influence on HLC values are listed in [Table R.7.1-17](#) (Staudinger and Roberts, 1996):

Table R.7.1-17 Conditions that have influence on HLC values

pH	Important for compound (substance) classes that dissociate to a significant extent in water because only nondissociated species undergo air-water exchange. For most natural waters ($6 < \text{pH} < 8$) the apparent HLC will be significantly less than the intrinsic HLC.
Compound Hydration	Important for aldehydes, which hydrate nearly completely in water, resulting in HLC apparent being several orders of magnitude lower than the intrinsic constant.
Compound concentration/ Complex mixtures effects	If a solution cannot be regarded as diluted (e.g. concentration approaching 10.0 mg/ml) HLC apparent will be lower than HLC values determined at lower concentrations.
Dissolved salts	If the ionic strength of a solution is high (e.g. seawater) the apparent HLC will be higher than the HLC determined in pure water.
Suspended solids /Dissolved Organic Matter (DOM)	If a compound is easily adsorbed (e.g. pesticides) the apparent HLC will be higher than the HLC determined in pure water.
Surfactants	Compounds with high K_{ow} are expected to have an effect on HLC by lowering its value. Recorded effects increase in direct proportion with K_{ow} .

It is worth noting that because of the complex nature of the water matrix the net effect of a possible combination of the parameters listed above may be more than the simple sum of individual effects (Staudinger and Roberts, 1996).

QSPR prediction of Henry's law constant

The prediction of HLC has been reviewed by Schwarzenbach *et al.* (1993), Reinhard and Drefahl (1999), Mackay *et al.* (2000) and Dearden and Schüürmann (2003). The most important approaches are:

- Ratio of water solubility (c_w) to vapour pressure (vp);
- Estimation using connectivity indices;
- Estimation using group and bond contribution methods.

The first method for estimating HLC is not strictly a QSAR method as it uses the water solubility (c_w) and vapour pressure (vp). It is not a highly accurate method, but neither is the measurement of HLC, especially for substances with very high or very low HLC values. vp/c_w

can be converted to the dimensionless form of HLC (ratio of concentrations in air and water, c_a/c_w) or K_{aw} by the following equation, which is valid at 25°C:

$$c_a/c_w = 40.874 \text{ vp}/c_w$$

Since both water solubility and vapour pressure can be calculated by QSAR methods, then this approach might in some circumstances be a QSAR based method. The method is limited to substances of low water solubility ($< 1.0 \text{ mol/L}$). If QSAR calculated values are used for vp and/or c_w , then the respective uncertainties must be considered. For miscible compounds or compounds with water solubility $> 1 \text{ mol/L}$ the vp/c_w method is not valid.

The second method is based on a combination of connectivity indices and calculated polarisability (Nirmalakhandan and Speece, 1988). A relatively narrow range of chemical types was used to develop the model, so it is not widely applicable. Moreover, Schüürmann and Rothenbacher (1992) found it to have poor predictive power.

Most prediction methods for HLC use a group or bond contribution approach, although some have used physicochemical properties (Dearden *et al.* 2000). The group and bond contribution methods were first used by Hine and Mookerjee (1974), who obtained, for a set of 263 diverse simple organic chemicals, a standard deviation of 0.41 log unit for the group contribution method and one of 0.42 for the bond contribution method. Cabani *et al.* (1981) claimed an improvement in the group contribution method over that of Hine and Mookerjee, whilst Meylan and Howard (1991) extended the bond contribution method and obtained, for a set of 345 diverse chemicals, a standard error of 0.34 log unit.

Evaporation rate

Evaporation rates generally have an inverse relationship to boiling points, i.e. the higher the boiling point, the lower the rate of evaporation. Knowledge of the evaporation rate of spills of volatile liquids can be useful in several respects. If it is known that a spill of a high vapour pressure liquid will evaporate completely in a short period of time, it may be preferable to isolate the area and avoid any intervention or clean-up. The evaporation rate also controls the atmospheric concentration of the vapour and hence the threat of explosion or fire. Data on the volatility properties of the liquid, its temperature, the wind speed, and the spill dimensions are used to calculate the evaporation rate and hence the fraction evaporated at any time.

The substance's tendency to partition into the atmosphere is controlled by the vapour pressure, which is essentially the maximum vapour pressure that a pure substance can exert in the atmosphere. This can be viewed as a kind of *solubility* of the chemical in the atmosphere. Using the ideal gas law ($PV=nRT$), the vapour pressure P in the pressure unit pascal (Pa) can be converted into a solubility (mol/m^3), where the gas constant R is $8.314 \text{ Pa}\cdot\text{m}^3/\text{mol}\cdot\text{K}$ and T is absolute temperature (K).

Conversion from vapour pressure into concentration in air under ambient temperature:

$$\% \text{ volume} = \text{vapour pressure (Pa)} / 101\,325 \times 100$$

$$\text{or ppm} = \text{vapour pressure (Pa)} / 101\,325 \times 1\,000\,000$$

Since the molar volume is the same for all ideal gases (equal volumes of all gases under the same conditions of temperature and pressure contain the same number of molecule) $\text{ppm} \equiv \text{volume (i.e. ml/m}^3\text{)}$. To convert to weight per unit volume:

$$X \text{ ppm} = X \times \text{MW} / 24.041 \text{ mg/m}^3, 1 \text{ mg/m}^3 = 24.041 / \text{MW ppm}$$

In the formulation of paints and related products, solvents are chosen based on their evaporation characteristics appropriate to the application technique and the curing

temperature. To a large extent the evaporation rate of a solvent determines where and how it can be used. In determining the evaporation rate of solvents, n-butyl acetate is used as the standard and is assigned an evaporation rate value of 1. Other solvents are assigned evaporation rate values that indicate how fast they evaporate in relation to n-butyl acetate. For instance, a solvent that evaporates three times as fast as n-butyl acetate would be assigned a value of 3, whereas a solvent that evaporates half as fast as n-butyl acetate would be assigned a value of 0.5.

The rate of evaporation is determined using ASTM D3539-87. A known volume of liquid is spread on a known area of filter paper that is suspended from a sensitive balance in a cabinet. Dry air or nitrogen at 25 °C is passed through the cabinet at a known rate. The loss of weight is determined and plotted against time.

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R.7.2 Skin corrosion/irritation, serious eye damage/eye irritation and respiratory tract corrosion/irritation

R.7.2.1 Introduction

Irrespective of whether a substance can become systemically available, changes at the site of first contact (skin, eye, mucous membrane/ gastro-intestinal tract, or mucous membrane/ respiratory tract) can be caused by exposure to a substance. These changes are considered local effects. A distinction in local effects can be made between those observed after single and those after repeated exposure. In this guidance document, the focus will be on local effects after single ocular, dermal or inhalatory exposure. However, wherever possible, use should also be made of existing repeated dose data insofar as they may contain valuable information for the purpose of assessing and classifying effects after single ocular, dermal or inhalatory exposure.

Substances causing local effects after single exposure can be further distinguished as irritant or corrosive substances, depending on the severity, reversibility or irreversibility of the effects observed. Corrosive substances are those which may destroy living tissues with which they come into contact. Irritant substances are non-corrosive substances which, through immediate contact with the tissue under consideration may cause inflammation (see Section [R.7.2.1.1](#) for complete definitions). These tissues are in the present context skin, eye (cornea, iris and conjunctiva) and mucous epithelia such as the respiratory tract. Criteria for classification of irritant and corrosive substances are given in Annex I to Regulation (EC) No 1272/2008 on Classification, Labelling and Packaging of substances and mixtures (CLP Regulation).

Certain substances may also cause irritant effects only after repeated exposure, for example organic solvents. This type of substance may have defatting properties (Ad-hoc Working group on Defatting substances, 1997). Substances that have a similar mode of action need to be considered for labelling with the supplemental statement EUH066 "*Repeated exposure may cause skin dryness or cracking*".

Information on the mechanisms underlying corrosion and irritation of skin, eye and respiratory tract is given in [Appendix R.7.2-1 Mechanisms of local toxicities: skin corrosion/irritation, serious eye damage/eye irritation and respiratory tract corrosion/irritation](#).

R.7.2.1.1 Definitions of skin corrosion/irritation, serious eye damage/eye irritation and respiratory tract corrosion/irritation

Definitions of skin corrosion/irritation, serious eye damage/eye irritation and respiratory tract corrosion/irritation can be found in the CLP Regulation ²⁷.

Skin irritation: Defined in Section 3.2.1.1 of Annex I to the CLP Regulation as "[...] *the production of reversible damage of the skin following the application of a test substance for up to 4 hours.*".

²⁷ Please note that the 8th Adaptation to Technical and Scientific Progress (ATP) of the CLP Regulation will apply from 1 February 2018. The amendment by the 8th ATP will take into account the 5th Revision of the Globally Harmonized System of Classification and Labelling of Chemicals (GHS), which was adopted in 2012 and contains in particular refined criteria for skin corrosion/irritation and serious eye damage/eye irritation.

Dermal concern after repeated exposure: Used for a substance which may cause skin dryness, flaking or cracking upon repeated exposure but which cannot be considered as skin irritant (see Section 1.2.4 of Annex II to the CLP Regulation).

Skin corrosion: Defined in Section 3.2.1.1 of Annex I to the CLP Regulation as "[...] *the production of irreversible damage to skin; namely, visible necrosis through the epidermis and into the dermis, following the application of a test substance for up to four hours. Corrosive reactions are typified by ulcers, bleeding, bloody scabs, and, by the end of observation at 14 days, by discolouration due to blanching of the skin, complete areas of alopecia, and scars. [...]*".

Eye irritation: Defined in Section 3.3.1.1 of Annex I to the CLP Regulation as "[...] *the production of changes in the eye following application of a test substance to the anterior surface of the eye, which are fully reversible within 21 days of application.*".

Serious eye damage: Defined in Section 3.3.1.1 of Annex I to the CLP Regulation as "[...] *the production of tissue damage in the eye, or serious physical decay of vision, following application of a test substance to the anterior surface of the eye, which is not fully reversible within 21 days of application. [...]*".

Respiratory tract irritation: There is no EU or OECD TG for respiratory tract irritation and testing for respiratory tract irritation is not a standard information requirement under REACH. Respiratory tract irritation is considered under the CLP Regulation (Table 3.8.1 of Annex I) as a transient target organ effect, i.e. an "[...] *effect which adversely alter[s] human function for a short duration after exposure and from which humans may recover in a reasonable period without leaving significant alteration of structure or function. [...]*". More specifically, respiratory tract irritation is often used to describe either or both of two different toxicological effects, *sensory irritation* and *local cytotoxic effects*. However, classification in STOT SE Category 3 for respiratory tract irritation is generally limited to local cytotoxic effects. "[...] *Respiratory irritant effects [are] characterised [by] localised redness, oedema, pruritis and/or pain and they impair function with symptoms such as cough, pain, choking, and breathing difficulties [...]*" (see Section 3.8.2.2.1 of Annex I to the CLP Regulation).

Respiratory tract corrosion: There is no EU or OECD TG for respiratory tract corrosion and testing for respiratory tract corrosion is not a standard information requirement under REACH. Respiratory tract corrosion is defined in Section 3.1.2.3.3 of Annex I to the CLP Regulation as "[...] *destruction of the respiratory tract tissue after a single, limited period of exposure analogous to skin corrosion; this includes destruction of the mucosa. [...]*".

Classification and labelling under the CLP Regulation:

Substances and mixtures causing skin corrosion/irritation, serious eye damage/eye irritation and/or respiratory tract corrosion/irritation can be further characterised by their classification under the CLP Regulation ²⁸.

Detailed information on the classification and labelling of substances and mixtures can be found in the [Guidance on the Application of the CLP criteria](#).

²⁸ Please note that the 8th Adaptation to Technical and Scientific Progress (ATP) of the CLP Regulation will apply from 1 February 2018. The amendment by the 8th ATP will take into account the 5th Revision of the Globally Harmonized System of Classification and Labelling of Chemicals (GHS), which was adopted in 2012 and contains in particular refined criteria for skin corrosion/irritation and serious eye damage/eye irritation.

a) For Skin effects

- **Skin corrosives** are classified in Category 1 with the Hazard statement H314 "*Causes severe skin burns and eye damage*". Further subcategorisation is defined based on the Draize skin corrosion *in vivo* test:
 - Subcategory 1A: Destruction of skin tissue occurs after exposure times ≤ 3 minutes and is observed within a period ≤ 1 hour after exposure,
 - Subcategory 1B: Destruction of skin tissue occurs after exposure times > 3 minutes and ≤ 1 hour and is observed within a period ≤ 14 days after exposure,
 - Subcategory 1C: Destruction of skin tissue occurs after exposure times > 1 hour and ≤ 4 hours and is observed within a period ≤ 14 days after exposure.
- **Skin irritants** are classified in Category 2 with the Hazard statement H315 "*Causes skin irritation*".

b) For Eye effects

- **Substances or mixtures causing serious eye damage** are classified in Category 1 with the Hazard statement H318 "*Causes serious eye damage*".
- **Substances or mixtures causing eye irritation** are classified in Category 2 with the Hazard statement H319 "*Causes serious eye irritation*".

c) For Specific Target Organ Toxicity with relevance to the respiratory tract

- **Substances or mixtures causing respiratory tract corrosion** are classified for Acute Toxicity by inhalation and labelled as EUH071 "*Corrosive to the respiratory tract*" if the corrosive effect causes the death of the animals within the criteria for Acute toxicity, or in Specific Target Organ Toxicity after Single Exposure (STOT SE) Category 1 (with the Hazard statement H370 "*Causes damage to the respiratory tract*") or Category 2 (with the Hazard statement H371 "*May cause damage to the respiratory tract*"), depending on the dose level required to cause the toxic effects.
- **Substances or mixtures causing respiratory tract irritation** *via* a local cytotoxic effect are classified in Specific Target Organ Toxicity after Single Exposure (STOT SE) Category 3 with the Hazard statement H335 "*May cause respiratory irritation*".

According to Section 1.2.6 of Annex II to the CLP Regulation, the Hazard statement EUH071 must also be applied to inhaled substances or mixtures classified for skin corrosion and not tested for acute inhalation toxicity.

Note that dermal and respiratory tract irritation following repeated exposure are not discussed in the present context, since this Guidance focuses on acute effects after single exposure. However, data from repeated exposure studies may be useful in certain cases (e.g. if the

substance was identified as a corrosive or strong irritant after the first application or for deriving quantitative information). Nevertheless, for the sake of completeness, both the definition of dermal irritation after repeated exposure as well as the related Hazard Statement EUH066 ("Repeated exposure may cause skin dryness or cracking") are given here. More guidance on local effects after repeated exposure can be found in Section [R.7.5](#) on repeated dose toxicity.

R.7.2.1.2 Objective of the guidance on skin corrosion/irritation, serious eye damage/eye irritation and respiratory tract corrosion/irritation

The general objectives are:

- a. to establish whether information from physical/chemical data, from non-testing methods (grouping, QSARs and expert systems), from *in vitro* studies, from animal studies or human experience provides evidence that the substance is, or is likely to be, corrosive.
- b. to establish whether information from physical/chemical data, from non-testing methods (grouping, QSARs and expert systems), from *in vitro* studies, from animal studies or human experience provides evidence of significant skin, eye or respiratory tract irritation.
- c. to establish if possible the time of onset and the extent and severity of the responses and information on reversibility.
- d. If possible to gather, in the process of hazard identification, any quantitative data on dose-response relationships that might allow the derivation of DNELs essential for a complete risk assessment.

If a risk assessment is necessary, both the severity of the identified hazard (in so far as it can be judged from the test data) and the probability of the occurrence of an acute corrosive or irritant response in humans must be assessed based on the likelihood of any exposure to the substance and in relation to the route, pattern and extent of the expected exposure.

Please note that there are currently no standard tests and no OECD TGs available for acute respiratory tract irritation and there is no testing requirement for respiratory tract irritation under the REACH Regulation. Consequently no testing and assessment strategy for respiratory tract corrosion/irritation is included in this guidance. Nevertheless, account should be taken of any existing and available data that provide evidence of the respiratory tract corrosion/irritation potential of a substance. For instance, acute inhalation studies including histopathological evaluation of the respiratory tract and/or examinations of nasal or bronchioalveolar lavage as well as repeated inhalation studies may provide important information for classification and labelling (See Section [R.7.2.12](#) for further details).

SKIN CORROSION/IRRITATION

R.7.2.2 Information requirements on skin corrosion/irritation

The information on skin corrosion/irritation that is required to be submitted for registration and evaluation purposes is specified in Annexes VI to XI to the REACH Regulation. According to Annex VI, the registrant should gather and evaluate all existing available information before considering further testing. This includes physico-chemical properties, (Q)SAR ((Quantitative) Structure-Activity Relationship), grouping, *in vitro* data, animal studies, and human data. For classified substances, information on exposure, use and risk management measures should also be collected and evaluated in order to ensure safe use of the substance.

If these data are inadequate for hazard and risk assessment, further testing should be carried out in accordance with the requirements of Annexes VII (≥ 1 tpa) and VIII (≥ 10 tpa) to the REACH Regulation.

R.7.2.2.1 Information requirements for quantities of ≥ 1 tpa (Annex VII to the REACH Regulation)

If new testing data are necessary, these must be derived from *in vitro* methods only. Annex VII does not foresee *in vivo* testing for skin corrosion/irritation.

The standard information requirements at this tonnage level for skin corrosion/irritation are specified in Section 8.1 in Column 1 of Annex VII as follows:

8.1.1. Skin corrosion, *in vitro*

8.1.2. Skin irritation, *in vitro*

Section 8.1 in Column 2 of Annex VII lists specific rules for adaptation according to which steps 8.1.1. and 8.1.2. do not need to be conducted. These rules are applicable when:

- the substance is a strong acid ($\text{pH} \leq 2.0$) or base ($\text{pH} \geq 11.5$) and the available information indicates that it should be classified as skin corrosive (Category 1), or
- the substance is spontaneously flammable in air or in contact with water or moisture at room temperature, or
- the substance is classified as acutely toxic by the dermal route (Category 1), or
- an acute toxicity study by the dermal route does not indicate skin irritation up to the limit dose level (2000 mg/kg body weight) (Please see footnote d to [Figure R.7.2-2](#) for further information).

If results from one of the two studies under point 8.1.1 or 8.1.2 already allow a conclusive decision on the classification of the substance or on the absence of skin irritation potential, the second study need not be conducted.

The *in vitro* methods that can be used to fulfil the standard information requirements of REACH Annex VII are detailed in Sections [R.7.2.3.1](#) and [R.7.2.4.1](#) of this Guidance, under “*In vitro* data”. In case an existing good quality *in vivo* skin irritation study is available, its results can be used to fulfil the standard information requirement, however an adaptation argument for not providing the *in vitro* study/ies will need to be submitted.

Guidance on the application of these rules is given in the testing and assessment strategies described in Sections [R.7.2.6](#) and [R.7.2.11](#) of this Guidance.

R.7.2.2.2 Information requirements for quantities of ≥ 10 tpa (Annex VIII to the REACH Regulation)

As specified in Section 8.1 of Column 2 of Annex VIII to the REACH Regulation, for substances manufactured or imported in quantities of ≥ 10 tpa *in vivo* testing must be considered only if the *in vitro* studies under Sections 8.1.1 and 8.1.2 of Annex VII are not applicable for the substance, or the result(s) of these studies are not adequate for classification and risk assessment.

Section 8.1 of Annex VIII specifies the conditions under which an *in vivo* study for skin irritation/corrosion is still required. For substances manufactured or imported in quantities of ≥ 10 tpa *in vivo* testing must be considered only if the *in vitro* studies under Sections 8.1.1 and 8.1.2 of Annex VII are not applicable for the substance, or the result(s) of these studies are not adequate for classification and risk assessment.

The study does not need to be conducted if:

- the substance is a strong acid ($\text{pH} \leq 2.0$) or base ($\text{pH} \geq 11.5$), or
- the substance is spontaneously flammable in air or in contact with water or moisture at room temperature, or
- the substance is classified as acutely toxic by the dermal route (Category 1), or
- an acute toxicity study by the dermal route does not indicate skin irritation up to the limit dose level (2000 mg/kg body weight) (Please see footnote d to [Figure R.7.2-2](#) for further information).

Guidance on the application of these rules is given in the testing and assessment strategies described in Sections [R.7.2.6](#) and [R.7.2.11](#) of this Guidance.

It should be noted that the conditions of acceptance by ECHA of implementation of any of the adaptation rules laid down in Annex XI are strict, and whenever an adaptation argument is being used (e.g. use of (Q)SARs, read-across or non-validated *in vitro* test methods), scientific justification, solid documentation and readiness for risk assessment and Classification and Labelling must be provided by registrants. For detailed information on these rules, see Annex XI to the REACH Regulation.

R.7.2.3 Information sources on skin corrosion/irritation

R.7.2.3.1 Non-human data on irritation/corrosion

Non-testing data on irritation/corrosion

Physico-chemical properties

Relevant information can be inferred from basic physico-chemical characteristics of a substance (e.g. extreme pH). Extreme pH values may indicate the potential of a substance to cause skin corrosion:

If the $\text{pH} \leq 2$ or $\text{pH} \geq 11.5$, then consider the substance to be corrosive to the skin (Category 1) when the pH is used as the sole basis for the classification decision (see also Section [R.7.2.4.1](#) of this Guidance).

Grouping, (Q)SARs and expert systems ²⁹

In REACH Annex XI two types of non-testing methods are mentioned which can be used for adaptation of standard information requirements, either as standalone (where possible) or in combination with other information (in the context of a *Weight-of-Evidence* assessment):

- qualitative and quantitative Structure-Activity-Relationships (SARs/QSARs, section 1.2, including expert systems, generally incorporating multiple (Q)SARs, expert rules and data) on the one hand, and
- grouping of substances and read-across approaches ³⁰ on the other.

The adaptation of standard information requirements can be applied for the assessment of skin corrosion/irritation, if it provides relevant and reliable data for the substance of interest. As specified in Annex XI of the REACH regulation, the use of non-testing methods needs to be justified and sufficiently documented. In the case of QSARs and expert systems, registrants need to prepare property predictions by completion of a QSAR Prediction Reporting Format (QPRF). The QPRF is a harmonised template for summarising and reporting substance-specific predictions generated by (Q)SAR models. For filling a data gap under REACH, it is also necessary to provide information on the prediction model employed following a QSAR Model Reporting Format (QMRF) document. The QMRF is a harmonised template for summarising and reporting key information on (Q)SAR model validity, including the results of any validation studies. The information is structured according to the OECD (Q)SAR validation principles (for further information see <http://www.oecd.org/env/ehs/risk-assessment/validationofqsarmodels.htm>). The JRC QSAR Model Database is an inventory of information on available QMRFs, freely accessible online (<https://eurl-ecvam.jrc.ec.europa.eu/databases/jrc-qsar-model-database>). More detailed guidance on QSAR models, their use and reporting formats, including the QMRF, is provided in Section R.6.1 of Chapter R.6 of the [Guidance on IR&CSA](#).

In general, there are several different ways in which non-testing methods can be used in the context of an Integrated Approach to Testing and Assessment (IATA) (OECD, 2014b), e.g.:

- for direct prediction of corrosion/irritation potential or the absence thereof,
- as part of a *Weight-of-Evidence* scheme (where the information from non-testing methods alone is not sufficient for a decision), or
- in order to decide how best to proceed with further (*in vitro*) testing (i.e. *via* a top-down or bottom-up approach). For further information see Section [R.7.2.6.2](#).

In the case of skin corrosion and irritation, many of the models have a mechanistic basis, which provides additional information on the relevance of the model.

²⁹ Further information can be found in *Chapter R.6 QSAR and grouping of chemicals* of the [Guidance on IR&CSA](#), the OECD Guidance on Grouping of Chemicals, Second Edition (OECD, 2014a), the new OECD Guidance on an Integrated Approach for Testing and Assessment (IATA) for skin corrosion and irritation (OECD, 2014b) and the JRC report on Alternative methods for regulatory toxicology (Worth, 2014).

³⁰ The relevant terminology is not always used consistently. With reference to the ECHA Guidance on QSAR and grouping, the terms category approach and analogue approach are used to describe techniques for grouping of substances, whilst the term read-across is reserved for a technique to fill data gaps, i.e. to transfer knowledge from one or more substances called source(s) to another substance with data gaps, named target substance.

- SAR and read-across on skin irritation and corrosion:

SARs and read-across are treated together in this section because the existence of a SAR (structural alert or set of fragments) provides one means of justifying read-across. In fact, structural alerts are substructures in the substance that are considered to reflect some kind of chemical or biochemical reactivity that underlies the toxicological effect. The occurrence of a structural alert for a substance suggests the presence of an effect, based on the notion that structural analogues that have exhibited corrosion (or irritation) potential can be used to predict a corrosive or irritant effect for the substance of interest, or to tailor further testing and assessment, as indicated in the OECD IATA for skin corrosion/irritation (OECD, 2014b).

Knowledge on structural alerts for skin irritation/corrosion is always evolving (in particular where new classes of substances are introduced into the market). Therefore predictions based on read-across may also be possible for chemically similar substances if it can be shown that their similarity reflects reactive substructures able to react with skin tissue, even if that substructure has so far not been coded into a structural alert in any of the available literature or software models.

Negative data from structural analogues may also be used to make predictions in certain cases. The absence of one of the known structural alerts for irritation and corrosion alone does not prove absence of effect, as knowledge of structural alerts for irritation and corrosion might be incomplete. For instance, other substructures (not yet identified as structural alerts) or other properties of the substance may be responsible for a corrosive or irritant effect. As an example, irritant contact dermatitis may occur indirectly, such as in the case of exposure to organic solvents with defatting properties. Substances that have a similar mechanism of action need to be considered for the supplemental labelling '*Repeated exposure may cause skin dryness or cracking*' (EUH066) (Ad-hoc Working group on Defatting Substances, 1997).

An example of a simple SAR is the use of the hydroperoxide group (R-O-O-H) as an alert for corrosivity, which is mechanistically based on the fact that hydroperoxides are both acidic and oxidisers. Another SAR is the peroxide group (R₁-O-O-R₂), based on the fact that peroxides decompose easily and thus have a low thermal stability. The radicals formed by breaking the O-O bond are reactive and may be the cause of irritation or corrosion.

A variety of SARs (including hydroperoxides) for predicting the presence of irritation or corrosion have been described by Hulzebos *et al.* (2001, 2003, 2005), and some of these have been incorporated into the BfR (German Federal Institute for Risk Assessment) rule-base, and the SICRET tool (Walker *et al.*, 2005, see [Appendix R.7.2-2](#)). The BfR alerts ("inclusion rules") for corrosion and irritation have more recently been incorporated into the Toxtree software (https://eurl-ecvam.jrc.ec.europa.eu/laboratories-research/predictive_toxicology/qsar_tools/toxtree) and into the OECD QSAR Toolbox (<http://www.qsartoolbox.org/>).

- QSARs and expert systems for skin corrosion and irritation:

An overview of available (Q)SARs for skin corrosion and irritation is provided in [Table R.7.2-1](#). QSARs and expert systems for skin corrosion and irritation have been described in several reviews (Hulzebos *et al.*, 2001, 2003, 2005; Patlewicz *et al.*, 2003; Gallegos Saliner *et al.*, 2006, 2008). A comparison of the predictive capacities of three popular commercial tools is also available (Mombelli 2008). A few examples are presented in [Appendix R.7.2-2](#), including literature-based QSAR models, and expert systems.

Most of the QSARs reported in the literature have been developed from small data sets of specific groups of substances, although in some cases more diverse and larger datasets were also examined. In general, it has been suggested that basic physico-chemical parameters such as acidity, basicity, hydrophobicity, and molecular size as well as electrophilic reactivity, are useful to predict the toxic potential of homologous substances. In contrast, models intended to

predict the toxic potential of heterogeneous groups of substances emphasise the commonality of structural features.

A number of models are coded into expert systems, which are computer programs that guide hazard assessment by predicting toxicity endpoints of certain chemical structures based on the available information. Expert systems can be based on an automated rule-induction system (e.g. TOPKAT, HazardExpert and MultiCASE), or on a knowledge-based system (e.g. DEREK Nexus or the BfR- former DSS ³¹). More details on available expert systems are reported in [Appendix R.7.2-2](#).

Not all of the models were developed with EU regulatory purposes in mind, so it is important to assess in each case whether the endpoint or effect being predicted corresponds to the regulatory endpoint of interest. The rule-base at the heart of the former BfR DSS has been developed to predict EU regulatory endpoints, however predictions refer to the former Dangerous Substance Directive (DSD) classification/labelling system used in the EU before the CLP regulation came into force, and in borderline cases the results of the prediction may not fully reflect the correct CLP classification. More details on this model are reported in [Appendix R.7.2-2](#).

It should also be noted that the criteria for classification as skin irritant Category 2 based on the mean score for erythema/eschar or for oedema in the *in vivo* test have changed from ≥ 2 under DSD to ≥ 2.3 under CLP. Consequently predictions as skin irritant Cat 2 from models developed based on the DSD criteria should be interpreted with caution since they may lead to overprediction and should not be used for direct classification under CLP. These models can however be argued to be "conservative" and therefore acceptable for predicting no classification under CLP.

Based on the BfR rule-base, the freely downloadable OECD QSAR Toolbox software contains two profilers relevant for corrosion/irritation, which encode both the "inclusion rules" (structural alerts predicting corrosion/irritation potential) and the "exclusion rules" ("IF...THEN NOT..." rules predicting the absence of irritation/corrosion potential) due to certain physico-chemical properties. The use in combination with other profilers (e.g. for skin metabolism) and data for analogues allows for the prediction of skin corrosion/irritation for new chemicals through read-across or category approaches. More details on the Toolbox specific contents for skin corrosion and irritation are reported in [Appendix R.7.2-2](#).

In the case of classification models for skin corrosion, where it is not indicated in the supporting documentation whether the predicted classification should be Skin Corrosive Category 1A, 1B or 1C, Category 1 prediction without further sub-categorisation should be used. Very few models are available (see Gallegos Saliner *et al.*, 2006, 2008 for review). Available models tend to focus on defined chemical classes (e.g. acids, bases, phenols) and might be useful as an alternative to *in vitro* testing for such classes. For classification and labelling, the BfR rule-base provides information that is the closest to the regulatory goal, since the system was designed to predict former EU Risk Phrases for skin irritation (R38) and corrosion (R34, R35) under the Dangerous Substance Directive (DSD). However, in borderline cases and as highlighted above, the prediction may not fully reflect the correct classification under CLP.

³¹ Distribution of the BfR expert system "Decision Support System for Local Lesions" (DSS) mentioned in previous versions of this guidance has been discontinued. However, the rule-base for skin and eye irritation/corrosion included in this system has been incorporated into software tools such as the OECD QSAR Toolbox or Toxtree (cf. below).

Table R.7.2–1 Overview of available (Q)SARs for skin corrosion/irritation. See [Appendix R.7.2–2](#) for more information on these models.

Category of model or source	Reference or name of the model	Applicability domain
Literature models	Barratt (<i>et al.</i>) (1995a, 1996 a,b,c); Whittle <i>et al.</i> (1996)	Diverse local models for acids, bases , phenols, neutral organic and electrophiles
	Hayashi <i>et al.</i> (1999)	Phenols
	Kodithala <i>et al.</i> (1999)	Phenols, esters, and alcohols
	Nangia <i>et al.</i> (1996)	Bases
	Smith <i>et al.</i> (2000 a,b)	Esters
	Gerner <i>et al.</i> (2004); Hulzebos <i>et al.</i> (2005); Walker <i>et al.</i> (2004)	New Chemicals Database, organic chemicals with no significant hydrolysis potential and purity > 95%
	Golla <i>et al.</i> (2009)	Organic chemicals from diverse classes
Data repositories	Danish QSAR database (http://qsar.food.dtu.dk/ , also included in the OECD QSAR Toolbox)	Industrial chemicals, pesticides, etc.
Computerised models	PaDEL-DDPredictor (http://padel.nus.edu.sg/software/padelddpredictor/) (Liew and Yap, 2013)	Calculated by the model based on the range of descriptors
	BfR rule-base, free (included in the OECD QSAR Toolbox and Toxmatch, Toxtree, ToxPredict and Ambit)	EU New chemicals (NONS) database, organic chemicals with no significant hydrolysis potential and purity > 95%
	ACD/Percepta	Organic chemicals
	Derek Nexus, commercial	Organic chemicals and some metals
	HazardExpert, commercial	Organic chemicals
	MolCode, commercial	Organic chemicals
	MultiCASE, commercial	Organic chemicals
	TOPKAT, commercial	Organic chemicals

Review papers	Hulzebos <i>et al.</i> (2001, 2003, 2005)	N.A.
	Patlewicz <i>et al.</i> (2003)	N.A.
	Gallegos Saliner <i>et al.</i> (2006, 2008)	N.A.
	Mombelli (2008)	N.A.

Abbreviation: N.A. = not applicable.

Testing data on skin corrosion/irritation

The internationally accepted test methods for skin corrosion/irritation as described in the Annex to the EU Test Methods (TM) Regulation (Council Regulation (EC) No 440/2008) and in OECD TGs (available at http://www.oecd.org/env/ehs/testing/oecdguidelinesforhetestingofchemicals.htm#Test_Guide_lines) are: EU method B.4 (OECD TG 404), EU B.40 (OECD TG 430), EU B.40bis (OECD TG 431), OECD TG 435 and EU B.46 (OECD TG 439).

Please note that the latest version of an adopted test guideline should always be used when generating new data, independently of whether it is published by EU or OECD.

The testing strategy developed for skin corrosion/irritation (see Section [R.7.2.6](#) of this Guidance) emphasises the need to evaluate all available information (including physico-chemical properties) before undertaking any *in vivo* testing. This strategy employs screening elements designed to avoid, as far as possible, *in vivo* testing of corrosive and severely irritating substances. In particular, *in vitro* tests should usually be performed first, and it should be assessed whether *in vivo* testing can be completely avoided.

In vitro data

Accepted *in vitro* test methods to detect skin corrosion/irritation (i.e. Category 1 and 2 under CLP) and/or absence of effects (i.e. not classified under CLP) are listed in [Table R.7.2-2](#). More information on the specific scope and limitations of these tests is provided in Section [R.7.2.4.1](#) of this Guidance, under "Testing data on skin corrosion/irritation".

In [Table R.7.2-2](#), when the classification outcome in the column "Classification according to the CLP Regulation" is indicated as "Cat. 1B/1C" or "Cat. 1/Cat. 2", this means that the test method alone cannot differentiate between those (sub-)categories and more information is needed to conclude on the exact classification. For instance if the result of an *in vitro* skin irritation study according to B.46/OECD TG 439 is positive, it cannot be concluded whether the substance is either corrosive (Cat. 1) or irritant (Cat. 2) to the skin and therefore additional information on skin corrosion potential is needed e.g. by performing an *in vitro* skin corrosion study.

Table R.7.2–2 Accepted *in vitro* test methods for skin corrosion/irritation

	Test method	Validation status, regulatory acceptance	EU Test Methods/ OECD test guideline	Classification according to CLP Regulation	EURL ECVAM DB-ALM protocol Nr.
Skin corrosion					
	TER	Validated and regulatory acceptance	B.40/TG 430	Cat. 1 or non corrosive	115
	EpiDerm™ SCT	Validated and regulatory acceptance	B.40 bis/TG 431	Cat. 1, 1A, 1B/1C or non-corrosive	119
	EpiSkin™	Validated and regulatory acceptance	B.40 bis/TG 431	Cat. 1, 1A, 1B/1C or non-corrosive ³²	118
	SkinEthic™ RHE	Validated and regulatory acceptance	B.40 bis/TG 431	Cat. 1, 1A, 1B/1C or non-corrosive	-
	epiCS®	Validated and regulatory acceptance	B.40 bis/TG 431	Cat. 1, 1A, 1B/1C or non-corrosive	-
	Corrositex (<i>in vitro</i> membrane barrier test method)	Validated and regulatory acceptance	N.A./TG 435	Cat. 1, 1A, 1B and 1C or non-corrosive	116
Skin irritation					
	EpiDerm™ SIT	Validated and regulatory acceptance	B.46/TG 439	Cat. 1/Cat. 2 or NC	138
	EpiSkin™	Validated and regulatory acceptance	B.46/TG 439	Cat. 1/Cat. 2 or NC	131
	SkinEthic™ RHE	Validated and regulatory acceptance	B.46/TG 439	Cat. 1/Cat. 2 or NC	135
	LabCyte EPI-MODEL24 SIT	Validated and regulatory acceptance	B.46/TG 439	Cat. 1/Cat. 2 or NC	-

Abbreviations: N.A. = not available; NC = not classified; RHE = Reconstructed Human Epidermis; SCT = Skin Corrosion Test; SIT = Skin Irritation Test; TER=Transcutaneous electrical resistance.

³² The EpiSkin SOP allows for differentiating between the 3 sub-categories and OECD GD 203 suggests the use of this method to distinguish 1B from 1C before *in vivo* testing is considered. However, OECD TG 431 currently only permits the use of EpiSkin to distinguish 1A from 1B/1C.

Further test method developments may occur and the registrants are advised to follow the latest updates through e.g. the EURL ECVAM website (<https://eurl-ecvam.jrc.ec.europa.eu/>) and ECHA's test methods webpage (<http://echa.europa.eu/support/testing-methods-and-alternatives>) for potential new test guidelines and test guideline updates.

Animal data

Annex I to the CLP Regulation defines skin corrosion/irritation as local toxic effects, and, as such, an assessment of skin corrosion/irritation is normally part of the acute testing phase of a toxicity programme and it is an early requirement of all regulatory programmes. Testing for skin corrosion/irritation has, historically, used animal models and a variety of test methodologies depending upon, for example, the laboratory undertaking the test and the area and intended application. An IATA, which aims at minimisation of animal testing and instead largely relies on internationally approved *in vitro* tests, has been adopted by the OECD in 2014 as Guidance Document 203 (OECD, 2014b). Thereby, animal models have become unnecessary in most cases when testing for this endpoint. This is in line with one of the objectives of the REACH Regulation, as described in Articles 13(1) and 25(1), on that animal testing should be undertaken only as a last resort, i.e., where a substance falls outside of the applicability domain of the available *in vitro* methods or the results are not conclusive.

In cases in which *in vivo* testing may be necessary, current approaches for skin corrosion/irritation testing *in vivo* are covered by the Acute Dermal Irritation/Corrosion test method (EU B.4/OECD TG 404). This guideline requires a tiered approach, whereby existing and relevant data are evaluated first. The guideline also recommends that testing in animals should only be conducted if determined to be necessary after consideration of available alternative methods. The *in vivo* test uses one animal (the rabbit is the preferred species); in the absence of severe effects this is followed by a further testing of up to two animals (in a sequential manner up to a total maximum of three animals). When two animals are used, if both exhibit the same response, no further testing is needed.

Both EU and OECD methods use the scoring system developed by Draize (1944). The EU criteria for classification are based on the mean tissue scores obtained over the first 24-72 hour period after exposure and on the reversibility or irreversibility of the effects observed. Skin irritants (Category 2) cause significant inflammation of the skin (erythema and/or oedema) but this effect is transient, i.e. the affected sites are repaired within the observation period of the test.

A corrosive substance causes full thickness destruction of the skin tissue and is classified as Skin corrosive (Category 1) and sub-classified in subcategory 1A, 1B or 1C depending upon the exposure time (3 min, 1 hour, and 4 hours, respectively) and observation time (1 hour, 14 days, and 14 days, respectively).

For existing animal data, the use of methods other than those specified in the Annex to the EU Test Methods Regulation, or corresponding OECD methods may be accepted on a case-by-case basis.

In addition to the EU B.4/OECD TG 404 mentioned above, further animal data may be available e.g. from:

- Acute dermal toxicity test (EU B.3/OECD TG 402)
- Skin sensitisation tests (EU B.6/OECD TG 406, EU B.42/OECD TG 429, and OECD TG 442A and 442B)

Section [R.7.2.6](#) of this Guidance provides comments on how to use information from these test in a testing and assessment strategy for skin corrosion/irritation. Additional *in vivo* tests may

also provide relevant information (see paragraph 37 of the OECD Guidance Document 203 (OECD, 2014b)) although the reporting and scoring of the irritation in these tests may not be sufficient in all cases to allow a final conclusion to be drawn.

R.7.2.3.2 Human data on skin corrosion/irritation

Existing human data include historical data that should be taken into account when evaluating intrinsic hazards of substances. *New* testing in humans for hazard identification purposes is not acceptable for ethical reasons.

Existing data can be obtained from case reports, poison information centres, medical clinics, occupational experience, epidemiological studies and volunteer studies. Their quality and relevance for hazard assessment should be critically reviewed. However, in general, human data can be used to determine a corrosive or irritating potential of a substance. Good quality and relevant human data have precedence over other data. However, absence of incidence in humans does not necessarily overrule *in vitro* data or existing animal data of good quality that are positive.

R.7.2.4 Evaluation of information on skin corrosion/irritation

R.7.2.4.1 Non-human data on skin corrosion/irritation

Non-testing data on skin corrosion/irritation

In 2014, the OECD approved an IATA for skin corrosion/irritation. The IATA includes description of various types of data that can be used in the assessment of these hazards, including the types of information presented below. The IATA has a modular approach, whereby the *domain, role in IATA, strengths, weaknesses and limitations* of each type of data are given in a tabular form. It is also explained with flow diagrams how the data can then be integrated. Detailed guidance is given on the *Weight-of-Evidence* approach and on how quality, adequacy, coverage and consistency of data is assessed within a *Weight-of-Evidence* approach (OECD, 2014b).

Physico-chemical properties

According to the current EU and OECD guidelines, substances should not be tested on animals for skin corrosion/irritation if they can be predicted to be corrosive to the skin (Category 1) from their physico-chemical properties. In particular, substances exhibiting strong acidity ($\text{pH} \leq 2.0$) or alkalinity ($\text{pH} \geq 11.5$) in solution are predicted to be corrosive to the skin and should not be tested on animals. Testing with *in vitro* methods can nevertheless be performed, especially if skin corrosion sub-categorisation is required. It should also be noted that although prediction of skin corrosion based on pH extremes shows a very high specificity ($> 90\%$), and therefore a low number of false positives (Worth *et al.*, 1998), it cannot be ruled out that some substances may be overpredicted if classification is based solely on pH data. However, substances that have other pH values will need to be considered further for their potential for skin corrosion/irritation. This model is included in the OECD IATA for skin corrosion and irritation (OECD, 2014b). Several studies have investigated and confirmed the usefulness of pH as a predictor of corrosion (Worth and Cronin, 2001) and as an element in tiered testing strategies (Worth, 2004).

Where extreme pH is the only basis for classification of a substance as corrosive, it may also be important to take into consideration the acid/alkaline reserve, i.e. a measure of the buffering capacity of that substance (Young *et al.*, 1988; Botham *et al.*, 1998; Young and How, 1994). However, it should be noted that for pure substances the sensitivity of pH for identifying skin corrosivity may actually be significantly reduced when combined with

acid/alkaline reserve information (Worth *et al.*, 1998). The buffering capacity should not be used alone to exonerate from classification of the substance as corrosive. Indeed, when the acid/alkaline reserve suggests that the substance might be non-corrosive, further *in vitro* testing should be considered (see Section 3.2.2.2 of Annex I to the CLP Regulation).

Grouping, (Q)SARs and expert systems

Guidance has been developed by the former ECB (Worth *et al.*, 2005) on how to apply (Q)SARs for regulatory use. Guidance on how to assess the validity and suitability of (Q)SAR models and adequacy of their predictions is given in Section R.6.1 of Chapter R.6 of the [Guidance on IR&CSA](#) and in the OECD Guidance document on the validation of (Q)SAR models (OECD, 2007). Essentially, the determination of whether a (Q)SAR result may be used to replace a test result can be broken down into three main steps:

1. an evaluation of the scientific validity (relevance and reliability) of the model,
2. an assessment of the applicability of the model to the chemical of interest and the reliability of the individual model prediction,
3. an assessment of the adequacy of the information for making the regulatory decision, including an assessment of completeness, i.e. whether the information is sufficient to make the regulatory decision, and if not, what additional (experimental) information is needed.

The assessment of model validity needs to be performed along the lines of the OECD principles for (Q)SAR validation (OECD, 2007), e.g. in terms of a defined endpoint, an unambiguous algorithm, a defined applicability domain, the statistical characteristics ("goodness-of-fit"), and mechanistic interpretation.

The following questions, *inter alia*, should be addressed when assessing the reliability of an individual prediction:

- i. Is the chemical of interest within the scope of the model, according to the defined applicability domain of the model?
- ii. Is the defined applicability domain suitable for the regulatory purpose?
- iii. How well does the model predict chemicals that are similar to the chemical of interest?
- iv. Is the model estimate reasonable, taking into account other information?

The mechanism of skin corrosion and irritation involves toxicodynamic and toxicokinetic parameters. Some models predict skin corrosion and irritation based on toxicodynamic properties only (e.g. acidity or basicity, electrophilicity, other reactivity, surfactant activity, solvating membranes). Such models have to be additionally evaluated to check whether they also take account of toxicokinetic parameters related to the potential of a substance to cross relevant outer membranes of the skin (stratum corneum) and to be active in the living tissue underneath; alternatively these models have to be used in combination with data covering such toxicokinetic parameters. Conversely some models predict (the absence of) corrosion and irritation solely from e.g. physico-chemical properties considered to illustrate the toxicokinetic behaviour of a substance. Such models should be evaluated to check whether they also take account of the activity of the substance (toxicodynamics), in particular for its potential corrosivity (whereby the corrosive action itself may lead to membrane destruction and subsequent tissue damage).

For example, the BfR rule-base implemented in Toxtree and the OECD QSAR Toolbox contains both physico-chemical exclusion rules and structure-based inclusion rules (structural alerts). Evaluations of these rules for the prediction/exclusion of skin corrosion/irritation (Rorije and

Hulzebos, 2005, on the physico- chemical exclusion rules; Gallegos Saliner *et al.*, 2007, on the structural alerts) have been carried out in accordance with the OECD principles for (Q)SAR validation (see [Appendix R.7.2-2](#)). However, inclusion and exclusion rules were evaluated separately, and not used in combination in these works.

When applied, these two sets of rules may sometimes provide contradictory information, i.e. a structural alert might indicate corrosion/irritation potential, while at the same time, based on physico-chemical properties, absence of effect is predicted. In such cases, it is recommended to consider additional information (e.g. on skin permeability or on the behaviour of chemically similar substances). In other cases, applicability of one (or more) of the physico-chemical exclusion rules might indicate absence of a corrosion/irritation potential of the target substance, while no structural alert for corrosion/ irritation is triggered. Given that the absence of any known structural alert is not equivalent to the absence of a potential effect, in such a situation the substance should still be examined for potentially reactive substructures (and examining the behaviour of chemical analogues would still be beneficial).

While these considerations apply to the use of the BfR rule-base for direct classification/non-classification, less certainty might be required e.g. for a decision on further *in vitro* testing: where the exclusion rules suggest the absence of an effect, a bottom-up approach could be followed, i.e. a test for irritation and not one for corrosion might be initiated (see Section [R.7.2.6.2](#)).

There is no other model available which sufficiently describes the absence of effects. Neutral organics³³ are expected not to be irritants; however their defatting potential should be discussed. Predicted absence of reactivity needs to be described in sufficient detail or be substantiated with other information.

Testing data on skin corrosion/irritation

In vitro data

There are EU and OECD adopted test guidelines (see Section [R.7.2.3.1](#)), according to which substances can be classified as skin corrosives, skin irritants, or not classified.

Annex VII to the REACH Regulation requires information from the *in vitro* tests specified below for skin corrosion/irritation, and not from animal tests. Guidance on how *in vitro* data can be used to fulfil Annex VII requirements is given in Section [R.7.2.6](#) of this document.

Data from the following types of test can be used for Annex VII requirements:

- **For skin irritation:**
 - **Reconstructed human epidermis (RHE) tests** (EU B.46/OECD TG 439): These tests are considered scientifically valid for the prediction of irritant (Category 2) and non-irritant (no category) substances for Annex VII purposes, and also Annex VIII according to the rules laid down in Annex XI (see Section [R.7.2.6](#) of this Guidance).

³³ By definition a neutral organic is a chemical which does not have potential reaction centres, even after skin metabolism.

The specific scope and limitations of these tests are:

- They discriminate skin irritants (Category 2) from substances not classified for skin irritation (no Category) under CLP. However, they cannot discriminate skin irritants (Category 2) from skin corrosives (Category 1). The latter discrimination needs to be addressed with an *in vitro* skin corrosion test;
- Cell viability in these models is measured by the MTT (3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide, Thiazolyl blue) assay. If a test substance acts directly on the MTT (e.g. is a direct MTT-reducer), is naturally coloured, or becomes coloured during tissue treatment, additional controls should be used to detect and correct for test substance interference with the viability measurement technique. Detailed description of how to correct for direct MTT reduction and interferences by colouring agents is available in the Standard Operating Procedures (SOPs) for the four validated test methods and referenced in the OECD and EU TGs ³⁴;
- This test method may not be applicable to all groups of chemical classes. For example metals or inorganic metal compounds were not included in the validation study and there is experience that some metals (e.g. cobalt) may give a false positive result;
- They do not allow testing of gases and aerosols.

- For skin corrosion:

- **Transcutaneous electrical resistance (TER) test method** (EU B.40/OECD TG 430)
- **Reconstructed Human Epidermis (RHE) test method** (includes more than one protocol) (EU B.40 bis/OECD TG 431)
- ***In vitro* membrane barrier test method** (OECD TG 435)

All the above-mentioned tests allow for the discrimination of skin corrosives (Category 1) from non-corrosive substances.

The specific scope and limitations of these tests are:

- None of them allows testing of gases and aerosols;
- Only the ***in vitro* Membrane Barrier** test method for skin corrosion is accepted to discriminate between skin corrosive subcategories 1A, 1B and 1C and non-corrosives;
- The ***in vitro* Membrane Barrier** test method has a limited applicability domain (only acids, bases and acid derivatives). In addition, test materials not causing detectable changes in the detection system (e.g. typically $4.5 < \text{pH} < 8.5$) cannot be tested;

³⁴ A revision of OECD TG 439 including the use of HPLC/UPLC-spectrophotometry as an alternative way to measure MTT formazan is currently under discussion at the OECD with a high probability of adoption in April 2015. If this revision is accepted, it will reduce the limitation of these test methods towards strongly coloured substances.

- The **RHE** test method can be used to distinguish subcategory 1A from subcategories 1B and 1C. The protocol of **EpiSkin**, which is one of the four validated methods included in the RHE test guideline, also allows for the discrimination of subcategory 1B from subcategory 1C and, according to the OECD IATA (OECD, 2014b), this information may be used in a *Weight-of-Evidence* assessment;
- TER cannot be used to subcategorise skin corrosive substances;
- The use of the **RHE** test method may not be applicable to all groups of chemical classes. For example there is reasonable doubt on the adequacy of this model for certain groups of fatty amine derivatives where RHE assays did not predict corrosivity, whereas these substances were corrosive in *in vivo* rabbit studies (Houthoff *et al.*, 2014). Furthermore, metals or inorganic metal compounds were not included in the validation study and there is experience that some metals (e.g. cobalt) may give a false positive result.

In relation to cell viability measurement by the MTT assay in **RHE** models, the same limitations as those specified above for the *in vitro* skin irritation test (EU method B.46/OECD TG 439) apply.

- [Quality aspects of existing *in vitro* data:](#)

For quality assessment of existing *in vitro* data that will form the basis for later possible *Weight-of-Evidence* considerations, see Section R.4.4 of Chapter R.4 of the [Guidance on IR&CSA](#), and for aspects that need to be taken into account in such a *Weight of Evidence* see Section R.5.2.1.2 of Chapter R.5 of the [Guidance on IR&CSA](#).

[Animal data](#)

Well-reported studies, particularly if conducted in accordance with the principles of GLP, can be used to identify substances which would be considered to cause, or not to cause, skin corrosion or skin irritation. There may be a number of skin corrosion/irritation studies already available for an existing substance, none of which are fully equivalent to an OECD TG or an EU test method such as those in the Annex to the EU Test Methods Regulation. If the results from such a batch of studies are consistent, they may, together, provide sufficient information on the skin corrosion/irritation potential of the substance.

If the results from a variety of studies are unclear, based on the criteria given below for evaluation of the data, the registrant will need to decide which of the studies is/are most reliable, relevant for the endpoint in question and adequate for classification purposes.

Particular attention should be given to the persistence of irritation effects, even those which do not lead to classification. Effects such as erythema, oedema, fissuring, scaling, desquamation, hyperplasia and opacity which do not reverse within the test period may indicate that a substance will cause persistent damage to the human skin.

Data from studies other than skin corrosion/irritation ones (e.g. other toxicological studies on the substance in which local responses of skin have been reported) may provide useful information though they may not be well reported in relation to, for example, the basic requirements for information on skin irritation. However, it should be noted that skin reactions and symptoms are not systematically scored in e.g. acute and sub-acute dermal toxicity studies since these studies are not specifically designed to address skin corrosion/irritation.

- [Quality Aspects of existing *in vivo* data:](#)

Data from **existing** irritation studies in animals must be taken into account before further testing is considered. A quality assessment of any such studies should be done using, for

example, the system developed by Klimisch *et al.* (1997), as described in Section R.4.2 of Chapter R.4 of the [Guidance on IR&CSA](#), and a judgement will need to be made as to whether any further testing is required. Some examples to note are:

- i. Was the animal species used the rabbit or was it another species such as the rat or the mouse? The rat and the mouse are not as sensitive as the rabbit for irritation testing.
- ii. How many animals were used? Current methodology requires a maximum of 3 animals tested in a sequential manner (with 1 animal being sufficient if skin corrosion effects are observed in the first tested animal, or 2 animals being sufficient if consistent effects are observed in the first and the second tested animals) but 6 were frequently used in the past (See Section 3.2.2.3.2.2 of the [Guidance on the Application of the CLP criteria](#) for the evaluation of results from tests that have been conducted with more than 3 animals).
- iii. How many dose levels were used? If dilutions were included, what solvent was used (as this may have influenced absorption)? Which dose volume was used?
- iv. Which exposure period was used? Single or repeated exposure?
- v. The method used to apply the substance to the skin should be noted i.e. whether occluded or semi-occluded and whether the application site was washed after treatment.
- vi. Check the observation period used post-exposure. Shorter periods than those in the current guideline may be adequate for non-irritants but may require a more severe classification for irritants when the observation period is too short to measure full recovery.

Irritation scores from old reports, reports produced for regulatory submission in the USA or in publications may be expressed as a Primary Irritation Score. Without the original data it is not always possible to convert these scores accurately into the scoring system used in the EU. For extremes, i.e. where there is either no irritation or severe irritation, it may not be necessary to look further, but average irritation scores pose a problem and expert judgement may be required to avoid repeat testing.

Observations such as those above can all be used to assess whether the existing animal test report available can be used to reliably predict the irritation potential of a substance, thus avoiding further testing.

R.7.2.4.2 Human data on skin corrosion/irritation

Well-documented *existing* human data from different sources can often provide very useful information on skin corrosion/irritation, sometimes for a range of exposure levels. Often the only useful information available on irritation is obtained from human experience (e.g. occupational settings). The usefulness of all human data on irritation will depend on the extent to which the effect, and its magnitude, can be reliably attributed to the substance of interest.

The quality and relevance of existing human data for hazard assessment should be critically reviewed. For example, in occupational studies with mixed exposure it is important that the substance causing skin corrosion or skin irritation has been accurately identified. There may also be a significant level of uncertainty in human data due to poor reporting and lack of specific information on exposure.

Examples of how existing human data can be used in hazard classification for irritation are provided in an ECETOC monograph (ECETOC, 2002).

Human data on local skin effects may be obtained from existing data on single or repeated exposure. The exposure could be of accidental nature or prolonged, for example in occupational settings. The exposure is usually difficult to quantify. When looking at the effects, corrosivity is characterised by destruction of skin tissue, namely visible necrosis through the epidermis and into the dermis. Corrosive reactions are typified by ulcers, bleeding and bloody scabs. After recovery the skin will be discoloured due to blanching of the skin and will present complete areas of alopecia and scars.

In addition to human data on local skin effects (which originate from clinical and occupational studies, poison information centres, case reports and retrospective epidemiological studies) existing human data from skin irritation human patch testing (HPT) might also be available. HPT is a controlled study involving the exposure of small patches of skin of human volunteers to substances for which skin corrosion and other unacceptable toxicological hazards can be excluded. HPT data have been compiled for example by Jírová *et al.* (2010), Basketter *et al.* (2012), as well as Ishii *et al.* (2013). Testing with human volunteers to obtain primary hazard data on skin corrosion/irritation for regulatory purposes is discouraged. Available good quality data should nevertheless be considered as appropriate and used for Classification and Labelling decision making. It should however be noted that the CLP Regulation does not contain clear criteria for classification for skin irritation based on human data.

R.7.2.4.3 Exposure considerations for skin corrosion/irritation

Exposure-based waiving from testing is not applicable to the endpoints of skin corrosion/irritation. Exposure-based waiving from testing as specified in Annex XI (3) of the REACH Regulation only applies to tests listed in Sections 8.6 and 8.7 of Annex VIII, Annex IX and Annex X according to the REACH text.

R.7.2.4.4 Remaining uncertainty on skin corrosion/irritation

Usually it is possible to unequivocally identify (or accept) a substance as being corrosive, whatever type of study provides the information.

There may be a significant level of uncertainty in human data on irritant effects (e.g. because of poor reporting, lack of specific information on exposure, subjective or anecdotal reporting of effects, small number of subjects).

Data from studies in animals and from *in vitro* tests performed according to internationally accepted test methods will usually give relevant information on the skin corrosion/irritation potential of a substance. In general, it is assumed that substances which cause skin corrosion/irritation in EU or OECD TG-compliant studies in animals or *in vitro* will cause skin corrosion/irritation in humans, and those which are not irritant in EU or OECD TG-compliant studies will not be irritant in humans (Please note that in general test animals are considered to be more sensitive to skin corrosion/irritation effects than humans (e.g. OECD, 2014b)). It should be borne in mind that one of the limitations of the *in vivo* corrosion/irritation studies is the subjective grading of the lesions. Moreover, inconsistent results from a number of similar studies increase the uncertainty in deriving data from animal or *in vitro* studies.

The scope of the *in vitro* tests for corrosion/irritation has also some limitations, as explained in Section [R.7.2.4.1](#) under "Testing data on corrosion/irritation". In addition inconsistent results from two or more *in vitro* tests could add to the overall uncertainty in interpretation of the data.

R.7.2.5 Conclusions on skin corrosion/irritation

R.7.2.5.1 Concluding on suitability for Classification and Labelling

In order to conclude on Classification and Labelling according to the CLP Regulation, all the available information needs to be taken into account and consideration should be given to both the [Guidance on the Application of the CLP criteria](#) and the various remarks (related to Classification and Labelling) made throughout this guidance document ³⁵.

R.7.2.5.2 Concluding on suitability for Chemical Safety Assessment

A dose-response assessment is difficult to make for skin corrosion/irritation simply because up to now most data have been generated for undiluted substances in accordance with test guidelines and traditional practice (which continues today). From a risk characterisation perspective it is therefore advisable to use the outcome of the classification procedure, i.e. a substance that is classified is assumed to be sufficiently characterised. However, a complete risk assessment requires both hazard and dose-response data and for local effects the concentrations is often the determinative dose metric. Consequently, if dose-response data are available, they must be taken into account ([Figure R.7.2-1](#)). For instance, dose-response information might be available from sub-acute or sub-chronic dermal toxicity studies (as such studies require a determination of a non-irritant dose in the dose selection), or from human experience and may in certain cases be determined using *in vitro* studies. However, when information is used from existing dermal toxicity studies (e.g. repeated dose), it should be noted that the test conditions do not reflect the test conditions used in the *in vivo* skin corrosion/irritation study: e.g. test material is applied in dilution vs. neat, vehicles/solvents are often used, exposure duration is different and test material application areas differ (see Module 5 of the OECD IATA (OECD, 2014b)).

Guidance on the possibilities for derivation of DNELs for skin corrosion/irritation is given in Appendix R.8-9 of Chapter R.8 of the [Guidance on IR&CSA](#).

R.7.2.5.3 Information not adequate

A *Weight-of-Evidence* approach comparing available adequate information with the tonnage-triggered information requirements under REACH may result in the conclusion that the requirements are not fulfilled. In order to proceed to further information gathering, the testing and assessment strategy described in Section [R.7.2.6](#) below is recommended.

R.7.2.6 Testing and assessment strategy for skin corrosion/irritation

The OECD has approved an IATA for skin corrosion/irritation (OECD, 2014b), which includes a description of various types of data that can be used in the assessment of these hazards. The IATA has a modular approach, whereby the *domain, role in IATA, strengths, weaknesses and limitations* of each type of data are given in a tabular form. Some parts of the IATA provide more detailed scientific background than the present document. Furthermore, the IATA gives

³⁵ Please note that the 8th Adaptation to Technical and Scientific Progress (ATP) of the CLP will apply from 1 February 2018. The amendment by the 8th ATP will take into account the 5th Revision of the Globally Harmonized System of Classification and Labelling of Chemicals (GHS), which was adopted in 2012 and contains in particular refined criteria for skin corrosion/irritation and serious eye damage/eye irritation.

detailed guidance on the *Weight-of-Evidence* approach. At the *Weight-of-Evidence* step, all existing information is integrated and assessed in order to decide whether further *in vitro* testing of the substance (or *in vivo* testing as a last option if *in vitro* testing is not possible or not conclusive) is necessary. While the OECD IATA provides slightly more detailed guidance than the testing and assessment strategy below, there is no conceptual difference between the two.

R.7.2.6.1 Objective / General principles

The following testing and assessment strategy is recommended for developing adequate and scientifically sound data for assessment/evaluation and classification of the skin corrosive and skin irritating properties of substances. For existing substances with insufficient data, this strategy can also be used to decide which additional data, beside those already available, are needed. The testing and assessment strategy is aimed at the identification of skin corrosion/irritation by using different elements where appropriate, depending on the information available. A basic principle of the strategy is that the results of one study or from an information source are evaluated before another study is initiated. The strategy seeks to ensure that the data requirements are met in the most efficient and humane manner so that animal usage and costs are minimised.

The different elements provided in [Figure R.7.2-1](#) describe information sources that can be used to conclude on a substance's hazard potential towards skin. The elements described in [Figure R.7.2-2](#) can be rearranged as appropriate, especially those in Part 1. This may be particularly helpful in cases where a conclusion can be drawn from certain elements without having to consider all of them. If judged relevant, elements in Part 1 can be omitted and *in vitro* testing can be performed immediately.

[Figure R.7.2-2](#) is divided into three parts whereby Part 1 aims at evaluating existing information that may be available on the substance. In Part 2 existing information and relevant data should be assessed in order to consider whether there is enough information available to conclude on the substance hazard properties within a *Weight-of-Evidence* analysis, in case it is not possible to make a conclusion based on single elements described in Part 1. In case no conclusion can be drawn from Parts 1 and 2, new data should be generated in Part 3 by first performing relevant *in vitro* testing. Only in case no conclusion can be drawn based on the *in vitro* testing, must *in vivo* testing be conducted (for substances at or above 10 tonnes per annum only).

Some guidance for testing is provided by the specific rules for adaptation from standard information requirements, as described in column 2 of Annexes VII-X to the REACH Regulation, together with some general rules for adaptation from standard information requirements in Annex XI.

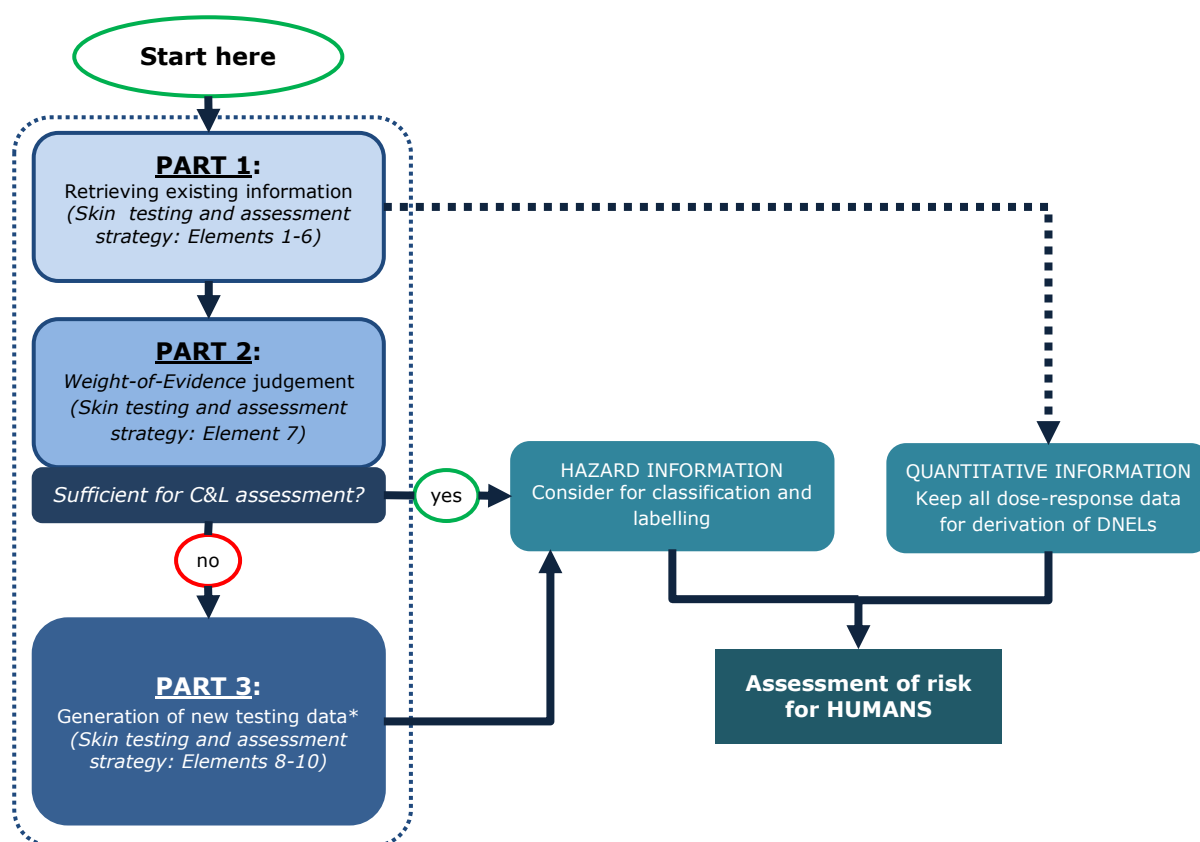
Risk assessment of the skin corrosion/irritation potential of a substance is normally made in a qualitative way provided that the substance has been classified as being corrosive or irritant to the skin. Existing test guidelines do not contain dose-response assessment, consequently a quantitative analysis will often not be possible. Therefore, hazard identification and appropriate classification is the key determinant in the information gathering strategy below. As a consequence, the use of *Assessment Factors* is of limited use in order to take into account uncertainty of data. However, the registrant is encouraged to keep and use all quantitative data that might be encountered in the process of retrieving hazard information in the context of the present testing strategy and to perform a complete risk assessment, comprising assessment of qualitative hazard as well as quantitative information.

It is recommended that the testing and assessment strategy be followed until element 6 ([Figure R.7.2-1](#) and [Figure R.7.2-2](#) in all cases and thereafter the *Weight-of-Evidence* analysis be performed. Clearly, all information sources/elements can be rearranged as appropriate, i.e. not all elements will necessarily be accompanied by data but it is important that all potential

data sources are explored prior to starting the *Weight-of-Evidence* analysis. While it is recommended that this approach be followed, other approaches may be deemed more appropriate and efficient on a case-by-case basis. For example, in case there is no existing data and it is anticipated that generation of “pre-testing data” would be non-conclusive, it may be appropriate to directly proceed to the information generation part. Furthermore, prior to performing any new *in vivo* test, the use of *in vitro* methods must be fully exploited (see Articles 13(1) and 25(1) of the REACH Regulation).

If the substance is not classified for skin corrosion/irritation, no risk assessment for this endpoint is performed, regardless of the exposure. Please note that there are no options for exposure-based waiving for these endpoints in the REACH Regulation.

The following flow chart ([Figure R.7.2–1](#)) gives an overview of a possible approach for defining a testing and assessment strategy for skin corrosion and irritation.



*Generation of new testing data according to Annex VII to VIII to the REACH Regulation and with due observation of the rules for adaptation of the standard testing regime laid down in Annex XI.³⁶

Figure R.7.2–1 Overview of the testing and assessment strategy for skin corrosion/irritation

³⁶ Please note that the information requirements in REACH Annexes VII and VIII in relation to skin corrosion/irritation and serious eye damage/eye irritation are currently under revision. This revision is expected to strengthen the role of *in vitro* methods and to remove the standard information requirement for an *in vivo* study at the Annex VIII level. As a consequence, once the new REACH Annexes come into force, an *in vivo* study would only be required where a substance falls outside of the applicability domain of the available *in vitro* methods or the results obtained from such methods would not allow a conclusive decision on (non-)classification and risk assessment.

R.7.2.6.2 Testing and assessment strategy for skin corrosion/irritation

Recommended approach

The testing and assessment strategy presented here comprises three parts (see [Figure R.7.2-2](#)): Part 1 (elements 1 to 6) is about retrieving existing information, Part 2 (element 7) represents a *Weight-of-Evidence* analysis and expert judgement, and Part 3 (elements 8 to 10) is about the generation of new information by testing.

In Part 1, existing and available information from the literature and databases is gathered and considered in the strategy approach. The order of the different elements, i.e. 1 to 6, is only indicative and they may be arranged as appropriate. This may be especially helpful in cases where a reliable conclusion can be drawn from certain elements without having to consider all of them. For instance, if the substance has an extreme pH (≤ 2.0 or ≥ 11.5) skin corrosivity is considered implicit (element 1c) and therefore the substance should be classified as skin corrosive (Category 1) according to CLP and further testing is not required. At the end of Part 1, and if no final conclusion could be derived directly from one or several of the available pieces of information, all the information collected should be analysed using a *Weight-of-Evidence* approach (element 7).

In the information generation part (elements 8 to 10), new information on the corrosion/irritation potential of substances is produced by means of *in vitro* (elements 8 and 9) or, as a last resort (see Articles 13(1) and 25(1) of the REACH Regulation), *in vivo* testing (element 10). Therefore, before concluding the *Weight-of-Evidence* analysis in element 7 and *in vitro* testing (elements 8 and 9), new *in vivo* tests should not be conducted. More information on how to use the *in vitro* methods for skin corrosion/irritation within the testing strategy can be found in the following paragraphs.

While it is recommended that this approach be followed, other approaches may be more appropriate and efficient on a case-by-case basis. For example, in case there is no existing data and it is anticipated that compilation of data at elements 1-7 would be non-conclusive, it may be appropriate to directly proceed to the information generation part.

Figure R.7.2–2 Testing and assessment strategy for evaluating the skin corrosion/irritation potential of substances (footnotes a to h are detailed below the figure).

Element	Information	Conclusion ³⁷
Existing data on physico-chemical properties		
1a	Is the substance spontaneously flammable in air or in contact with water or moisture at room temperature? →	YES: No testing required (Column 2 adaptation in section 8.1 of Annexes VII and VIII)
1b	Is the substance an organic hydroperoxide or an organic peroxide? →	YES: Consider classifying as: ■ corrosive (Skin Corrosive Cat. 1B) if the substance is a hydroperoxide, or ■ irritant (Skin Irritant Cat. 2) if the substance is a peroxide. OR Provide evidence supporting deviating classification or non-classification ³⁸ .
1c	Is the pH of the substance ≤ 2.0 or ≥ 11.5 ? ^a →	YES: Consider classifying as corrosive (column 2, section 8.1. of Annexes VII and VIII) if pH is used as the sole basis for classification decision. Where classification is based upon consideration of pH alone, subcategorisation is not possible and therefore Skin Corrosive Cat.1 should be applied.
1d	Are there other physical or chemical properties that indicate that the substance is corrosive/irritant? →	YES: Use this information for <i>Weight-of-Evidence</i> analysis (Element 7).
Existing human data		
2	Are there adequate existing human data ^b which provide evidence that the substance is a corrosive or irritant? →	YES: Consider classifying accordingly.

³⁷ Please note that the 8th Adaptation to Technical and Scientific Progress (ATP) of the CLP Regulation will apply from 1 February 2018. The amendment by the 8th ATP will take into account the 5th Revision of the Globally Harmonized System of Classification and Labelling of Chemicals (GHS), which was adopted in 2012 and contains in particular refined criteria for skin corrosion/irritation and serious eye damage/eye irritation.

³⁸ Information on e.g. *in vitro* testing may provide evidence on a more suitable classification, if there is some doubt on the correct classification.

Existing animal data from corrosion/irritation studies		
3	Are there data from existing studies <i>on corrosion and irritation</i> in laboratory animals, which provide sound conclusive evidence that the substance is a corrosive, irritant or non-irritant? →	YES: Consider classifying accordingly (either Skin Corrosive Cat. 1, 1A, 1B, 1C or Skin Irritant Cat. 2) or consider no classification.
Existing data from general toxicity studies via the dermal route and from sensitisation studies		
4a	Is the substance classified as acutely toxic by the dermal route (Category 1)? ^c →	YES: The substance will be classified for acute dermal toxicity (column 2 adaptation in section 8.1 of Annexes VII and VIII). No new testing for skin irritation/corrosion is needed in this case.
4b	Has the substance proven to be a corrosive, irritant or non-irritant in a suitable acute dermal toxicity test? ^d →	YES: If test conditions are consistent with OECD TG 404, consider classifying accordingly (Skin Corrosive Cat. 1, 1A, 1B, 1C or Skin Irritant Cat. 2) or consider no classification.
4c	Has the substance proven to be a corrosive or an irritant in sensitisation studies or after repeated exposure? ^e →	YES: This information cannot be used for considering a concrete classification conclusion but must be used exclusively within the integrated <i>Weight-of-Evidence</i> judgement.
Existing/new (Q)SAR data and read-across		
5a	Are there structurally related substances (suitable "read-across" or grouping), which are classified as corrosive to the skin (Skin Corrosive Cat. 1), or do suitable (Q)SAR methods indicate corrosion potential of the substance? ^f →	YES: Consider classifying as Skin Corrosive Cat. 1.
5b	Are there structurally related substances (suitable "read-across" or grouping), which are classified as irritant to the skin (Skin Irritant Cat. 2), or indicating that the substance is non-irritant, or do suitable (Q)SAR methods indicate irritant or non-irritant potential of the substance? ^f →	YES: Consider classifying accordingly.
Existing in vitro data		
6a	Has the substance demonstrated corrosive properties in an EU/OECD adopted <i>in vitro</i> test? Data from <i>in vitro</i> test methods that have been validated and are considered scientifically valid but are not yet adopted by EU and/or OECD may also be used if the provisions defined in Annex XI are met. →	YES: Consider classifying as corrosive. If discrimination between Skin Corrosive Cat. 1A, 1B and 1C is not possible, Cat. 1 must be chosen. If a negative result is obtained and there is no existing data from (an) <i>in vitro</i> skin irritation study(ies), the

		irritation potential must be determined, e.g. with an <i>in vitro</i> skin irritation test.
6b	Has the substance demonstrated irritant or non-irritant properties in an EU/OECD adopted <i>in vitro</i> test? Data from <i>in vitro</i> test methods that have been validated and are considered scientifically valid but are not yet adopted by EU and/or OECD may also be used if the provisions defined in Annex XI are met. →	YES: Consider classifying accordingly (Skin Irritant Cat. 2) or consider no classification. If a positive result is obtained and there is no existing data from (an) <i>in vitro</i> skin corrosion study(ies), the corrosion potential must be determined e.g. with an <i>in vitro</i> skin corrosion test (Element 8).
6c	Are there data from (a) non-validated suitable <i>in vitro</i> test(s), which provide sound conclusive evidence that the substance is corrosive/ irritant? ⁹ →	YES: Consider classifying accordingly (Skin Corrosive Cat 1, 1A, 1B, 1C or Skin Irritant Cat. 2).

Weight-of-Evidence analysis

7	The “elements” described above may be arranged as appropriate. Taking all available existing and relevant data mentioned above (Elements 1-6) into account, is there sufficient information to make a decision on whether classification/labelling is necessary, and – if so – how to classify and label? →	YES: Classify accordingly (Skin Corrosive Cat. 1, 1A, 1B, 1C or Skin Irritant Cat. 2) or consider no classification. If discrimination between Skin Corrosive Cat 1A, 1B and 1C is not possible, Cat. 1 must be chosen.
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New *in vitro* tests for corrosivity ⁹

8	Does the substance demonstrate corrosive properties in (an) EU/OECD adopted <i>in vitro</i> test(s) for skin corrosion? → Data from <i>in vitro</i> test methods that have been validated and are considered scientifically valid but are not yet adopted by EU and/or OECD may also be used if the provisions defined in Annex XI are met.	YES: Classify accordingly (Skin Corrosive Cat. 1A, 1B or 1C). If discrimination between Cat. 1A, 1B and 1C is not possible, Cat. 1 must be chosen. If a negative result is obtained, the irritation potential of the substance must be determined, e.g. with an <i>in vitro</i> skin irritation test (Element 9), in order to determine if the substance should be classified as Skin Irritant Cat. 2 or not classified.
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New *in vitro* tests for irritation ⁹

9	Does the substance demonstrate irritating or non-irritating properties in (an) EU/OECD adopted <i>in vitro</i> test(s) for skin irritation? Data from <i>in vitro</i> test methods that have been validated and are considered scientifically valid but are not yet adopted by EU and/or OECD may also be used if the provisions defined in Annex XI are met. →	YES: Classify accordingly (Skin Irritant Cat. 2) or consider no classification. If a positive result is obtained and there is no existing data from (an) <i>in vitro</i> skin corrosion study(ies), the corrosion potential must be determined e.g. with an <i>in vitro</i> skin corrosion test (Element 8).
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		If a conclusion on skin corrosion/irritation cannot be drawn by using <i>in vitro</i> testing, <i>in vivo</i> testing should be performed (at Annex VIII level only).
<i>New in vivo test for corrosion/irritation as a last resort (Annex VIII to the REACH Regulation)^h</i>		
10	Does the substance demonstrate corrosive or irritant properties in an EU/OECD adopted <i>in vivo</i> test? →	YES: Classify accordingly (Skin Corrosive Cat. 1, 1A, 1B, 1C or Skin Irritant Cat. 2). NO: No classification needed.

Notes to the information scheme on skin corrosion/irritation:

a) Note that if the buffering capacity suggests that the substance may not be corrosive, further data are needed to confirm this, preferably using an appropriate *in vitro* test method.

b) Data from case reports, occupational experience, poison information centres, HPTs or from clinical studies.

c) If the substance is classified as fatal in contact with skin ($LD_{50} \leq 50$ mg/kg bw), further testing for skin corrosion/irritation would result in severe suffering or death of the animal. Thus, further testing is not required and sufficient labelling (warning) is provided by the Hazard statement H310 "*Fatal in contact with skin*" and the GHS Pictogram GHS06 with the signal word "Danger". The classification of a substance as fatal in contact with skin requires strict risk management measures and hence, since all contact with the skin must be avoided, there is no need to investigate the skin corrosion/irritation potential further. In case existing information on skin corrosion/irritation is available, it should be included in the registration dossier and used for classification and labelling for skin corrosion/irritation.

d) Has the substance proven to be either an irritant or a corrosive in an acute dermal toxicity test carried out with **rabbits** with the undiluted test substance (liquids) or with a suitable suspension (solids)? In case of signs of skin corrosion, classify as Skin Corrosive (subcategorisation as 1A, 1B or 1C, where possible). In all other cases: calculate or estimate the amount of test substance per cm² and compare this to the test substance concentration of 80 µl or 80 mg/cm² employed in the EU B.4/OECD TG 404 for dermal corrosion/irritation test with rabbits. If in the same range and adequate scoring of skin effects is provided, classify or not as Skin Irritant Category 2. In case conclusive negative data was obtained in rabbits, stop. If not in the same range and inadequate scoring of skin effects, use for *Weight-of-Evidence* analysis and proceed.

In case the test was performed in other species, which may be less sensitive (e.g. **rat**), evaluation must be made with caution. Usually, the rat is the preferred species for toxicity studies within the EU. The limit dose level of 2000 mg/kg bw of a solid is normally applied as a 50% suspension in a dose volume of 4 ml/kg bw onto a skin surface area of about 5x5 cm. Assuming a mean body weight of 250 g, a dose of 1 ml of the suspension will be applied to an area of 25 cm², i.e. 20 mg test substance per cm². In case of an undiluted liquid, 0.5 ml is applied to 25 cm², i.e. 20 µl/cm². Considering the fact that the rat skin is less sensitive compared to rabbit skin, much lower exposures are employed and, in general, the scoring of dermal effects is performed less accurately, the results of dermal toxicity testing in rats will

not be adequate for classification with respect to skin irritation. Only in case of evidence of skin corrosivity in the rat dermal toxicity test can the test substance be classified as Skin Corrosive Category 1. All other data should be used for *Weight of Evidence*.

e) Regarding data from skin sensitisation studies, the skin of guinea pigs is less sensitive than that of rats which is, in turn, less sensitive than that of rabbits. Only in case of evidence of skin corrosivity in the sensitisation test (Maximisation or Buhler) with the neat material or dilutions of solids in water, physiological saline or vegetable oil, should the test substance be classified as Skin Corrosive Category 1. However, care should be exercised when interpreting findings from guinea pig studies, particularly from maximisation protocols, as intradermal injection with adjuvant readily causes necrosis. All other data should be used for *Weight of Evidence* only. Information on irritant properties from skin sensitisation tests cannot be used to conclude on a specific classification regarding acute skin irritation but may be used in a *Weight-of-Evidence* analysis. In general, irritation data from the Local Lymph Node Assay are not usable. The test substance is applied to the dorsum of the ear by open topical application, and specific vehicles for enhancement of skin penetration are used.

f) Conclusion on no classification can be made if the *in silico* model has been shown to predict adequately the absence of the classified effect and if it also fulfils the requirements of Annex XI to the REACH Regulation. Prediction of the absence of the classified effect can be made either by triggering an exclusion rule in the BfR system (to be checked on a case-by-case basis), or based on a negative prediction in a classification QSAR that was trained on both positive and negative substances. The suitability of the model (reliability, relevance) should be very carefully checked to make sure that the prediction is fit for purpose, and the applicability of the model to the substance should also be justified (e.g. fulfilment of the conditions of Section 1.3 of Annex XI to the REACH Regulation should be checked). For read-across, generation of new *in vivo* data should be avoided.

g) New *in vitro* testing should be performed following a top-down or bottom-up approach. Please see the following paragraph "How to use the *in vitro* methods for skin corrosion/irritation within the strategy". While it may be appropriate to use information from non-validated *in vitro* tests if already existing, it is highly recommended to adhere to the test protocols whose scientific validity has been established by formal validation and which, ideally, have been officially adopted by the European Commission and/or by the OECD. Data obtained from suitable non-validated suitable *in vitro* tests can only be used according to the criteria set out in Annex XI, section 1.4 of the REACH Regulation, i.e. only positive results can be accepted.

h) *In vivo* testing should not be conducted in case the substance falls under the scope of the specific *in vitro* tests performed, and there are no substance-specific limitations on use of those tests. *In vivo* testing must only be considered in cases where *in vitro* studies are not applicable, or the results of these studies are not adequate for classification and risk assessment.

How to use the *in vitro* methods for skin corrosion/irritation within the testing and assessment strategy

For skin corrosion and irritation no single *in vitro* test method can fully replace the *in vivo* test (EU TM B.4 / OECD TG 404) across the full range of skin responses. However, the *in vitro* methods specified in Section [R.7.2.3.1](#) and [R.7.2.4.1](#) may replace the *in vivo* test depending on the outcome of the study or when combined within a tiered testing strategy.

Certain steps need to be taken before any testing (*in vitro* or *in vivo*) is conducted as described in the introductory paragraph of Annex VII to the REACH Regulation, i.e. assessment of all available data e.g. existing *in vitro*, *in vivo* and human data (see [Figure R.7.2-2](#)).

If a conclusion on classification cannot be made based on existing information, the following test(s) need(s) to be performed:

- 1) Skin corrosion, *in vitro*
- 2) Skin irritation, *in vitro*

New *in vitro* testing should be performed following a top-down or bottom-up approach, based on presumed properties ([Figure R.7.2–3](#)). The top-down approach should be used when available information suggests that the substance may be irritant or corrosive to the skin. The bottom-up approach, on the other hand, should be followed only when available information suggests that the substance may not be irritant to the skin.

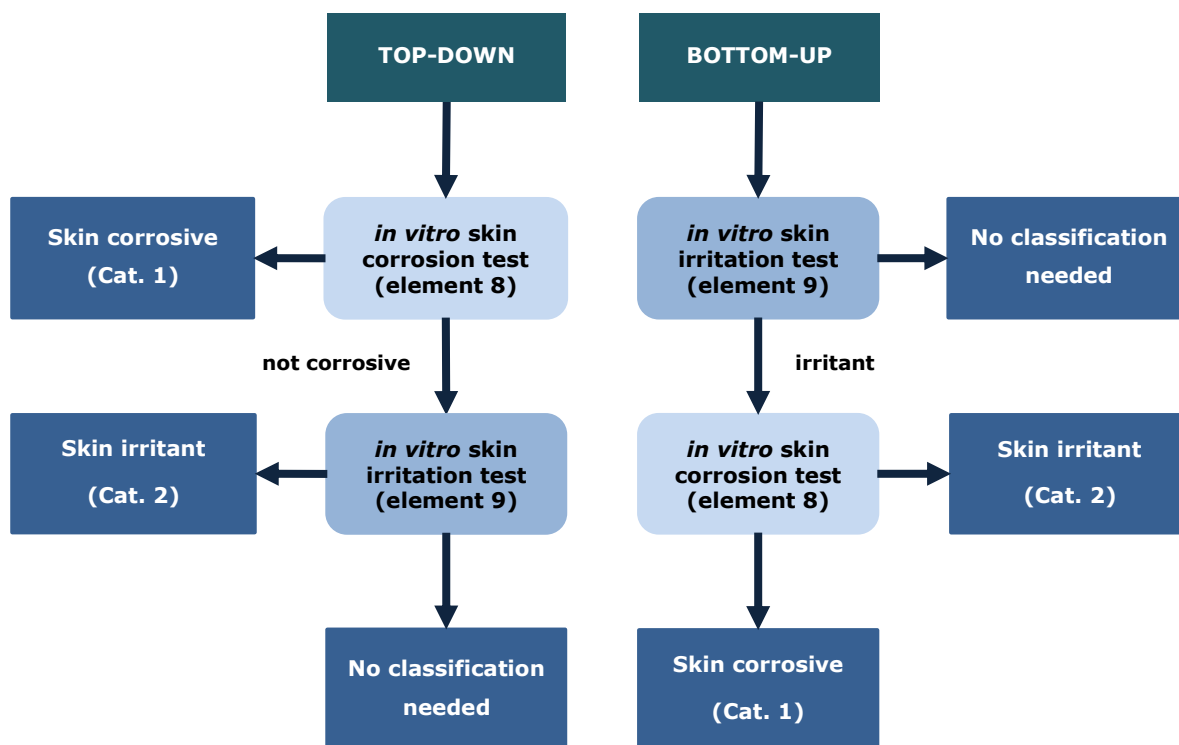


Figure R.7.2–3 Schematic presentation of Top-down and Bottom-up approaches for Skin Corrosion/irritation.

After these steps, no new *in vivo* testing is necessary (for any tonnage level) unless:

- a) the substance does not fall under the scope and applicability domain of the specific *in vitro* tests performed, and there are no substance-specific limitations to using those tests, and
- b) the Registrant cannot use the results of the *in vitro* test(s) performed for classification and risk assessment.

It is important to note that it is the responsibility of the registrant to ensure that the chosen test method is suitable for the substance in order to obtain adequate information from the *in vitro* studies. For most substances, the use of adopted EU or OECD TGs for skin corrosion/irritation purposes will provide results that will have regulatory acceptance under REACH.

SERIOUS EYE DAMAGE/EYE IRRITATION

R.7.2.7 Information requirements for serious eye damage/eye irritation

The information on serious eye damage/eye irritation that is required to be submitted for registration and evaluation purposes is specified in Annexes VI to XI to the REACH Regulation. According to Annex VI, the registrant should gather and evaluate all existing available information before considering further testing. This includes physico-chemical properties, (Q)SAR ((Quantitative) Structure-Activity Relationship), grouping, *in vitro* data, animal studies, and human data. For classified substances, information on exposure, use and risk management measures should also be collected and evaluated in order to ensure safe use of the substance.

If these data are inadequate for hazard and risk assessment, further testing should be carried out in accordance with the requirements of Annexes VII (≥ 1 tpa) and VIII (≥ 10 tpa) to the REACH Regulation.

R.7.2.7.1 Information requirements for quantities of ≥ 1 tpa (Annex VII to the REACH Regulation)

If new testing data are necessary, these must be derived from *in vitro* methods only. Annex VII does not foresee *in vivo* testing for serious eye damage/eye irritation.

The standard information requirements at this tonnage level for serious eye damage/eye irritation are specified in Section 8.2 in Column 1 of Annex VII as follows:

8.2.1. Serious eye damage/eye irritation, *in vitro*

Section 8.2 in Column 2 of Annex VII lists specific rules for adaptation according to which step 8.2.1 is not necessary. These rules are applicable when:

- the substance is classified as corrosive to the skin, leading to classification as "serious eye damage (Category 1)", or
- the substance is classified as a skin irritant and the available information indicates that it should be classified as an eye irritant (Category 2), or
- the substance is a strong acid ($\text{pH} \leq 2.0$) or base ($\text{pH} \geq 11.5$) and the available information indicates that it should be classified as "serious eye damage (Category 1)", or
- the substance is flammable in air or in contact with water or moisture at room temperature.

In addition, in Section 8.2.1 of Column 2, the REACH Regulation specifies that "*If results from a first in vitro study do not allow a conclusive decision on the classification of a substance or on absence of eye irritation potential, (an)other in vitro study(ies) for this endpoint shall be considered.*"

The *in vitro* methods that can be used to fulfil the standard information requirements for REACH Annex VII are detailed in Sections [R.7.2.8.1](#) and [R.7.2.9.1](#) of this Guidance, under "*In vitro* data". In case an existing good quality *in vivo* eye irritation study is available, its results

can be used to fulfil the standard information requirement, however an adaptation argument for not submitting the *in vitro* study would need to be submitted.

Guidance on the application of these rules is given in the testing and assessment strategies described in Sections [R.7.2.6](#) and [R.7.2.11](#) of this Guidance.

R.7.2.7.2 Information requirements for quantities of ≥ 10 tpa (Annex VIII to the REACH Regulation)

As specified in Section 8.2 of Column 2 of Annex VIII to the REACH Regulation, for substances manufactured or imported in quantities of ≥ 10 tpa *in vivo* testing must be considered only if the *in vitro* study(ies) under Section 8.2.1 in Annex VII is (are) not applicable for the substance, or the results of this (these) study(ies) are not adequate for classification and risk assessment.

Section 8.2 of Annex VIII specifies the conditions under which an *in vivo* study for serious eye damage/eye irritation is still required. For substances manufactured or imported in quantities of ≥ 10 tpa, *in vivo* testing must only be considered if the *in vitro* studies under Section 8.2.1 of Annex VII are not applicable for the substance, or the result(s) of these studies are not adequate for classification and risk assessment.

The study does not need to be conducted if:

- the substance is classified as corrosive to the skin, or
- the substance is a strong acid ($\text{pH} \leq 2.0$) or base ($\text{pH} \geq 11.5$), or
- the substance is spontaneously flammable in air or in contact with water or moisture at room temperature.

Guidance on the application of these rules is given in the testing and assessment strategy for serious eye damage/eye irritation described in Section [R.7.2.11](#) of this Guidance.

It should be noted that the conditions of acceptance by ECHA of implementation of any of the adaptation rules laid down in Annex XI are strict, and whenever an adaptation argument is being used (e.g. use of (Q)SARs, SARs, read-across or non-validated *in vitro* test methods), scientific justification, solid documentation and readiness for risk assessment and Classification and Labelling must be provided by registrants. For detailed information on these rules, see Annex XI to the REACH Regulation.

R.7.2.8 Information sources on serious eye damage/eye irritation

R.7.2.8.1 Non-human data on serious eye damage/eye irritation

Non-testing data on serious eye damage/eye irritation

Physico-chemical properties

Relevant information can be inferred from basic physico-chemical characteristics of a substance (e.g. extreme pH). *Extreme* pH values may indicate the potential of a substance to cause skin corrosion or serious eye damage:

If the pH is ≤ 2 or pH ≥ 11.5 , then consider the substance to be corrosive to the skin (Category 1) and to cause serious eye damage (Category 1) when pH is used as the sole basis for the classification decision (See also Sections [R.7.2.4.1](#) and [R.7.2.9.1](#) of this Guidance).

Grouping, (Q)SARs and expert systems ³⁹

In REACH Annex XI two types of non-testing methods are mentioned which can be used for adaptation of standard information requirements, either as standalone (where possible) or in combination with other information (in the context of a *Weight-of-Evidence* assessment):

- qualitative and quantitative Structure-Activity-Relationships (SARs/QSARs, section 1.2, including expert systems, generally incorporating multiple (Q)SARs, expert rules and data) on the one hand, and
- grouping of substances and read-across approaches ⁴⁰.

The adaptation of standard information requirements can be used for the assessment of serious eye damage/eye irritation, if it provides relevant and reliable data for the substance of interest. As specified in Annex XI to the REACH Regulation, the use of non-testing methods needs to be justified and sufficiently documented. In the case of QSARs and expert systems, registrants need to prepare property predictions by completion of a QSAR Prediction Reporting Format (QPRF). The QPRF is a harmonised template for summarising and reporting substance-specific predictions generated by (Q)SAR models. For filling a data gap under REACH, it is also necessary to provide information on the prediction model employed following a QSAR Model Reporting Format (QMRF) document. The QMRF is a harmonised template for summarising and reporting key information on (Q)SAR model validity, including the results of any validation studies. The information is structured according to the OECD (Q)SAR validation principles (for further information see <http://www.oecd.org/env/ehs/risk-assessment/validationofqsarmodels.htm>). The JRC QSAR Model Database is an inventory of information on available QMRFs, freely accessible online (<https://eurl-ecvam.jrc.ec.europa.eu/databases/jrc-qsar-model-database>). More detailed guidance on QSAR models, their use and reporting formats, including the QMRF, is provided in Section R.6.1 of Chapter R.6 of the [Guidance on IR&CSA](#).

In general, there are several different ways in which non-testing methods can be used in the context of an IATA (an IATA for serious eye damage and eye irritation is currently under development by the OECD), e.g.:

- for direct prediction of serious eye damage/eye irritation potential or the absence thereof,
- as part of a *Weight-of-Evidence* scheme (where the information from non-testing methods alone is not sufficient for a decision), or

³⁹ Further information can be found in *Chapter R.6 QSAR and grouping of chemicals* of the [Guidance on IR&CSA](#), the OECD Guidance on Grouping of Chemicals, Second Edition (OECD, 2014a), the new OECD Guidance on an Integrated Approach for Testing and Assessment (IATA) for skin corrosion and irritation (OECD, 2014b) and the JRC report on Alternative methods for regulatory toxicology (Worth, 2014).

⁴⁰ The relevant terminology is not always used consistently. With reference to the ECHA Guidance on QSAR and grouping, the terms category approach and analogue approach are used to describe techniques for grouping of substances, whilst the term read-across is reserved for a technique to fill data gaps, i.e. to transfer knowledge from one or more substances called source(s) to another substance with data gap, named target substance.

- in order to decide how best to proceed with further (*in vitro*) testing (i.e. *via* a top-down or bottom-up approach). For further information see Section [R.7.2.11.2](#).
- SARs and read-across for serious eye damage and eye irritation:

In principle, the same considerations apply as with the use of SARs and read-across for skin corrosion/irritation (see Section [R.7.2.3.1](#)). Structural alerts for serious eye damage/eye irritation have been described in the literature, e.g. in Gerner *et al.* (2005).

The occurrence of structural analogues that exhibit serious eye damage (or eye irritation) potential can also be used to predict the effect in the substance of interest and adapt the respective information requirements. Negative data from structural analogues may also be used to make predictions in certain cases, however, absence of one of the known structural alerts for irritation and corrosion alone does not prove absence of effect, as knowledge of structural alerts for irritation and corrosion might be incomplete. For instance, other substructures (not yet identified as structural alerts) or other properties of the substance may be responsible for a corrosive or irritant effect.

- QSARs and expert systems for serious eye damage and eye irritation:

An overview of available (Q)SARs for serious eye damage/eye irritation is provided in [Table R.7.2-3](#). An extensive review of the state-of-the-art was published by the former ECB (Gallegos Saliner *et al.* 2006, 2008). In [Appendix R.7.2-3](#) some examples are given to illustrate currently available models and the techniques that have been used to develop them. Examples of models based on classical regression and classification techniques, together with more innovative approaches, are collected in [Appendix R.7.2-3](#).

The most widely used expert systems for assessing eye irritation are the same as those used for assessing skin corrosion and irritation. Details on automated rule-induction systems (e.g. TOPKAT and MultiCASE), and on knowledge-based systems (e.g. DEREK Nexus, and the BfR rule-base) are reported in [Appendix R.7.2-3](#).

The freely downloadable OECD QSAR Toolbox software contains two profilers relevant for serious eye damage/eye irritation based on the BfR rule-base, which encode "inclusion rules" (structural alerts predicting serious eye damage/eye irritation potential) with a suggestion that exclusion of serious eye damage/eye irritation potential might be possible based on certain physico-chemical properties. The use in combination of profilers and data for analogues could allow for the prediction of serious eye damage/eye irritation for new substances through a read-across or category approach. More details on the OECD QSAR Toolbox specific contents for skin irritation and corrosion are reported in [Appendix R.7.2-3](#).

Not all of the models were developed with EU regulatory purposes in mind, so it is important to assess in each case whether the endpoint or effect being predicted corresponds to the regulatory endpoint of interest. The BfR model for the prediction of serious eye damage/eye irritation has been developed to predict EU regulatory endpoints, however predictions refer to the former DSD classification/labelling system used in the EU before the CLP Regulation came into force, and in borderline cases the results of the prediction may not fully reflect the correct CLP classification. More details on this model are reported in [Appendix R.7.2-3](#).

It should also be noted that the criteria for classification of a substance as eye irritant Category 2 based on the mean score for corneal opacity and conjunctival redness in the *in vivo* test have changed from ≥ 2 and ≥ 2.5 , respectively, under DSD to ≥ 1 and ≥ 2.0 , respectively, under CLP. Consequently predictions as eye irritant Cat 2 from models developed based on the DSD criteria should be interpreted with caution since they may lead to underprediction and should not be used for direct classification under CLP.

In the case of classification models for serious eye damage/eye irritation, the classification criteria used in model development should be compared with the EU classification criteria, to assess the relevance of the model. Where it is not indicated in the supporting literature whether the predicted classification should be Category 1 (Serious eye damage) or Category 2 (Eye irritation), the category chosen should be supported with expert judgement.

Table R.7.2–3 Overview of available (Q)SARs for serious eye damage/eye irritation. See [Appendix R.7.2–3](#) for more information on these models.

Category of model or source	Reference or name of the model	Applicability domain
Literature models	Solimeo <i>et al.</i> (2012)	Not available
	Abraham <i>et al.</i> (2003)	Pure bulk liquids
	Gerner <i>et al.</i> (2005)	Based on Physico-chemical values
	Barratt (1995b, 1997)	Neutral organic chemicals
Computerised models	PaDEL-DDPredictor (http://padel.nus.edu.sg/software/padelddpredictor/) (Liew and Yap, 2013)	Calculated by the model based on the range of descriptors
	BfR rule-base, free (included in the OECD QSAR Toolbox and Toxmatch, Toxtree, ToxPredict and Ambit)	EU New chemicals (NONS) database, organic chemicals with no significant hydrolysis potential and purity >95%
	ACD/Percepta, commercial	Organic chemicals
	Derek Nexus, commercial	Organic chemicals and some metals
	HazardExpert, commercial	Organic chemicals
	MolCode, commercial	Organic chemicals
	MultiCASE, commercial	Organic chemicals
	TOPKAT, commercial	Organic chemicals
Review papers	Patlewicz <i>et al.</i> , 2003	N.A.
	Gallegos Saliner <i>et al.</i> (2006, 2008)	N.A.

Abbreviation: N.A. = not applicable.

Testing data on serious eye damage/eye irritation

The internationally accepted test methods for serious eye damage/eye irritation as described in the Annex to the EU Test Methods (TM) Regulation (Council Regulation (EC) No 440/2008) and in OECD TGs (available at http://www.oecd.org/env/ehs/testing/oecdguidelinesforthetestingofchemicals.htm#Test_Guide_lines) are: EU B.5 (OECD TG 405), EU B.47 (OECD TG 437), EU B.48 (OECD TG 438), OECD TG 460, OECD TG 491 and OECD TG 492.

At the OECD there are currently three additional draft TGs under discussion regarding the eye hazard, i.e. EpiOcular™ EIT, Short-time exposure (STE) test method and Cytosensor® microphysiometer (CM) test method. Additional test methods may become available for addressing the eye hazard, therefore the reader is advised to check the OECD website and ECHA's test methods webpage (<http://echa.europa.eu/support/testing-methods-and-alternatives>) to check the current status of these test methods.

Please note that the latest version of an adopted test guideline should always be used when generating new data, independently of whether it is published by the EU or OECD.

The testing and assessment strategy developed for serious eye damage/eye irritation (see Section [R.7.2.11](#) of this Guidance) emphasises the need to evaluate all available information (including physico-chemical properties) before undertaking any *in vivo* testing. This strategy employs screening elements designed to avoid, as far as possible, *in vivo* testing of corrosive and severely irritating substances. In particular, *in vitro* tests should usually be performed first, and it should be assessed whether *in vivo* testing can be completely avoided.

In vitro data

Accepted *in vitro* test methods to detect serious eye damage (Category 1 under CLP) and/or absence of effects requiring classification for serious eye damage/eye irritation (i.e. not classified under CLP) are listed in [Table R.7.2-4](#). More information on the specific scope and limitations of these tests is provided in Section [R.7.2.9.1](#) under "Testing data on serious eye damage/eye irritation".

Table R.7.2-4 Accepted *in vitro* test methods for serious eye damage/eye irritation

Test method	Validation status, regulatory acceptance	EU Test Method /OECD test guideline	Classification according to CLP Regulation	EURL ECVAM DB-ALM protocol Nr.
Serious eye damage / eye irritation				
BCOP	Validated and regulatory acceptance	B.47 / OECD TG 437	Cat. 1 or NC	98, 124
ICE	Validated and regulatory acceptance	B.48 / OECD TG 438	Cat. 1 or NC	80
FL	Validated and regulatory acceptance	N.A. / OECD TG 460	Cat. 1	71

	STE	Validated and regulatory acceptance	N.A. / OECD TG 491	Cat. 1 or NC	N.A.
	RhCE	Validated and regulatory acceptance	N.A. / OECD TG 492	NC	N.A.
	CM ⁴¹	Validated and considered to be scientifically valid	N.A. / OECD draft TG available and being considered for adoption	Cat. 1 or NC	130
	Ocular Irritation® Assay ⁴²	Validated	N.A. / N.A.	Cat. 1	157
Test methods currently with limited application under REACH					
	IRE ⁴³	validated	N.A. / N.A.	Cat. 1	85
	HET-CAM ⁴³	Validated	N.A. / N.A.	Cat. 1	47, 96

NOTE: During the validation exercise EURL ECVAM concluded that the SkinEthic™ Human Corneal Epithelium (HCE) is not sufficiently sensitive for identifying substances not classified for serious eye

⁴¹ The CM test method was validated by EURL ECVAM and considered to be scientifically valid (<https://eurl-ecvam.jrc.ec.europa.eu/validation-regulatory-acceptance/topical-toxicity/eye-irritation>; section 1.2) and was also reviewed by ICCVAM (<http://ntp.niehs.nih.gov/?objectid=807EF83B-92CC-9A6C-3FFE8725DF1F9F5D>); A draft OECD Test Guideline is available at: <http://www.oecd.org/env/ehs/testing/section4healtheffects.htm>.

⁴² The Ocular Irritation® Assay has undergone an external prospective and retrospective validation study co-sponsored by In Vitro International (the method developer) and INT.E.G.RA (Esques *et al.*, 2014) and appears to be a suitable test method for the identification of substances causing serious eye damage (CLP Category 1) and not requiring classification for the eye hazard. The test method is also proposed by the developer to be suitable for the identification of substances not classified for serious eye damage/eye irritation based on the outcome of a validation study. However, an independent peer-review of the validation study is still pending and therefore the final applicability of the test method still needs to be confirmed. Therefore conclusions on classification cannot be drawn from negative results before the scientific validity of the test method to correctly identify substance not requiring classification for serious eye damage/eye irritation has been confirmed.

⁴³ Concerning the IRE and HET-CAM test methods, ICCVAM validation assessments in 2007 and 2010 that these test methods were not sufficiently accurate for regulatory use or that there was not sufficient data, especially for Category 2 chemicals, to make a final conclusion on their validity and recommended additional studies (<http://ntp.niehs.nih.gov/pubhealth/evalatm/test-method-evaluations/ocular/in-vitro/index.html> & <http://ntp.niehs.nih.gov/pubhealth/evalatm/test-method-evaluations/ocular/in-vitro-test-methods/index.html>). The Manual of Decisions of the Competent Authorities (EC, 2009) concluded that there is enough evidence available to conclude that the test methods are able to detect substances causing severe damage to eyes. Positive results can therefore be used for classification purposes i.e. leading to a classification of Category 1 for serious eye damage and labelling with H318 "Causes serious eye damage" according to CLP.

damage/eye irritation (the test method produced an unacceptable number of false negative results in the validation study) and recommended optimisation and further validation of the test method by the developer (EURL ECVAM, 2015).

Abbreviations: BCOP = Bovine Corneal Opacity and Permeability; CM = Cytosensor Microphysiometer; FL = Fluorescein Leakage; HET-CAM = Hen's Egg Test on Chorioallantoic Membrane; ICE = Isolated Chicken Eye; IRE = Isolated Rabbit Eye; N.A. = not available; NC = not classified; RhCE = Reconstructed human Cornea-like Epithelium Test Method; STE = Short-Time Exposure.

The test methods indicated in [Table R.7.2-4](#) above are either organotypic assays (BCOP, ICE, IRE and HET-CAM), cytotoxicity and cell function based assays (CM, FL and STE), reconstructed human cornea-like epithelium assays (RhCE), or *in chemico* assays (Ocular Irritation®). These test methods are mainly concerned with modelling the immediate effects of substances on the cornea. *In vivo* eye irritation endpoints which may not be covered by the above-mentioned optimised protocols are the following:

- i. persistence/reversibility of effects
- ii. discolouration on the cornea ⁴⁴

Concerning persistence and reversibility of effects, the OECD TGs for BCOP (OECD TG 437) and ICE (OECD TG 438) and the OECD GD 160 (OECD, 2011) state that histopathological examination of the corneas may be potentially useful when a more complete characterization of corneal damage is needed. Some evidence has been published showing that histopathology may support the identification of irreversible effects produced by non-extreme pH detergent and cleaning products when used in combination with the ICE test method (Cazelle *et al.*, 2014). However, more work is still needed to assess the usefulness of the histopathological evaluation concerning identification of irreversible effects.

There are currently no validated *in vitro* eye irritation test methods available that could be used for the direct identification of Eye irritants Category 2 under CLP.

Additional test methods currently under development to assess different ranges of eye irritation potential are e.g. the *Ex Vivo* Eye Irritation Test (EVEIT) and the Porcine Cornea Reversibility Assay (PorCORA). The EVEIT and PorCORA test methods are organotypic assays which use either isolated rabbit or porcine corneas, respectively, and have been proposed to be able to discriminate between reversible and irreversible (persistent) effects by directly monitoring the recovery process in excised corneas kept in culture for several days following chemical exposure (Frentz *et al.*, 2008; Spöler *et al.*, 2010; Piehl *et al.*, 2010, 2011).

Testing and Assessment strategies combining different test methods according to their applicability domain and capacity to classify in the different ranges of serious eye damage/eye irritation (from those listed in table R.7.2-4 and those mentioned in the previous paragraphs) still need to be developed to facilitate the identification of Category 2 substances on the basis of methods that currently can only be used to directly identify Category 1 and/or not classified substances.

Further test method developments may occur and the registrants are advised to follow the latest updates through e.g. EURL ECVAM website (<https://eurl-ecvam.jrc.ec.europa.eu/>) and ECHA's test methods webpage (<http://echa.europa.eu/support/testing-methods-and-alternatives>) for potential new test guidelines and test guideline updates.

⁴⁴ Current *in vitro* TGs (listed in table R.7.2-4 above) do not cover discoloration of the cornea, but some test methods may give indications about this effect.

Animal data

Annex I to the CLP Regulation defines serious eye damage/eye irritation as local toxic effects, and, as such, an assessment of serious eye damage/eye irritation is normally part of the acute testing phase of a toxicity programme and it is an early requirement of all regulatory programmes. Testing for serious eye damage/eye irritation has, historically, used animal models and a variety of test methodologies depending upon, for example, the laboratory undertaking the test, the area and intended application. However, in line with one of the objectives of the REACH Regulation, as described in Articles 13(1) and 25(1) and Annex VI, animal testing should be undertaken only as a last resort after i) considering all existing available test data and ii) generating information whenever possible by means of alternative methods to animal testing such as *in vitro* methods, QSAR models, grouping or read-across.

In cases in which *in vivo* testing is necessary, current approaches for serious eye damage/eye irritation testing *in vivo* are covered by the Acute Eye Irritation/Corrosion test method (EU B.5/OECD TG 405). This guideline recommends a tiered approach, whereby existing and relevant data are evaluated first. The guideline also recommends that testing in animals should only be conducted if determined to be necessary after consideration of available alternative methods. The *in vivo* test uses one animal (the rabbit is the preferred species); in the absence of severe effects this is followed by a further testing of up to two animals (a total maximum of three animals).

Both EU and OECD methods use the scoring system developed by Draize (1944). The EU criteria for classification are based on the mean tissue scores obtained over the first 24-72 hour period after exposure and on the reversibility or irreversibility of the effects observed. Currently, *irritants* (Category 2 Eye irritants) cause significant inflammation of the eye (conjunctiva redness/oedema, cornea and/or iris) but this effect is transient, i.e. the affected sites are repaired within the observation period of the test. A substance causing considerable damage to the cornea and/or iris is classified in Category 1 for Serious Eye Damage. The criteria for classification in Category 1 for Serious Eye Damage include persistence of effects (effects on the cornea, iris or conjunctiva that are not expected to be reversed or have not fully reversed within an observation period of normally 21 days, i.e. with a score >0), irreversible staining of the eye and/or criteria for the degree of severity.

For existing data, the use of methods other than those specified in the Annex to the EU Test Methods Regulation, or corresponding OECD methods, such as the rabbit Low Volume Eye Test (LVET) (Griffith *et al.*, 1980) may be accepted on a case-by-case basis (see also ESAC, 2009).

R.7.2.8.2 Human data on serious eye damage/eye irritation

Existing human data include historical data that should be taken into account when evaluating intrinsic hazards of substances. *New* testing in humans for hazard identification purposes is not acceptable for ethical reasons.

Existing data can be obtained from case reports, poison information centres, medical clinics, occupational experience, epidemiological studies and volunteer studies. Their quality and relevance for hazard assessment should be critically reviewed. However, in general, human data can be used to determine a corrosive or irritating potential of a substance. Good quality and relevant human data have precedence over other data. However, absence of incidence in humans does not necessarily overrule *in vitro* data or existing animal data of good quality that are positive.

R.7.2.9 Evaluation of information on serious eye damage/eye irritation

R.7.2.9.1 Non-human data for serious eye damage/eye irritation

Non-testing data on serious eye damage/eye irritation

Physico-chemical properties

According to the current EU and OECD guidelines, substances should not be tested on animals for serious eye damage/eye irritation if they can be predicted to be corrosive to the skin (Category 1 of CLP) or cause serious eye damage (Category 1 of CLP) from their physico-chemical properties. In particular, substances exhibiting strong acidity ($\text{pH} \leq 2.0$) or alkalinity ($\text{pH} \geq 11.5$) in solution are predicted to be corrosive to the skin or cause serious eye damage, and should not be tested on animals. Testing with *in vitro* methods can nevertheless be performed to confirm classification decisions (see Section 3.3.2.3 of Annex I to the CLP Regulation).

A substance known or predicted to be corrosive to the skin can be considered to cause Serious Eye Damage (Category 1). However, no conclusion can be made regarding serious eye damage/eye irritation potential when the pH has an intermediate value (when $2.0 < \text{pH} < 11.5$). Where extreme pH is the only basis for classification of a substance as "serious eye damage", it may also be important to take into consideration the acid/alkaline reserve, i.e. a measure of the buffering capacity (Young *et al.*, 1988,; Young and How, 1994). However, the buffering capacity should not be used alone to exonerate from classification of the substance as corrosive. Indeed, when the acid/alkaline reserve suggests that the substance may not cause serious eye damage, further *in vitro* testing should be considered (see Section 3.3.2.3 of Annex I to the CLP Regulation).

Grouping, (Q)SARs and expert systems

Guidance has been developed by the former ECB (Worth *et al.*, 2005) on how to apply (Q)SARs for regulatory use. Guidance on how to assess the validity and suitability of (Q)SAR models and adequacy of their predictions is given in Section R.6.1 of Chapter R.6 of the [Guidance on IR&CSA](#). Essentially, the determination of whether a (Q)SAR result may be used to replace a test result can be broken down into three main steps:

1. evaluation of the scientific validity (relevance and reliability) of the model,
2. assessment of the applicability of the model to the chemical of interest and the reliability of the individual model prediction,
3. assessment of the adequacy of the information for making the regulatory decision, including an assessment of completeness, i.e. whether the information is sufficient to make the regulatory decision, and if not, what additional (experimental) information is needed.

The assessment of model validity needs to be performed along the lines of the OECD principles for (Q)SAR validation (OECD, 2007), e.g. in terms of a defined endpoint, an unambiguous algorithm, a defined applicability domain, the statistical characteristics ("goodness-of-fit"), and mechanistic interpretation.

The following questions, *inter alia*, should be addressed when assessing the reliability of an individual prediction:

- i. Is the chemical of interest within the scope of the model, according to the defined applicability domain of the model?

- ii. Is the defined applicability domain suitable for the regulatory purpose?
- iii. How well does the model predict chemicals that are similar to the chemical of interest?
- iv. Is the model estimate reasonable, taking into account other information?

The mechanism of serious eye damage/eye irritation involves toxicodynamic and toxicokinetic parameters. Some models predict serious eye damage and eye irritation based on toxicodynamic properties only (e.g. acidity or basicity, electrophilicity, other reactivity, surfactant activity, solvating membranes). Such models have to be additionally evaluated to check whether they also take account of toxicokinetic parameters related to the potential of a substance to cross relevant outer membranes of the eye (cornea) and to be active in the living tissue underneath; alternatively, these models have to be used in combination with data covering such toxicokinetic parameters. Conversely models predict (the absence of) serious eye damage/eye irritation solely from e.g. physico-chemical properties considered to illustrate the toxicokinetic behaviour of a substance. Such models should be evaluated to check whether they also take account of the activity of the substance (toxicodynamics), in particular for its potential to cause serious eye damage (whereby the corrosive action itself may lead to membrane destruction and subsequent tissue damage).

For example, the BfR rule-base implemented in Toxtree and the OECD QSAR Toolbox contains both physico-chemical exclusion rules and structure-based inclusion rules (structural alerts). Evaluations of these rules for the prediction/exclusion of eye irritation (Tsakovska *et al.*, 2005, on structural alerts; Tsakovska *et al.*, 2007, on physico-chemical exclusion rules) have been carried out in accordance with the OECD principles for (Q)SAR validation (see [Appendix R.7.2-3](#)). However, inclusion and exclusion rules were evaluated separately, and not used in combination in these works.

When applied, these two sets of rules may sometimes provide contradictory information, i.e. a structural alert might indicate serious eye damage/eye irritation potential, while at the same time, based on physico-chemical properties, absence of effect is predicted. In such cases, it is recommended to consider additional information (e.g. on the behaviour of chemically similar substances). In other cases, applicability of one (or more) of the physico-chemical exclusion rules might indicate absence of serious eye damage/eye irritation potential of the target substance, while no structural alert for serious eye damage/eye irritation is triggered. Given that the absence of any known structural alert is not equivalent to the absence of a potential effect, in such a situation the substance should still be examined for potentially reactive substructures (and examining the behaviour of chemical analogues would still be beneficial).

While these considerations apply to the use of the BfR rule-base for direct classification/non-classification, less certainty might be required e.g. for a decision on further *in vitro* testing: where the exclusion rules suggest the absence of an effect, a bottom-up approach could be followed, i.e. a test for eye irritation and not one for serious eye damage might be initiated (see Section [R.7.2.11.2](#)).

There is no other model available which sufficiently describes the absence of effects. Neutral organics⁴⁵ are expected not to be irritants. Predicted absence of reactivity needs to be described in sufficient detail or be substantiated with other information.

⁴⁵ By definition a neutral organic is a chemical which does not have potential reaction centres, even after skin metabolism.

Testing data on serious eye damage/eye irritation

In vitro data

There are EU and OECD adopted test guidelines (see Section [R.7.2.8.1](#)), according to which substances can be classified as causing serious eye damage or not classified.

Annex VII to the REACH Regulation requires information from *in vitro* tests for serious eye damage/eye irritation, and not from animal tests. Guidance on how *in vitro* data can also be used to fulfil Annex VIII requirements, is given in Section [R.7.2.11](#) of this document.

Data from the following types of tests can be used:

- **Bovine Corneal Opacity and Permeability (BCOP) test method** (EU B.47/OECD TG 437): The specific scope and limitations are:
 - This test is recommended for identifying substances that cause serious eye damage, i.e. substances to be classified in Eye Damage Category 1 under CLP, without further testing, and also recommended to identify substances that do not require classification for eye irritation or serious eye damage i.e. leading to non-classification under CLP, without further testing;
 - If, as a result of testing, the substance is neither classified as "Eye Damage Category 1" nor identified as not requiring classification under CLP, further testing/evaluation is required;
 - This test may result in false positive Category 1 predictions (serious eye damage) for alcohols and ketones and false negative predictions (underpredicted Category 1 substances) for substances that would be classified as "Eye Damage Category 1" *in vivo* based on persistence of effects only (i.e., that do not meet Category 1 classification criteria based on the mean scores obtained from the first 3 observation days but show persistent effects at the 21st observation day) (Adriaens *et al.*, 2014; OCED, 2013). See also Section [R.7.2.8.1](#) for "*In vitro* test methods for serious eye damage/eye irritation";
 - This test does not allow testing of gases and aerosols.
- **Isolated Chicken Eye (ICE) test method** (EU B.48/OECD TG 438): The specific scope and limitations are:
 - This test is recommended for identifying substances that cause serious eye damage, i.e. substances to be classified in Eye Damage Category 1 under CLP, without further testing, and also recommended to identify substances that do not require classification for eye irritation or serious eye damage i.e. leading to non-classification under CLP, without further testing;
 - If, as a result of testing, the substance is neither classified as "Eye Damage Category 1" nor identified as not requiring classification under CLP, further testing/evaluation is required;
 - Similar limitations in relation to false positive and false negative predictions, as specified for the BCOP assay above, apply to this test method as well;
 - This test does not allow testing of gases and aerosols.

- **Fluorescein leakage (FL) test method** (OECD TG 460): The specific scope and limitations are:
 - This test is recommended for identifying substances that cause serious eye damage, i.e. substances to be classified in Eye Damage Category 1 under CLP, without further testing;
 - This test is not recommended for the identification of substances which should be classified as "Eye irritants Category 2" or of substances which should not be classified for serious eye damage and eye irritation;
 - This test is only applicable to water soluble substances and/or where the toxic effect is not affected by dilution;
 - Its applicability domain does not include strong acids and bases, cell fixatives and highly volatile substances;
 - If, as a result of testing, the substance is not classified as "Eye Damage Category 1" under CLP, further testing/evaluation is required.

- **Short Term Exposure (STE) test method** (OECD TG 491): The specific scope and limitations are:
 - This test is recommended for identifying substances that cause serious eye damage, i.e. substances to be classified in Eye Damage Category 1 under CLP, without further testing, and also recommended to identify substances that do not require classification for eye irritation or serious eye damage i.e. leading to non-classification under CLP, without further testing;
 - If, as a result of testing, the substance is neither classified as "Eye Damage Category 1" nor identified as not requiring classification under CLP, further testing/evaluation is required;
 - Its applicability domain does not include highly volatile substances with a vapor pressure over 6 kPa and solids (substances or mixtures) other than surfactants and mixtures composed only of surfactants.

- **Reconstructed human Cornea-like Epithelium (RhCE) Test Method** (OECD TG 492): The specific scope and limitations are:
 - This test is recommended for identifying substances not requiring classification for eye irritation or serious eye damage.
 - This test is not recommended for the identification of eye irritants (Category 2) or substances causing serious eye damage (Category 1).
 - Substances absorbing light in the same range as MTT formazan and substances able to directly reduce the vital dye MTT (3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide, Thiazolyl blue) to MTT formazan may interfere with the tissue viability measurements. Additional controls should be used to detect and correct for test substance interference with the viability measurement technique (see OECD TG for further details).
 - This test does not allow testing of gases or aerosols.

Beside results from the test methods mentioned above, a positive outcome (Serious Eye Damage Category 1) from one of five *in vitro* assays (i.e. the IRE, HET-CAM, CM, STE, Ocular Irritation™ assay) is also accepted in the EU to classify a substance as "Eye Damage Category 1" under CLP. A negative outcome, i.e. leading to non-classification according to CLP, can also be accepted for fulfilling the information requirement on the basis of test data obtained with the CM test method, in case the substance falls into the applicability domain of the test method.

Currently, there are no validated *in vitro* methods available for the direct identification of Category 2 Eye irritants.

- [Quality Aspects of existing *in vitro* data:](#)

For quality assessment of existing *in vitro* data that will form the basis for later possible *Weight-of-Evidence* considerations, see Section R.4.4 of Chapter R.4 of the [Guidance on IR&CSA](#), and for aspects that need to be taken into account in such a *Weight of Evidence* see Section R.5.2.1.2 of Chapter R.5 of the [Guidance on IR&CSA](#).

[Animal data](#)

Well-reported studies, particularly if conducted in accordance with the principles of GLP, can be used to identify substances which would be considered to cause, or not to cause serious eye damage or eye irritation. There may be a number of serious eye damage/eye irritation studies already available for an existing substance, none of which are fully equivalent to an OECD TG or an EU test method such as those in the Annex to the EU Test Methods Regulation. If the results from such a batch of studies are consistent, they may, together, provide sufficient information on the serious eye damage/eye irritation potential of the substance.

If the results from a variety of studies are unclear, based on the criteria given below for evaluation of the data, the registrant will need to decide which of the studies are most reliable, relevant for the endpoint in question and adequate for classification purposes.

Particular attention should be given to the persistence of irritation effects, even those which do not lead to classification. Effects such as persistent corneal opacity, discolouration of the cornea by a dye substance, adhesion, pannus, and interference with the function of the iris or other effects that impair sight which do not reverse within the test period may indicate that a substance will cause persistent damage to the human eye.

Data from studies other than skin corrosion/irritation studies (e.g. other toxicological studies on the substance in which local responses of skin have been reported) may provide useful information though they may not be well reported in relation to, for example, the basic requirements for information on skin irritation.

Data from studies other than serious eye damage/eye irritation studies (e.g. other toxicological studies on the substance in which local responses of the eye have been reported) may provide useful information though they may not be well reported in relation to, for example, the basic requirements for information on eye irritation. More notably, eye reactions and symptoms are not systematically scored in studies not specifically designed to address serious eye damage/eye irritation.

- [Quality Aspects of existing *in vivo* data:](#)

Data from **existing** irritation studies in animals must be taken into account before further testing is considered. A quality assessment of any such reports should be done using, for example, the system developed by Klimisch *et al.* (1997), as described in Section R.4.2 of

Chapter R.4 of the [Guidance on IR&CSA](#), and a judgement will need to be made as to whether any further testing is required. Some examples to note are:

- i. Was the animal species used the rabbit or was it another species such as the rat or the mouse? Normally the rabbit is used for eye irritation testing.
- ii. How many animals were used? Current methodology requires a maximum of 3 animals tested in a sequential manner (with 1 or 2 animals being sufficient if serious eye damage/irreversible effects are observed in the first or second tested animal, respectively) but 6 were frequently used in the past (see Section 3.3.2.3.2.2 of the [Guidance on the Application of the CLP criteria](#) for the evaluation of results from tests that have been conducted with more than 3 animals).
- iii. How many dose levels were used? If dilutions were included, what solvent was used (as this may have influenced absorption)? Which dose volume was used?
- iv. Check the observation period used post exposure. Shorter periods than in the current guideline may be adequate for non-irritants but may require a more severe classification for irritants when the observation period is too short to measure full recovery.
- v. Was initial pain noted after instillation of the test substance onto the eye? Was the substance washed out from the eye? Was fluorescent staining used?
- vi. How was the test material applied onto the eye?

Irritation scores from old reports, reports produced for regulatory submission in the USA or in publications may be expressed as a Maximum Average Score (MAS). Without the original data it is not always possible to convert these scores accurately into the scoring system used in the EU. For extremes, i.e. where there is either no irritation or severe irritation, it may not be necessary to look further, but average irritation scores pose a problem and expert judgement may be required to avoid repeat testing.

Observations such as those above can all be used to assess whether the existing animal test report available can be used reliably to predict the irritation potential of a substance, thus avoiding further testing.

- [Specific considerations:](#)

A refinement of the classical Draize test is the rabbit low volume eye test (LVET). The test protocol deviates from OECD TG 405 in that in the LVET, 10 µl is directly applied onto the cornea. The grading scale and the data interpretation in the LVET is exactly the same as those used in OECD TG 405. The validity of the LVET was reviewed by EURL ECVAM between 2006 and 2009 *via* retrospective validation for the detergent and cleaning products applicability domain (for further details, see <https://eurl-ecvam.jrc.ec.europa.eu/validation-regulatory-acceptance/topical-toxicity/eye-irritation>). Anatomical and physiological considerations for rabbit and human eyes indicate that a dose volume of 10 µl is appropriate (A.I.S.E. 2006): the tear volume in both rabbit and man is approximately the same (~ 7-8 µl), and after blinking, the volume capacity in the human eye is ~10 µl after blinking. Furthermore the use of direct cornea exposure mimics human exposure scenarios that can be reasonably expected (e.g. accidental ocular exposure during household use) and for the specific use domain of household detergents and cleaning products as well as their main ingredients (i.e. surfactants) as used in these products. These considerations suggest that the LVET is also potentially a suitable test to demonstrate toxicological effects on man of potential eye hazards of substances. The LVET has been used in industry for the safety evaluation of single substances (Griffith *et al.*, 1980) and detergent and cleaning products (Freeberg *et al.*, 1984; Freeberg *et al.* 1986a,b; Cormier *et*

al., 1995; Roggeband *et al.*, 2000), and has shown to be a very good predictor of the effects in man. It still overpredicts, but less than the classical Draize test of OECD TG 405.

After peer review, the LVET was not recommended for prospective use, i.e. to generate new data but it was acknowledged that existing LVET data of the limited use domain mentioned above may be used for purposes of classification and labeling decisions. Moreover, it was recognised that existing LVET data of this limited use domain may be used as supplementary data for future validation studies. No additional testing should however be performed to further develop or validate the LVET test. It was also pointed out that the LVET has a tendency to classify in lower hazard categories when compared to OECD TG 405. Nevertheless, it was acknowledged that these data may still be useful on a case-by-case basis, with respect to test data for household detergents, cleaning products and surfactants used in such products (ESAC, 2009).

In summary, available data from the LVET on substances should be considered and must be carefully evaluated. For the classification of substances it must be taken into account that the test has a limited applicability domain (detergent and cleaning products). Consequently, within the applicability domain of household detergents, cleaning products and their main ingredients, positive LVET data (be it Category 2 or Category 1) can be used for the appropriate classification for either serious eye damage or eye irritation, but negative data from LVET as a *stand alone method* (in the absence of any other information) are not conclusive for *no classification*.

R.7.2.9.2 Human data on serious eye damage/eye irritation

Well-documented *existing* human data of different sources can often provide very useful information on serious eye damage/eye irritation, sometimes for a range of exposure levels. Often the only useful information available on irritation is obtained from human experience (e.g. occupational settings). The usefulness of all human data on irritation will depend on the extent to which the effect, and its magnitude, can be reliably attributed to the substance of interest. Experience has shown that it is difficult to obtain useful data on substance-induced eye irritation, but data may be available on human ocular responses to certain types of mixtures (e.g. Freeberg *et al.*, 1986a).

The quality and relevance of existing human data for hazard assessment should be critically reviewed. For example, in occupational studies with mixed exposure it is important that the substance causing serious eye damage or eye irritation has been accurately identified. There may also be a significant level of uncertainty in human data due to poor reporting and lack of specific information on exposure.

Examples of how existing human data can be used in hazard classification for irritation are provided in an ECETOC monograph (ECETOC, 2002).

Substances causing Serious eye damage Category 1 give more severe corneal opacity and iritis than Eye irritants Category 2. Category 1 substances induce considerable tissue damage which can result in serious physical decay of vision. It is recognised that such severe lesions usually do not reverse within 21 days (relates to animals) (see Section 3.3 of Annex I to the CLP Regulation). In contrast, the effects of Category 2 substances are reversible within 21 days. In humans, an ophthalmic examination by a physician would reveal a decay of vision. If it is not transient but persistent it implies classification in Category 1. If the discrimination between Category 1 and Category 2 is not obvious, then Category 1 might be chosen, however other types of information may be generated e.g. by performing *in vitro* testing, to support the conclusion (for further information, see Section 3.3 of the [Guidance on the Application of the CLP criteria](#)).

R.7.2.9.3 Exposure considerations for serious eye damage/eye irritation

Exposure-based waiving from testing is not applicable to the endpoint of serious eye damage/eye irritation. Exposure-based waiving from testing as specified in Annex XI (3) of the REACH Regulation only applies to tests listed in Sections 8.6 and 8.7 of Annex VIII, Annex IX and Annex X according to the REACH text.

R.7.2.9.4 Remaining uncertainty on serious eye damage/eye irritation

Usually it is possible to unequivocally identify (or accept) a substance as causing serious eye damage, whatever type of study provides the information.

There may be a significant level of uncertainty in human data on irritant effects (e.g. because of poor reporting, lack of specific information on exposure, subjective or anecdotal reporting of effects, small numbers of subjects).

Data from studies in animals and from *in vitro* tests performed according to internationally accepted test methods will usually give relevant information on the serious eye damage/eye irritation potential of a substance. In general, it is assumed that substances which cause serious eye damage/eye irritation in EU or OECD TG-compliant studies in animals or *in vitro* will cause serious eye damage/eye irritation in humans, and those which are not irritant in EU or OECD TG-compliant studies will not be irritant in humans (Please note that in general test animals are considered to be more sensitive to serious eye damage/eye irritation than humans (e.g. Adriaens *et al.*, 2014)). It should be borne in mind that some of the limitations of the *in vivo* serious eye damage/eye irritation study include its high variability, the variable exposure being dependent on the physico-chemical properties of the test substance, and the subjective grading of the lesions (Adriaens *et al.*, 2014; Cormier *et al.*, 1996; Prinsen, 2006; Marzulli and Ruggles, 1973; Weil and Scala, 1971). Moreover, inconsistent results from a number of similar studies increases the uncertainty in deriving data from animal or *in vitro* studies.

The scope of the *in vitro* tests for serious eye damage/eye irritation has also some limitations, as explained in Section [R.7.2.9.1](#) under "Testing data on serious eye damage/eye irritation". In addition inconsistent results from two or more *in vitro* tests could add to the overall uncertainty in interpreting the data.

R.7.2.10 Conclusions on serious eye damage/eye irritation

R.7.2.10.1 Concluding on suitability for Classification and Labelling

In order to conclude on Classification and Labelling according to the CLP Regulation, all the available information needs to be taken into account and consideration should be given to both the [Guidance on the Application of the CLP criteria](#) and the various remarks (related to Classification and Labelling) made throughout this guidance document ⁴⁶.

⁴⁶ Please note that the 8th Adaptation to Technical and Scientific Progress (ATP) of the CLP Regulation will apply from 1 February 2018. The amendment by the 8th ATP will take into account the 5th Revision of the Globally Harmonized System of Classification and Labelling of Chemicals (GHS), which was adopted in 2012 and contains in particular refined criteria for skin corrosion/irritation and serious eye damage/eye irritation.

R.7.2.10.2 Concluding on suitability for Chemical Safety Assessment

A dose-response assessment is difficult to make for serious eye damage/eye irritation simply because up to now most data have been generated for undiluted substances in accordance with test guidelines and traditional practice (which continues today). From a risk characterisation perspective it is therefore advisable to use the outcome of the classification procedure, i.e. a substance that is classified is assumed to be sufficiently characterised. However, a complete risk assessment requires both hazard and dose-response data and for local effects the concentration is often the determinative dose metric. Consequently, if dose-response data are available, they must be taken into account (see [Figure R.7.2-4](#)).

Guidance on the possibilities for derivation of DNELs for serious eye damage/eye irritation is given in Appendix R.8-9 of Chapter R.8 of the [Guidance on IR&CSA](#).

R.7.2.10.3 Information not adequate

A *Weight-of-Evidence* approach comparing available adequate information with the tonnage-triggered information requirements under REACH may result in the conclusion that the requirements are not fulfilled. In order to proceed to further information gathering the testing and assessment strategy described in Section [R.7.2.11](#) below is recommended.

R.7.2.11 Testing and assessment strategy for serious eye damage/eye irritation

R.7.2.11.1 Objective / General principles

The following testing and assessment strategy is recommended for developing adequate and scientifically sound data for assessment/evaluation and classification of the serious eye damage and eye irritation properties of substances. For existing substances with insufficient data, this strategy can also be used to decide which additional data, beside those already available, are needed. The testing and assessment strategy is aimed at the identification of serious eye damage/eye irritation by using different elements where appropriate, depending on the information available. A basic principle of the strategy is that the results of one study or from an information source are evaluated before another study is initiated. The strategy seeks to ensure that the data requirements are met in the most efficient and humane manner so that animal usage and costs are minimised.

The different elements provided in the [Figure R.7.2-4](#) describe information sources that can be used to conclude on a substance's hazard potential towards the eye. The elements described in [Figure R.7.2-5](#) can be rearranged as appropriate, especially those in Part 1. This may be particularly helpful in cases where a conclusion can be drawn from certain elements without having to consider all of them. If judged relevant, elements in Part 1 can be omitted and *in vitro* testing can be performed immediately.

[Figure R.7.2-5](#) is divided into three parts whereby Part 1 aims at evaluating existing information that may be available on the substance. In Part 2 existing information and relevant data should be assessed in order to consider whether there is enough information available to conclude on the substance's hazard properties within a *Weight-of-Evidence* analysis, in case it is not possible to make a conclusion based on single elements described in Part 1. In case no conclusion can be drawn from Parts 1 and 2, new data should be generated in Part 3 by first performing relevant *in vitro* testing. *In vivo* testing must only be conducted in case no conclusion can be drawn based on the *in vitro* testing (and only for substances at or above 10 tonnes per annum only).

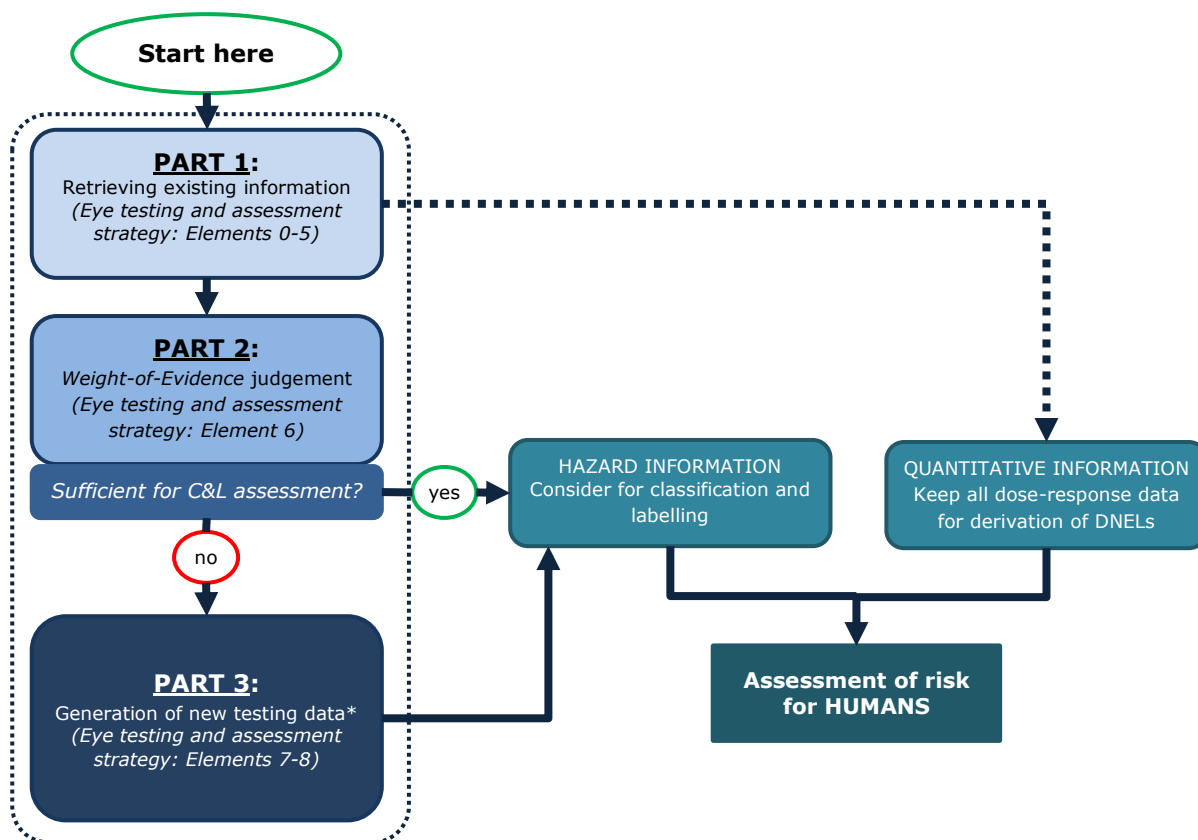
Some guidance for testing is provided by the specific rules for adaptation from standard information requirements, as described in column 2 of Annexes VII-X to the REACH Regulation, together with some general rules for adaptation from standard information requirements in Annex XI.

Risk assessment of the serious eye damage/eye irritation potential of a substance is normally made in a qualitative way provided that the substance has been classified as causing serious eye damage/eye irritation. Existing test guidelines do not contain dose-response assessment, consequently a quantitative analysis will often not be possible. Therefore, hazard identification and appropriate classification is the key determinant in the information gathering strategy below. As a consequence, the use of *Assessment Factors* is of limited use in order to take into account uncertainty of data. However, the registrant is encouraged to keep and use all quantitative data that might be encountered in the process of retrieving hazard information in the context of the present testing strategy and to perform a complete risk assessment, comprising assessment of qualitative hazard as well as quantitative information.

It is recommended that the testing and assessment strategy be followed until element 5 ([Figure R.7.2-4](#) and [Figure R.7.2-5](#)) in all cases and thereafter the *Weight-of-Evidence* analysis be performed. Clearly, all information sources/elements can be rearranged as appropriate, i.e. not all elements will necessarily be accompanied by data but it is important that all potential data sources are explored prior to starting the *Weight-of-Evidence* analysis. While it is recommended that this approach be followed, other approaches may be more appropriate and efficient on a case-by-case basis. For example, in case there is no existing data and it is anticipated that generation of "pre-testing data" would be non-conclusive, it may be deemed appropriate to directly proceed to the information generation part. Furthermore, prior to performing any new *in vivo* test, the use of *in vitro* methods must be fully exploited (see Articles 13(1) and 25(1) of the REACH Regulation).

If the substance is not classified for serious eye damage/eye irritation, no risk assessment for this endpoint is performed, regardless of the exposure. Please note that there are no options for exposure-based waiving for these endpoints in the REACH Regulation.

The following flow chart ([Figure R.7.2-4](#)) gives an overview of a possible approach for defining a testing and assessment strategy for serious eye damage/eye irritation.



*Generation of new testing data according to Annex VII to VIII to the REACH Regulation and with due observation of the rules for adaptation of the standard testing regime laid down in Annex XI.

Figure R.7.2–4 Overview of the testing and assessment strategy for serious eye damage/eye irritation

R.7.2.11.2 Testing and assessment strategy for serious eye damage/eye irritation

Recommended approach

The testing and assessment strategy for serious eye damage/eye irritation (see [Figure R.7.2–5](#)) is completely analogous in structure to that for skin corrosion/irritation. The testing and assessment strategy consists of three parts: Part 1 (elements 0 to 5) is about retrieving existing information, Part 2 (element 6) represents a *Weight-of-Evidence* analysis and expert judgement (element 6), and Part 3 is about generation of new information by testing (elements 7 to 8).

In Part 1, existing and available information from the literature and databases is gathered and considered in the strategy approach. The order of the different elements, i.e. 0 to 5, is only indicative and they may be arranged as appropriate. This may be particularly helpful in cases where a reliable conclusion can be drawn from certain elements without having to consider all of them. For instance, if the substance is classified as corrosive to the skin or has an extreme pH (≤ 2.0 or ≥ 11.5) serious eye damage is considered implicit (element 1c) and therefore the substance should be classified as causing serious eye damage (Category 1) according to CLP and further testing is not required. At the end of Part 1 and if no final conclusion could be

derived directly from one or several of the available pieces of information, all the information collected should be analysed using a *Weight-of-Evidence* approach (element 6).

In the information generation part (elements 7 to 8), new information on the serious eye damage/eye irritation potential of substances is generated by means of *in vitro* (element 7) or, as a last resort (see Articles 13(1) and 25(1) of the REACH Regulation), *in vivo* testing (element 9). Therefore, before concluding the *Weight-of-Evidence* analysis in element 6 and *in vitro* testing (elements 7a and 7b), new *in vivo* tests should not be conducted. More information on how to use the *in vitro* methods for serious eye damage/eye irritation within the testing strategy can be found in the following paragraphs.

While it is recommended that this approach be followed, other approaches may be more appropriate and efficient on a case-by-case basis. For example, in case there is no existing data and it is anticipated that compilation of data at elements 0-6 would be non-conclusive, it may be appropriate to directly proceed to the information generation part.

Figure R.7.2–5 Testing and assessment strategy for evaluating the serious eye damage/eye irritation potential of substances (footnotes a to f are detailed below the figure).

Element	Information	Conclusion ⁴⁷
Conclusion of the information strategy on skin corrosion/irritation		
0	Is the substance classified as a skin corrosive? →	YES: When assigned Skin Corrosive Cat. 1, 1A, 1B or 1C, the risk of severe damage to eyes is considered implicit (Serious Eye Damage Cat. 1) (Column 2 adaptation of Annexes VII and VIII).
Existing data on physico-chemical properties		
1a	Is the substance spontaneously flammable in air or in contact with water or moisture at room temperature? →	YES: No testing required (Column 2 adaptation of Annexes VII and VIII).
1b	Is the substance an organic hydroperoxide or an organic peroxide? →	YES: Consider classifying for: ■ When assigning a Skin Corrosive Cat. 1B classification for a hydroperoxide, the risk of serious eye damage is considered implicit. Consider classifying as Serious Eye Damage Cat. 1, or ■ When assigning a Skin Irritant Cat. 2 classification for a peroxide, the risk of eye irritation is considered implicit. Consider classifying as Eye Irritant Cat. 2. OR Provide evidence supporting deviating classification or non-classification ⁴⁸ .
1c	Is the pH of the substance ≤ 2.0 or ≥ 11.5? ^a →	YES: Consider classifying as Serious Eye Damage Cat. 1 (column 2 adaptation in section 8.2 of Annexes VII and VIII) if pH is used as the sole basis for classification decision.
1d	Are there other physical or chemical properties that indicate that the substance causes serious eye damage or eye irritation? →	YES: Use this information for <i>Weight-of-Evidence</i> analysis (Element 6).

⁴⁷ Please note that the 8th Adaptation to Technical and Scientific Progress (ATP) of the CLP will apply from 1 February 2018. The amendment by the 8th ATP will take into account the 5th Revision of the Globally Harmonized System of Classification and Labelling of Chemicals (GHS), which was adopted in 2012 and contains in particular refined criteria for skin corrosion/irritation and serious eye damage/eye irritation.

⁴⁸ Information on e.g. *in vitro* testing may provide evidence on more suitable classification, if there is some doubt on the correct classification.

Existing human data		
2	Are there adequate existing human data ^b which provide evidence that the substance has the potential to cause serious eye damage or eye irritation? →	YES: Consider classifying accordingly (Serious Eye Damage Cat. 1 or Eye Irritant Cat. 2).
Existing animal data from serious eye damage/eye irritation studies		
3	Are there data from existing studies on serious eye damage/eye irritation in laboratory animals, which provide sound conclusive evidence that the substance is seriously damaging to the eye, eye irritant or non-irritant? →	YES: Consider classifying accordingly (Serious Eye Damage Cat. 1 or Eye Irritant Cat. 2) or consider no classification.
Existing/new (Q)SAR data and read-across		
4	Are there structurally related substances (suitable "read-across" or grouping), which are classified as causing serious eye damage/eye irritation, or indicating that the substance is non-irritant, or do valid (Q)SAR methods indicate serious eye damage/eye irritation or non-irritation of the substance? ^c →	YES: Consider classifying accordingly.
Existing in vitro data		
5a	Has the substance demonstrated serious eye damage, eye irritation or non-irritating properties in an EU/OECD adopted <i>in vitro</i> test? Data from <i>in vitro</i> test methods that have been validated and are considered scientifically valid but are not yet adopted by EU and/or OECD may also be used if the provisions defined in Annex XI are met. →	YES: Consider classifying accordingly (Serious Eye Damage Cat. 1 or Eye Irritant Cat. 2) or consider no classification. If discrimination between Serious Eye Damage Cat. 1 and Eye Irritant Cat. 2 is not possible, Serious Eye Damage Cat. 1 must be chosen.
5b	Are there acceptable data from (a) non-validated suitable <i>in vitro</i> test(s), which provide sound evidence that the substance causes serious eye damage/eye irritation? ^d →	YES: Consider classifying accordingly (Serious Eye Damage Cat. 1 or Eye Irritant Cat. 2). If discrimination between Serious Eye Damage Cat. 1 and Eye Irritant Cat. 2 is not possible, Serious Eye Damage Cat. 1 must be chosen.
Weight-of-evidence analysis		
6	The "elements" described above may be arranged as appropriate. Taking all available existing and relevant data mentioned above (Elements 0 – 5) into account, is there sufficient information to make a decision on whether classification/labelling is necessary, and – if so – how to classify and label? →	YES: Classify accordingly (Serious Eye Damage Cat. 1 or Eye Irritant Cat. 2) or consider no classification.

New <i>in vitro</i> tests for serious eye damage/eye irritation (Annex VII to the REACH Regulation) ^e		
7a	Does the substance demonstrate serious eye damage, eye irritation or non-irritant properties in (an) EU/OECD adopted <i>in vitro</i> test(s) for the eye hazard characterisation?^e→ Data from <i>in vitro</i> test methods that have been validated and are considered scientifically valid but are not yet adopted by EU and/or OECD may also be used if the provisions of Annex XI are met.	YES: Classify accordingly (Serious Eye Damage Cat. 1 or Eye Irritant Cat. 2) or consider no classification. If discrimination between Serious Eye Damage Cat. 1 and Eye Irritant Cat. 2 is not possible, Serious Eye Damage Cat. 1 must be chosen. If a conclusion on the eye hazard cannot be drawn by using <i>in vitro</i> testing, <i>in vivo</i> testing must be considered (at Annex VIII level only).
7b	Does the substance demonstrate serious eye damage or eye irritant properties in (a) non-validated suitable <i>in vitro</i> test(s) for serious eye damage/eye irritation? ^d →	YES: Classify as Serious Eye Damage Cat. 1 or Eye Irritant Cat. 2. If a conclusion on the eye hazard cannot be drawn by using <i>in vitro</i> testing, <i>in vivo</i> testing must be considered (at Annex VIII level only).
New <i>in vivo</i> test for serious eye damage/eye irritation as a last resort (Annex VIII to the REACH Regulation) ^f		
8	Does the substance demonstrate serious eye damage or eye irritation in an OECD adopted <i>in vivo</i> test? →	YES: Classify accordingly (Serious Eye Damage Cat. 1 or Eye Irritant Cat. 2). NO: No classification needed.

Notes to the information scheme on serious eye damage/eye irritation:

- a) Note that if the buffering capacity suggests the substance may not cause serious eye damage, further data are needed to confirm this, preferably using an appropriate *in vitro* test method.
- b) Data from case reports, occupational experience, poison information centres or from clinical studies.
- c) Conclusion on no classification can be made if the model has been shown to adequately predict the absence of the classified effect and if it fulfils the requirements of Annex XI to the REACH Regulation. Prediction of the absence of the classified effect can be made either by triggering an exclusion rule in the BfR system (to be checked on a case-by-case basis), or based on a negative prediction in a classification QSAR that was trained on both positive and negative substances. The suitability of the model (reliability, relevance) should be very carefully checked to make sure that the prediction is fit for purpose, and the applicability of the model to the substance should also be justified (e.g fulfilment of the conditions of Section 1.3 of Annex XI to the REACH Regulation should be checked). For read-across, generation of new *in vivo* data should be avoided.

d) Data obtained from non-validated suitable *in vitro* tests can only be used according to the criteria set out in section 1.4 of Annex XI to the REACH Regulation, i.e. only positive results can be accepted. However, there are already several EU/OECD adopted test methods which should be primarily used (see [Table R.7.2-4](#)).

e) New *in vitro* testing should be performed following a top-down or bottom-up approach. Please see the following paragraph "How to use the *in vitro* methods serious eye damage/eye irritation within the strategy". It is highly recommended to adhere to the test protocols whose scientific validity has been established by validation and which, ideally, have been officially adopted by the European Commission and/or by the OECD.

f) *In vivo* testing should not be conducted in case the substance falls under the scope of the specific *in vitro* test(s) performed, and there are no substance-specific limitations to using those tests. *In vivo* testing must be considered only in case *in vitro* studies are not applicable, or the results of these studies are not adequate for classification and risk assessment..

How to use the *in vitro* methods for serious eye damage/eye irritation within the testing and assessment strategy

For serious eye damage/eye irritation no single *in vitro* test method is currently able to fully replace the regulatory *in vivo* test, known as the Draize eye test (EU B.5/OECD TG 405) across the full range of ocular responses for different chemical classes. However, the *in vitro* test methods specified in Sections [R.7.2.8.1](#) and [R.7.2.9.1](#) may be used for partial replacement within a tiered testing strategy or as stand-alone test methods depending on the outcome of the study. Moreover, combinations of several alternative test methods may be able to fully replace the Draize eye test. Testing strategies such as the top-down or bottom-up approaches provide a means of incorporating existing information, QSAR predictions, read-across and grouping and *in vitro* test results.

Certain steps need to be taken before any testing (*in vitro* or *in vivo*) is conducted, as described in the introductory paragraph to Annex VII to the REACH Regulation, i.e. assessment of all available information which could be e.g. information from skin corrosion studies ([Figure R.7.2-5](#)).

If a conclusion on classification cannot be reached based on existing information, the next steps are:

- 1) One or more *in vitro* studies for serious eye damage/eye irritation should be performed, and the outcome can be:
 - a. In the case of a positive and definitive result from BCOP, ICE, FL, STE, CM or Ocular Irritation® the substance can be classified as causing "serious eye damage" (Cat. 1 of CLP), and no further *in vivo* test is necessary.
 - b. In addition, BCOP, ICE, STE, RhCE, or CM tests can also provide information on whether the substance does not require any classification for serious eye damage/ eye irritation. If the tests show that no classification is needed, no further *in vivo* testing is necessary.
 - c. For Annex VII information requirement, as no *in vivo* testing is foreseen, a *Weight-of-Evidence* approach may be needed in order to conclude on the eye hazard potential of the substance. The assessment should take all relevant pieces of information into account. This means that in case where the available *in vitro* test(s) for serious eye damage/eye irritation does (do) not enable a definitive conclusion on the classification for the eye hazard to be drawn, information obtained e.g. from skin irritation testing

should be considered. Thus, in case inconsistent *in vitro* results for serious eye damage/eye irritation are obtained, the *Weight-of-Evidence* including information on skin irritation (Category 2) may support the classification for eye irritation (Category 2), as a precautionary principle. See also the [Guidance on the Application of the CLP criteria](#);

d. At Annex VIII level, if neither of these conclusions can be made other *in vitro* study(ies) for this endpoint must be considered. If the *in vitro* studies are not suitable for the substance, or the results are not adequate for classification and risk assessment, a further test conducted *in vivo*, to assess the eye irritation potential must be considered, i.e. none of the *in vitro* methods described above can be used for the direct identification of eye irritants (Cat. 2 of CLP).

New *in vitro* testing should be performed following a top-down or bottom-up approach based on presumed properties (Scott *et al.*, 2010). The top-down approach (start with an *in vitro* test able to identify substances that are seriously damaging to the eye, i.e. classified as "Serious eye damage Cat. 1") should be used when all available collected information and the *Weight-of-Evidence* assessment result in a high *a-priori* probability of the substance being seriously damaging to the eye. The bottom-up approach, on the other hand (start with an *in vitro* test able to identify substances not requiring classification for serious eye damage/eye irritation, i.e. not classified) should be followed when all available collected information and the *Weight-of-Evidence* assessment result in a high *a-priori* probability of the substance being non-irritant to the eyes.

Note: Registrants must make sure that the substance falls within the scope and applicability domain of the specific *in vitro* tests performed, and that there are no substance-specific limitations to using those tests (see *in vitro* tests for serious eye damage/eye irritation and sections [R.7.2.8.1](#) and [R.7.2.9.1](#)). For most substances, the use of EU- or OECD-adopted test methods for the eye hazard characterisation will provide results that will have regulatory acceptance under REACH.

RESPIRATORY TRACT CORROSION/IRRITATION

R.7.2.12 Information sources on respiratory tract corrosion/irritation

The evaluation of respiratory tract corrosion/irritation potential can be based on expert judgement using evidence such as: human and animal experience, existing (*in vitro*) data, substance properties like pH values, volatility (Saturated Vapour Concentration (SVC)) or dustiness, information from similar substances or any other pertinent data.

R.7.2.12.1 Animal data

There are currently no EU or OECD adopted test guidelines that deal specifically with respiratory tract corrosion or irritation. Studies that could inform on the respiratory tract corrosion/irritation potential of the substance concerned are single or repeated inhalation exposure studies (information on (histo-)pathological changes).

Single inhalation exposure studies *in vivo* may provide information on nasal irritation such as rhinitis, whereas histopathological examination of respiratory tract tissues of animals repeatedly exposed by inhalation (28-day and 90-day inhalation studies) may provide information on inflammatory/cytotoxic effects such as hyperemia, edema, inflammation or mucosal thickening. Data from bronchoalveolar lavage may give additional information on the inflammatory response.

It is noteworthy that, while histopathology is not a standard element of the OECD TG 436 for Acute Inhalation Toxicity, TG 436 specifies that "*Additional examinations included a priori by design may be considered to extend the interpretive value of the study, such as... providing evidence of irritation by microscope examination of the respiratory tract. Examined organs may include those showing evidence of gross pathology in animals surviving 24 or more hours, and organs known or expected to be affected. Microscopic examination of the entire respiratory tract may provide useful information for test articles that are reactive with water, such as acids and hygroscopic test articles*".

Moreover, the data on local dermal or ocular corrosion/irritation might contain information that is relevant for the respiratory endpoint and this should be considered accordingly. It is for instance a reasonable precaution to assume that corrosive (and severely irritating) substances would also cause respiratory tract irritation or even corrosion when vaporised or in the form of an aerosol. Furthermore, information from cases where symptoms have been described associated with occupational exposures can be used on a case-by-case basis to characterise the respiratory tract corrosion/irritation potency of a substance. Existing and available information from acute and repeated dose inhalation toxicity studies may also be considered sufficient to show that the substance causes respiratory tract corrosion/irritation at a specific concentration level or range. The data need to be carefully evaluated with regard to the exposure conditions (sufficient documentation required). Possible confounding factors should be taken into account.

R.7.2.12.2 Human data

Existing human data include historical data that should be taken into account when evaluating intrinsic hazards of substances. *New* testing in humans for hazard identification purposes is not acceptable for ethical reasons.

Existing human data can be obtained from case reports, poison information centres, medical clinics, and occupational experience or from epidemiological studies or volunteer studies. Their quality and relevance for hazard assessment should be critically reviewed. However, in

general, human data can be used to determine a corrosive or irritating potential of a substance. Good quality and relevant human data have precedence over other data. However, absence of incidence in humans does not necessarily overrule existing good quality animal data that are positive.

Specifically with regard to respiratory tract irritation, there is a view in the occupational health literature that sensory irritation may be a more sensitive effect than overt tissue-damaging irritation, given that its biological function is to serve as an immediate warning against substances inhaled during a short period of time which could damage the airways, and that it triggers physiological reflexes that limit inhalation volumes and protect the airways. However, there is a lack of documented evidence to indicate that this is a generic position that would necessarily apply to all inhaled irritants.

R.7.2.13 Evaluation of information on respiratory tract corrosion/irritation

All data available should be evaluated to estimate a substance's potential to induce respiratory tract corrosion or irritation.

R.7.2.13.1 Animal data

The evaluation is based on data from inhalation studies (acute, repeated exposure):

- Clinical symptoms of dyspnoea or breathing difficulties,
- Histomorphology of the respiratory tract,
- Lavage examination (nasal, bronchoalveolar).

Useful information may be obtained from the single and repeated inhalation toxicity studies for classification and labelling as well as for DNEL derivation.

For derivation of a DNEL (acute - inhalation, local effects) information from animal studies with acute and/or repeated inhalation exposure may be used. This usually requires that in the study several exposure concentrations had been used that allow derivation of a No Observed Adverse Effect Concentration (NOAEC) and/or a Low Observed Adverse Effect Concentration (LOAEC) or a benchmark concentration (BMC) as starting points for DNEL derivation (Section R.8.2.1 and Appendix R.8-8 of Chapter R.8 of the [Guidance on IR&CSA](#)). In case such information is only available from repeated dose inhalation studies, derivation of a long-term DNEL (long-term - inhalation, local effects) might be more appropriate.

For classification and labelling purposes, the severity of the effects (reversible versus irreversible) and the target within the respiratory tract (upper versus lower respiratory tract) need to be considered.

In case animal studies show reversible effects (usually in the upper respiratory tract), the studies can be used as part of a *Weight-of-Evidence* evaluation for classification for STOT SE Category 3. Reversible respiratory tract effects may be clinical signs of toxicity like dyspnoea or rhinitis and histopathological effects like hyperemia, oedema, minimal inflammation or thickened mucous layer which may be reflective of the characteristic clinical symptoms described above.

In case the studies show significant changes, more than transient in nature, especially in the lower respiratory tract (bronchiolar and alveolar region), classification for STOT SE Category 1 or 2 might be considered, depending on the concentration at which the effects occur.

Significant changes to the respiratory tract may include necrosis, or other morphological changes that are potentially reversible but provide clear evidence of marked organ dysfunction. However, if such effects were only observed in inhalation studies with repeated exposure and the mode of action indicates that the significant damage to the respiratory tract is due to repeated exposure, classification for "Specific Target Organ Toxicity after Repeated Exposure (STOT RE), Category 1 or 2 might be more appropriate (see Section 3.9 of the [Guidance on the Application of the CLP criteria](#)).

For corrosive substances that may be acutely toxic, the additional labelling with EUH071 "*Corrosive to the respiratory tract*" should be considered (see Section 3.1 of the [Guidance on the Application of the CLP criteria](#)). It is presumed that corrosive substances will cause toxicity by inhalation exposure. The Hazard statement EUH071 must be assigned for substances that may be inhaled in addition to classification for acute inhalation toxicity, if data are available that indicate that the mechanism of toxicity is corrosivity. In cases where no acute inhalation test has been performed and the substance may be inhaled, this hazard statement must also be assigned. However, if corrosive substances are used in mixtures in sub-corrosive concentrations, it needs to be ensured that an appropriate classification for potential respiratory tract irritation is applied. For liquids the volatility/SVC, and for solids dustiness, if applicable, should be taken into consideration.

R.7.2.13.2 Human data

The evaluation is based on:

- Experience from occupational exposure;
- Published data on volunteers (objective measurements, psychophysical methods, and subjective reporting);
- Other data (e.g. from nasal lavage).

Consideration should be given to real-life human observational experience, if this is properly collected and documented (Arts *et al.*, 2006), e.g. data from well-designed workplace surveys or worker health monitoring programmes. For substances with an array of industrial uses and with abundant human evidence, the symptoms of respiratory tract irritation can sometimes be associated with certain concentrations of the irritants in the workplace air and might thus allow derivation of DNELs. However, the exposure details need to be well documented and due consideration should be given to possible confounding factors.

Data on sensory irritation of the airways may be available from volunteer studies. These include objective measurements of respiratory tract irritation such as electrophysiological responses, data from lateralization threshold testing, biomarkers of inflammation in nasal or bronchoalveolar lavage fluids. Including anosmics as subjects could exclude odour as a bias. Good quality and relevant human data have precedence over other data. However, absence of positive findings in humans does not necessarily overrule good quality animal data that are positive.

Human data demonstrating respiratory tract irritation are used primarily for classification for Specific Target Organ Toxicity after Single Exposure (STOT SE), Category 3 (H335: "*May cause respiratory irritation*") under CLP (see Section 3.8 of the [Guidance on the Application of the CLP criteria](#)).

Such effects are characterised by localised redness, oedema, pruritis and/or pain and they impair function with symptoms such as cough, pain, choking, and breathing difficulties. Subjective human observations could be supported by objective measurements of clear respiratory tract irritation (such as electrophysiological responses, biomarkers of inflammation in nasal or bronchoalveolar lavage fluids). Furthermore, the symptoms observed in humans

should also be typical of those that would be produced in the exposed population rather than being an isolated idiosyncratic reaction or response triggered only in individuals with hypersensitive airways. Ambiguous reports simply of 'irritation' must be excluded as this term is commonly used to describe a wide range of sensations including those such as smell, unpleasant taste, a tickling sensation, and dryness, which are outside the scope of classification for respiratory tract irritation.

R.7.2.14 Conclusions on respiratory tract corrosion/irritation

R.7.2.14.1 Concluding on suitability for Classification and Labelling

In order to conclude on Classification and Labelling according to the CLP Regulation, all the available information needs to be taken into account, and consideration should be given to both the [Guidance on the Application of the CLP criteria](#) and the various remarks (related to Classification and Labelling) made throughout this guidance document.

R.7.2.14.2 Concluding on suitability for Chemical Safety Assessment

A dose-response assessment might be possible. Animal studies, especially those with repeated inhalation exposure and several exposure concentrations, may be available that allow derivation of a NOAEC and/or a LOAEC as starting points for DNEL derivation.

Human data indicative of respiratory tract irritation that provide reliable quantitative information on the threshold for the irritative effects may also be used to derive DNEL (acute - inhalation, local effects) (see Section R.8.2.1 and Appendix R.8-8 of Chapter R.8 of the [Guidance on IR&CSA](#)).

R.7.2.15 References on skin corrosion/irritation, serious eye damage/eye irritation and respiratory tract corrosion/irritation

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Appendices R.7.2-1 to 3 to Section R.7.2

Appendix R.7.2–1 Mechanisms of local toxicities: skin corrosion/irritation, serious eye damage/eye irritation and respiratory tract corrosion/irritation

Content of Appendix R.7.2-1:

- Mechanisms of skin corrosion and irritation
- Mechanisms of serious eye damage/eye irritation
- Mechanisms of respiratory tract corrosion and irritation

MECHANISMS OF SKIN CORROSION AND IRRITATION

Clinically, different types of irritant contact dermatitis (ICD) exist, and have been classified on the basis of differences in morphology and mode of onset, as: acute irritant dermatitis (primary irritation); irritant reaction; delayed, acute irritant contact dermatitis; cumulative irritant dermatitis; traumatic irritant dermatitis, pustular and acneiform irritant dermatitis; non-erythematous irritant dermatitis; and subjective irritation (Lammintausta and Maibach, 1990).

Two different pathogenetic pathways may be involved in ICD. Acute ICD is characterised by an inflammatory reaction which mimics allergic contact dermatitis, with the release of inflammatory mediators and cytokines. Chronic ICD, on the other hand, is characterised by disturbed barrier function, associated with an increased epidermal turnover which leads clinically to lichenification (Berardesca and Distant, 1994).

The clinically relevant elements of skin irritation are a disturbance of the desquamation process, resulting in scaling or hyperkeratosis (chronic effects), i.e. epidermal events, and an inflammatory response with vasodilation and redness in combination with extravasation of water, which may be observed as papules, vesicles and/or bullae and oedema (acute effects), i.e. events essentially taking place in the dermis (Serup, 1995). The onset of irritation takes place at the stratum corneum level and later in the dermis, whereas early events in sensitisation occur in the dermis. Variations in the skin reactions are dependent on the degree of injury induced, as well as on the effects of an irritant substance on different cell populations. For example, pigmentary alterations are due to effects on melanocytes, whereas ulcerations are due to extensive keratinocyte necrosis (skin corrosion). The release of cytokines and mediators can be initiated by a number of cells, including living keratinocytes and those of the stratum corneum, which thus modulate inflammation and repair (Sondergard *et al.*, 1974; Hawk *et al.*, 1983; Barker *et al.*, 1991; Baadsgaard and Wang, 1991; Hunziker *et al.*, 1992; Berardesca and Distant, 1994).

The physico-chemical properties, concentration, volume and contact time of the irritant give rise to variations in the skin response. Furthermore, inter-individual differences exist, based on age, gender, race, skin colour and history of any previous skin disease. In the same individual, reactivity differs according to differences in skin thickness and skin sensitivity to irritation of the different body regions. Finally, a greater sensitivity to some irritants (DMSO, propylene glycol, SLS and soap) has been reported during winter, because of the reduced hydration state of the skin (Frosch and Pilz, 1995). Although clinically different types of irritant reactions can be observed, they are all based on cellular and biochemical mechanisms which induce the irritant response. It is not yet possible to conclude whether the observed clinical differences are actually due to differences in biochemical mechanisms, and further investigations are needed.

According to Barratt (1995a) and further elaborated by Walker *et al.* (2004), for organic substances, the mechanisms leading to skin irritation are normally described by a two-stage process where a substance first has to penetrate the *stratum corneum* and then trigger a biological response in deeper epidermal or dermal layers.

For strong inorganic acids and bases, no *stratum corneum* penetration is needed because they erode the *stratum corneum*. According to the Technical Guidance Document (TGD) supporting Commission Directive 93/67/EEC on risk assessment for new notified and existing substances (EC, 2003), the percutaneous absorption of acrylates, quaternary ammonium ions, heterocyclic ammonium ions and sulphonium salts is slow, since these substances are binding to macromolecules in the skin. As a result of binding, corrosion can occur as the *stratum corneum* is eroded. Reactivity can be caused by electrophiles and/or pro-electrophiles. Electrophiles contain atoms, such as N, O or halogens attached to a C-atom, which makes that specific C-atom positively charged and therefore reactive with electron-rich regions of peptides and proteins. This causes irritation *via* covalent binding to the skin.

At this time, the following mechanisms are proposed for inducing skin irritation or skin corrosion by affecting the structure and function of the *stratum corneum* :

1. Mechanisms of skin irritation:

- Reaction with skin proteins and interference with lipids in the *stratum corneum* by surface-active agents (denaturation of proteins, disruption of plasma membrane lipids),
- Dissolving of plasma membrane lipids and thus defatting and disintegration of skin by low molecular weight organic substances.

2. Mechanisms of skin corrosion:

- Erosion of the *stratum corneum* by most inorganic acids and bases and by strong organic acids with $\text{pH} \leq 2.0$ and bases with $\text{pH} \geq 11.5$, and
- Binding to skin components in the *stratum corneum* by cationic surfactants and percutaneous absorption of acrylates, quaternary ammonium ions, heterocyclic ammonium ions and sulphonium salts.

3. Mechanisms that may lead to both skin irritation and corrosion:

- Penetration of the *stratum corneum* by anionic or non-surfactant organic substances with sufficient hydrophobic and hydrophilic properties, and
- Elicitation of an inflammatory and/or cytotoxic response in the epidermis or dermis.

The severity of these responses may determine whether irritation or corrosion occurs.

MECHANISMS OF SERIOUS EYE DAMAGE AND EYE IRRITATION

Eye injury can be caused by many insults. These can be physical such as puncture by sharp objects. Eye injury can be caused by substances such as systemic drugs that can enter into the eye through the blood stream (e.g. Cyclosporine, vaccines, intravenous immunoglobulines, intravenous streptokinase). Various degrees of eye injury can also be caused by direct (topical) contact with substances or mixtures such as acids, alkalis, solvents or surfactants. These materials may come into contact with the eye intentionally, e.g. through the use of eye drops,

medications, products intended for use around the eyes, but also unintentionally, e.g. accidental spills and splashes of consumer products or accidental exposures in the workplace.

In general, substances or mixtures which come directly into contact with the eye may cause local effects on the frontal tissues and substructures of the eye, e.g. cornea, conjunctiva, iris, lachrymal system and eye lids. There are several modes of action by which topical substances and mixtures cause eye injury e.g. cell membrane lysis, saponification and coagulation (see [Table R.7.2-5](#)).

Table R.7.2-5 Categories of irritant substances and their typical mode of action in eye irritation.

Substance/mixtures	Mode of Action
Inert substances	May cause effect due to large size. Protrusions may cause direct puncture of the eye.
Acids	May react directly with cellular components e.g. eye proteins and cause coagulation, lysis or precipitation resulting in relatively localised injury.
Bases (Alkalis)	May actively disrupt the cell membrane lipids by alkaline action i.e. saponification. May penetrate to the deeper layers of the eye tissue. May react directly with cellular components and cause coagulation or lysis of the tissue.
Solvents	May cause membrane lysis by dissolving lipids in plasma membranes of epithelial and underlying cells resulting in loss of the cells affected and, as a result, tissue degradation, which might be transient, depending on the repair mechanisms (cell proliferation, tissue restoration). May also cause coagulation.
Lachrymators	May stimulate the sensory nerve endings in the corneal epithelium causing an increase in tearing.

The degree of eye injury is usually dependent on the characteristics (chemical category/class) and concentration of the substance or mixture. Acids and alkalis usually cause immediate irritation to the eyes. Other substances may cause eye injuries that start as mild but progress to be more severe at a later period e.g. substances that react with cellular constituents *via* alkylation or oxidative attack on macromolecules. An example of these types of substances are e.g. peroxides, mustards and bleaches (Scott *et al.*, 2010).

Upon exposure of the ocular surface to eye irritants, inflammation of the conjunctiva can be induced. This includes dilation of the blood vessels causing redness, increased effusion of water causing swelling (oedema/chemosis) and an increase in the secretion of mucus leading to an increase in discharge. Visual acuity can be impaired. Effects on the cornea may be more severe (e.g. destruction of the cornea, or persistent corneal opacity or discoloration of the cornea by a dye substance), or reversible where effects are limited to the epithelia. Irritants may also produce an increase in tear production and changes to the tear film integrity such as increased wetness. Iritis may result from direct irritation or become a secondary reaction to the corneal injury. Once the iris is inflamed, infiltration of fluids can follow which affects the ability to adjust the size of the pupil and decreases the reaction to light leading to decreased visual acuity. Due to the richness of nerves in the iris, irritation also causes subjective symptoms such as itching, burning and stinging.

Eye injury can be reversible or irreversible depending on the degree of damage and degree of repair. Damage to the corneal epithelium alone can repair quickly, often with no permanent eye damage. The cornea may still repair fairly well if the damage goes beyond the basement membrane into the superficial part of the stroma but the repair process may take days or even weeks to occur. Once the damage extends significantly into the stroma, corneal ulceration can occur due to the subsequent series of inflammatory processes. If damage extends to and beyond the endothelium, corneal perforation may occur which is irreversible and may cause permanent loss of vision. Eye injury can cause different degrees of functional loss e.g. increase of tear production, opacification of the cornea, oedema and so decrease visual acuity.

The body has its own defence mechanisms, e.g. sensing the pain, stinging and burning, and the eyelids will blink to avoid full exposure to the substance. Increased tear production and blinking of the eyes with the help of the drainage apparatus help to dilute or clear the causative agent. Such defence mechanisms are highly developed in man with rapid adverse blinking and profuse tear production resulting from exposure of the eye to a foreign material that is irritating. It is well reported in the literature that species differences occur in the rate of blinking and tear production mechanism that can influence how effectively foreign materials are removed from the eye.

MECHANISMS OF RESPIRATORY TRACT CORROSION AND IRRITATION

Corrosion of the respiratory tract includes destruction of the mucosa followed by proliferation of epithelial cells. Remodeling of tissue may occur with chronic injury if repair mechanisms are unable to keep pace. Mild epithelial or endothelial injury without basement membrane damage, severe inflammation, or persistence of the inciting agent may be resolved by simple cellular regeneration. With more severe damage, a significant inflammation component may be elicited which may be followed by tissue destruction or fibrosis. In some cases, persistence of the inciting agent within the tissue may lead to the development of a granulomatous disease, as observed with inhalation exposure to crystalline silica or carbon nanotubes (Harkema *et al.*, 2013).

Corrosive effects in the respiratory tract may be non-specific, e.g. induced by highly acidic or basic substances like sulphuric acid. However, acute necrosis and loss of olfactory epithelium may also be observed following inhalation or bloodborne exposure to toxicants that require metabolic activation by the P450 system, such as 3-methylfuran. Once the basement membrane is exposed, cytokines are released and inflammation takes place (Harkema *et al.*, 2013).

The term "respiratory tract irritation" is often used to indicate either or both of two different toxicological effects. These are i) cytotoxic effects in the affected tissue, and ii) sensory irritation.

Cytotoxic effects in the respiratory tract are comparable to dermal and eye irritation. These effects are characterised by inflammation (increased blood flow (hyperemia), local infiltration with white blood cells, swelling, oedema) and there may also be haemorrhage, and eventual necrosis and other pathological changes. The effects are in principle reversible. A recent publication has proposed the term "tissue irritation" for this kind of effects (Brüning *et al.*, 2014).

Chronic irritation can lead to repeated episodes of cell proliferation in the affected tissues, and this may increase the risk of tumour development. The nature of effects depends on the substance and its primarily targeted region; the severity of effects depends on the concentration and duration of exposure. In general, repeated exposure studies in animals focus on observing (histo)pathological evidence for tissue damage. In case overt tissue damage (mucosal erosion and ulceration) occurs, a non-specific cytotoxic action at the site of

contact along the respiration route can be assumed. Depending on the concentration and duration of exposure a severity gradient of lesions from anterior to posterior regions can be observed (in contrast to effects in certain mucosa types depending on the metabolic activation of the test substance) and, depending on the severity and the extent of the lesions, adjacent submucosal tissues can also be affected (e.g. by cartilage destruction). Such lesions are not fully reversible due to scar formation or replacement of the original mucosa, or may induce other serious health effects such as marked bleeding or persistent airway obstruction.

"Sensory irritation" refers to the local and central reflex interaction of a substance with the autonomic nerve receptors, which are widely distributed in the mucosal tissues of the eyes and upper respiratory tract. Three substance or substance-group specific target sites of sensory irritation generating different responses can be identified: a) nasal (and eye) irritation, i.e. interaction with the trigeminal nerve, b) pharyngeal irritation, i.e. interaction with the glossopharyngeal nerve, and c) larynx and lower respiratory tract, i.e. interaction with the vagus nerve.

Sensory irritation leads to unpleasant sensations such as pain, burning, pungency, and tingling. The severity depends on the airborne concentration of the irritant rather than on the duration of exposure. Sensory irritation is a receptor-mediated effect, and usually occurs almost immediately upon exposure to the inhaled irritant. It leads to reflex involuntary responses such as sneezing, lacrimation, rhinorrhea, coughing, vasodilatation of blood vessels in the nasal passages, and changes in the rate and depth of respiration. In humans, protective behavioural responses such as covering the nose and mouth can also occur. Sensory irritation is distinct from odour sensation, which is mediated *via* different nerve pathways (olfactory). However, there is evidence that odour perception and other cognitive influences can affect the perception of sensory irritation in humans.

In rodents, sensory irritation leads to a reflex reduction in the respiratory rate (breath-holding). This reflex effect on respiration can be measured experimentally (determination of the RD₅₀ value in the Alarie assay (Alarie, 1973)) although results may vary considerably depending on the species and strain of rodents, on the exposure duration (time should be long enough to induce changes), and results also show inter-laboratory variability. Investigations of the correlation between the results of the Alarie test and human data are difficult since the parameters examined in humans and mice are different and adequate human data to determine a human equivalent to the RD₅₀ is not available at the moment. The results of a study by Cometto-Muniz *et al.* (1994) indicate that RD₅₀ values in animals are not easily comparable with "nasal pungency thresholds" in humans.

As indicated, human data are mostly based on subjective experiences and need to be carefully controlled in order to prevent confounding by odour perception (Dalton, 2003; Doty *et al.*, 2004). Validated questionnaires have been developed for the investigation of sensory irritation responses in human volunteers. Emphasis was given to developing a spectrum of objective measurements (see review by Arts *et al.*, 2006). Compiling toxicological profiles for substances in the workplace demonstrates that sensory irritation often appears to be a very sensitive and relevant endpoint in human risk assessment. Accordingly, 40 % of the occupational exposure limit values (OELs) are based on the avoidance of sensory irritation (Dick and Ahlers 1998; Edling and Lundberg 2000; van Thriel *et al.*, 2006). This endpoint is related to the interaction of volatile substances with neuronal sensors located in mucous membranes of the respiratory tract and the eyes. In many cases, data from controlled human studies are either not available or inadequate, so OELs are predominantly derived from animal data investigating local effects in the respiratory tract. These effects are usually measured as tissue irritation. Comparison of human data on sensory irritation with data from subacute and subchronic inhalation studies in animals led to the proposal of a default assessment factor of 3 for extrapolating animal data concerning local irritating effects to humans (Brüning *et al.*, 2014).

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Appendix R.7.2-2 (Q)SARs and expert systems for skin corrosion and irritation

Content of Appendix R.7.2-2:

- Literature-based QSAR models
- Commercial models
- BfR rule-base
- OECD QSAR Toolbox

In principle, Annex XI to the REACH Regulation allows for an adaptation of the standard information requirements by using (Q)SARs, including the prediction of non-irritancy. However, for the endpoint skin corrosion/irritation, only very few of the currently available models are suitable for this purpose if used as stand-alone methods. Nevertheless, such models can still have merit when used as supporting information or in *Weight-of-Evidence* approaches and for positive prediction of skin corrosion/irritation.

LITERATURE-BASED QSAR MODELS

In the open scientific literature, (Q)SARs have been based on continuous (e.g. Primary Irritation Indices) or categorical (e.g. EU classifications) measures of skin irritation.

For defined classes of substances, categorical QSARs have been reported for discriminating between corrosives and non-corrosives (Barratt, 1996a, 1996b), and between skin irritants and non-irritants (Smith *et al.*, 2000a, 2000b). These studies did not actually provide a transparent algorithm for classifying chemicals, so they are of limited value for regulatory use. However, they illustrate the feasibility of developing such models.

A linear discriminant model for distinguishing between irritant and non-irritant liquid esters in human volunteers was reported by Smith *et al.* (2000a). As mentioned above the exact algorithm is not clear. In addition the primary irritation index for human irritation may need translation when these scores are considered for classification. However, the results could be informative for future model development for esters, since they indicate that irritant esters can be distinguished from non-irritants on the basis of a limited number of physico-chemical parameters.

For defined classes of substances, continuous QSARs for predicting the Primary Irritation Index (PII) have also been published (Barratt, 1996b; Hayashi *et al.*, 1999; Kodithala *et al.*, 2002). For example, the application of stepwise regression analysis to a set of 52 neutral and electrophilic organic substances produced the following model:

$$\text{PII} = 1.047 \log P - 0.244 \text{ MV} + 0.888 \text{ DM} + 0.353$$

$$N=52, r^2 = 0.422, r_{cv}^2 = 0.201, s=1.376, F=11.70$$

This equation indicates that the PII has a positive dependence on log P (logarithm of the octanol-water partition coefficient) and DM (dipole moment), and a negative dependence on MV (molecular volume). This model has a low goodness-of-fit (r^2) and a poor predictivity (as reflected by r_{cv}^2), so is not recommended for regulatory use. More research is needed into the development of models for predicting PII and it should be considered whether the information generated could be used in the setting of DNELs.

Some limited evidence indicates that the reactive effects of acids and bases can be predicted by using the acid/base dissociation constant (pKa), which can itself be predicted by using commercially available software products, such as the SPARC program. Evidence for the usefulness of pKa as a predictor of skin irritation for acids has been provided by Berner *et al.* (1988, 1989, 1990), whereas evidence for the usefulness of pKa as a predictor of skin irritation for bases has been provided by Nangia *et al.* (1996). Barratt also used pKa for predicting the effects of acids and bases (Barratt, 1995a). These studies did not address the question of how to use pKa where there are multiple functional groups in the substance of interest, and therefore multiple ionisation constants. Based on current knowledge, no clear recommendations can be made on how to use pKa information.

An overview on the available literature-based models for skin corrosion/irritation is given in [Table R.7.2–6](#).

Table R.7.2–6 Available literature-based models for skin corrosion/irritation.

Reference	Content
QSAR models	
Barratt (1996a) Quantitative structure-activity relationships (QSARs) for skin corrosivity of organic acids, bases and phenols: Principal components and neural network analysis of extended datasets.	This paper describes QSAR models relating skin corrosivity data of organic acids, bases and phenols to their log(octanol/water partition coefficient), molecular volume, melting point and pK(a).
Barratt (1996b) Quantitative structure-activity relationships for skin irritation and corrosivity of neutral and electrophilic organic chemicals.	This paper describes QSAR models derived by relating skin irritation and corrosivity data of neutral and electrophilic organic chemicals to their log(octanol/water partition coefficient) (logP), molecular volume, dipole moment and 1/molecular weight.
Barratt (1996c) The use of in vitro cytotoxicity measurements in (Q)SAR methods for the prediction of the skin corrosivity potential of acids.	This paper describes quantitative structure-activity relationships (QSAR) methods that relate the severity of skin corrosivity (designated by the EC risk phrases R34 and R35) of acids to parameters that model their skin permeability and cytotoxicity. Skin permeability was modelled by log(octanol/water partition coefficient), molecular volume and melting point, while the cytotoxicity of the acids was accounted for by their pKa values and the in vitro cytotoxicity of their sodium salts towards Swiss mouse embryo 3T3 cells.
Gerner <i>et al.</i> (2004) Quantitative structure-property relationships modeling of skin irritation.	This paper describes limit values for specific physico-chemical properties that are appropriate for identifying chemical substances that have no skin irritation or corrosion potential. These physicochemical properties include melting point, molecular weight, octanol/water partition coefficient, surface tension, vapour pressure, aqueous solubility and lipid solubility.
Golla <i>et al.</i> (2009) Quantitative structure-property relationships modeling of skin irritation.	This paper describes a skin irritation QSPR model based on rabbit Draize test data for 186 compounds, which included chemicals from diverse molecular classes. The effectiveness of using a combination of traditional, functional group and structural descriptors has been studied. The effects of molecular size, reactivity and skin penetration on skin irritation have been also analysed.

Hayashi et al. (1999) A quantitative structure-activity relationship study of the skin irritation potential of phenols.	This paper describes QSARs for skin irritation potential derived using twenty-four phenols, using the following descriptors: the absolute hardness calculated from HOMO and LUMO energy levels for reactivity, and log P for permeability. The selection of the descriptors was based on the hypothesis that skin irritation is induced by reaction of phenols with macromolecules present in epidermal and dermal levels of the skin.
Hulzebos et al. (2005) Use of structural alerts to develop rules for identifying chemical substances with skin irritation or skin corrosion potential.	This paper describes the identification and categorisation of structural alerts for acute skin lesions as irritation or corrosion or a combination of corrosion/irritation alerts.
Kodithala et al. (2002) Prediction of skin irritation from organic chemicals using membrane-interaction QSAR analysis.	This paper describes membrane-interaction QSAR analysis carried out for a training set of 22 hydroxy organic compounds for which the Draize skin irritation scores, PII, had been determined. Skin irritation potency is predicted to increase with (1) increasing effective concentration of the compound available for uptake into phospholipid-rich regions of a cellular membrane, (2) increasing binding of the compound to the phospholipid-rich regions of a cellular membrane, and (3) the chemical reactivity of the compound as reflected by the highest occupied molecular orbital (HOMO) and/or lowest unoccupied molecular orbital (LUMO) of the molecule.
Walker et al. (2004) (Q)SARs for Predicting Skin Irritation and Corrosion: Mechanisms, Transparency and Applicability of Predictions.	This paper describes previously-developed (Q)SARs for predicting skin irritation and corrosion, proposes mechanisms of skin irritation and corrosion, and discusses the transparency and applicability of predictions.
Walker et al. (2005) The Skin Irritation Corrosion Rules Estimation Tool (SICRET).	This paper describes the Skin Irritation Corrosion Rules Estimation Tool (SICRET) that was developed to allow others to estimate whether their chemicals are likely to cause skin irritation or skin corrosion. SICRET uses physicochemical property limits to identify chemicals with no skin corrosion or skin irritation potential.
Whittle (1996) Skin corrosivity potential of fatty acids: <i>In vitro</i> rat and human testing and (Q)SAR studies.	This paper investigates the corrosive potential of a series of fatty acids - propanoic acid (C3), butanoic acid (C4), hexanoic acid (C6), octanoic acid (C8), decanoic acid (C10) and dodecanoic acid (C12) - according to an <i>in vitro</i> skin corrosivity test (IVSCT) using both rat skin and human skin. The results are discussed in the context of a QSAR for the corrosivity of organic acids, with the putative mechanism that corrosivity is a function of the ability of the chemical to permeate the skin together with its cytotoxicity, expressed in this case as acidity (pK(a)).
Worth and Cronin (2001) The use of pH measurements to predict the potential of chemicals to cause acute dermal and ocular toxicity.	This paper presents a the development of classification models based on pH data for predicting the potential of chemicals to cause skin corrosion, skin irritation and eye irritation. The possible application of these models in the context of tiered testing strategies is discussed.

Reviews and evaluation of existing models	
Gallegos Saliner et al. (2006) Review of Literature-Based Models for Skin and Eye Irritation and Corrosion.	This report reviews the state-of-the-art of <i>in silico</i> and <i>in vitro</i> methods for assessing dermal and ocular irritation and corrosion. In this review, emphasis is placed on literature-based QSAR models for skin and eye irritation and corrosion as well as computer-based expert systems.
Gallegos Saliner et al. (2008) Review of (Q)SAR Models for Skin and Eye Irritation and Corrosion.	This paper reviews the state-of-the-art of <i>in silico</i> methods for assessing dermal and ocular irritation and corrosion. It is based on an in-depth review performed by the European Chemicals Bureau of the European Commission: Joint Research Centre. The most widely used <i>in silico</i> approaches are classified into methods to assess (1) skin irritation, (2) skin corrosion and (3) eye irritation. In this review, emphasis is placed on literature-based (Q)SAR models.
Gallegos Saliner et al. (2007) Evaluation of SARs for the prediction of skin irritation/corrosion potential: structural inclusion rules in the BfR decision support system.	This work evaluates the structural inclusion rules implemented in the Decision Support System for skin irritation and corrosion developed at the German Bundesinstitut für Risikobewertung (BfR) for predicting the absence of skin irritation and/or corrosion. The following assessments were performed: (a) a confirmation of the structural rules by rederiving them from the original training set (1358 substances), and (b) an external validation by using a test set of 200 chemicals not used in the derivation of the rules.
Mombelli (2008) An evaluation of the predictive ability of the QSAR software packages, DEREK, HAZARDEXPERT and TOPKAT, to describe chemically-induced skin irritation.	This paper reports the performance of the skin irritation module of three commercially-available software packages: DEREK, HAZARDEXPERT and TOPKAT. Their performances were tested on the basis of data published in the literature for 116 chemicals.
Rorije and Hulzebos (2005) Evaluation of (Q)SARs for the prediction of Skin Irritation/Corrosion Potential. Physicochemical exclusion rules.	This work evaluates the physical-chemical rule-base incorporated in the Decision Support System for skin irritation and corrosion developed at the German Bundesinstitut für Risikobewertung (BfR) for predicting the absence of skin irritation and/or corrosion. This evaluation includes 1) the compliance of the rule-base with the OECD principles on (Q)SARs, 2) the derivation of the (Q)SAR rules, 3) the external validation of these rules, including an assessment of the suitability of the dataset used for validation.

Further details on these models can be found in Chapter 3 of the JRC report "Alternative methods for regulatory toxicology - a state-of-the-art review" (Worth *et al.*, 2014).

COMMERCIAL MODELS

There is a number of software tools available that provide access to QSARs for skin corrosion/irritation.

TOPKAT, which is commercialised by Accelrys (<http://accelrys.com/solutions/scientific-need/predictive-toxicology.html>), incorporates models to discriminate severe irritants from non-severe irritants, as well as mild/moderate irritants from non-irritants. These models are based on work by Enslein *et al.* (1987). The algorithm of TOPKAT is not very transparent. The model predicts a probability of a weak/mild/moderate and severe irritation. It states that

probabilities <0.3 and >0.7 give sufficient certainty of the prediction. The model gives the sensitivity and specificity values of the specific classes such as acyclic etc., which are mostly around or above 90%. It also shows similar structures from the TOPKAT perspective including the experimental result. The TOPKAT predictions of weak/mild/moderate and severe irritation need to be translated to consider them for classification. The models indicate whether the prediction is in the applicability domain of the model.

There is a rule-base for irritation in **Derek Nexus** (Sanderson and Earnshaw, 1991; Combes and Rodford, 2004), which is developed and regularly updated by LHASA Ltd (<http://www.lhasalimited.org/products/derek-nexus.htm>). To predict toxicity, the program checks whether any alerts within the query structure match previously characterised toxicophores (substructure with potential toxic effect) in the knowledge base. The reasoning engine then assesses the likelihood of a structure being toxic, and a message indicating the nature of the toxicological hazard is provided together with relevant literature references. There are nine levels of confidence: certain, probable, plausible, equivocal, doubted, improbable, impossible, open, contradicted. The Derek Nexus rule-base has 25 structural alerts for the prediction of skin corrosion/irritation. There are some combined alerts for respiratory tract irritation and irritation of the gastrointestinal tract but these are not specific to skin corrosion or irritation. If Derek Nexus does not make a prediction of corrosion or irritation, it cannot be concluded that there is no effect – it could mean that none of the known alerts was found to be present in the substance of interest or it was outside the applicability domain of that specific alert. The Derek Nexus model is transparent in its algorithm, when the model is fired showing the structural alert and its limitations. The alert is supported with literature references and sometimes with example substances. The example substances are supposed to support the mechanistic reasoning. The Derek Nexus model can be used for positive identification of skin irritation. The confidence levels have to be taken into account for the purpose of classification. The Derek Nexus model cannot be used to predict non-corrosion/irritation as the model only contains alerts that detect the presence of corrosion/irritation.

HazardExpert is a rule-based software tool developed and commercialised by CompuDrug Chemistry Ltd. (<http://www.compudrug.com/hazardexpertpro>) for predicting the toxicity of organic substances in humans and in animals (Smithing and Darvas, 1992). HazardExpert uses a fragment-based approach to predict toxicokinetic effects and various human health effects, including membrane irritation. Since this endpoint is not clearly defined in HazardExpert, it is recommended not to use it directly for the assessment of skin or eye irritation. However, it could be used as supplementary information in a *Weight-of-Evidence* approach for positive prediction.

The Multiple Computer Automated Structure Evaluation (**MultiCASE**) program, developed by MultiCASE Inc. (http://www.multicase.com/case-ultra-models#skin_eye_tox_bundle), is an automated rule induction tool that automatically identifies molecular fragments likely to be relevant to the activity of molecules (Klopman, 1992; Klopman *et al.*, 1993). It also provides an indication of the importance of these fragments in relation to the potency of the molecules containing them. MultiCASE can be used to predict various human health endpoints, including eye irritation (Klopman *et al.*, 1993; Rosenkranz *et al.*, 1998). However, it is not clear how to relate the MultiCASE scoring system to Draize scores or regulatory classifications. In principle, the MultiCASE model can be used for positive and negative indications of skin irritation. The structural alert is provided as well as information on its internal validation. The MultiCASE model also indicates whether it is in the applicability domain of the model. The MultiCASE predictions of weak/mild/moderate and severe irritation need to be translated to consider them for classification.

ACD/Labs Percepta Predictors (<http://www.acdlabs.com/products/percepta/predictors.php>), developed by ACD/Labs, includes a module for skin and eye irritation. It estimates the potential of a compound to cause eye or skin irritation in a standard rabbit Draize test. The

predictions are reported as qualitative irritation categories (not irritating, slightly irritating, irritating, highly irritating, and corrosive). Probabilistic models are supplemented by an expert system that identifies Structural Alerts relevant to the irritation properties of compounds. Overall, 21 structural alerts were formulated for rabbit eye irritation, and 17 alerts for the rabbit skin irritation case. The categorisation of effect needs to be compared to the CLP cut-offs if application for REACH purposes is intended.

PaDEL-DDPredictor includes several models for skin and eye irritation and corrosion (<http://padel.nus.edu.sg/software/padelddpredictor/>). The models have been built on a training set of 1707 compounds using one and two dimensional descriptors. The final predictions rely on consensus models based on majority voting from base models predictions. The applicability domain is defined by the range of descriptors for compounds in the training set.

QSAR PREDICTION REPOSITORY

The Danish EPA (<http://qsar.food.dtu.dk/>) has developed an in-house MultiCASE model for predicting severe versus mild skin irritation based on 800 test results taken from RTECS (Registry of Toxic Effects of Chemical Substances), the HSDB (Hazardous Substances Data Bank) and the former official list of EU-classified substances (Annex I of Directive 67/548/EEC, now replaced by Annex VI to the CLP Regulation). It is not clear how the RTECS and HSDB classification criteria for irritation correlate with the EU criteria. Due to limitations in the information for assessing the reliability of the prediction, these predictions are difficult to use in the regulatory context.

BFR DECISION SUPPORT SYSTEM

A decision support system (DSS) developed by the German Federal Institute for Risk Assessment (BfR) uses physico-chemical exclusion rules to predict the absence of skin corrosion/irritation potential in combination with structural inclusion rules (SARs) to predict the presence of such potential (Gerner *et al.*, 2004; Hulzebos *et al.*, 2005; Walker *et al.*, 2004). The exclusion rules are based on physico-chemical properties such as molecular weight, aqueous solubility, and log K_{ow} , whereas the inclusion rules are based on sub-structural molecular features. The physico-chemical rules are assumed to implicitly take into account bioavailability (skin penetration) whereas the structural rules take reactivity into account. The physico-chemical and structural rule-bases are designed to predict the former EU risk phrases for skin irritation (R38) and skin corrosion (R34 and R35).

The exclusion rules have the following general form:

IF (physico-chemical property) A THEN predict the absence of toxic effect B

Example: IF Log K_{ow} < -3.1 THEN the substance does not need to be considered for classification

Some of the exclusion rules can be applied to all structures within the domain, whereas others only refer to a subset containing certain elements.

The structural inclusion rules take the following general form:

IF (substructure) A THEN predict the occurrence of toxic effect B.

Example: IF *Chlorosilane alert is present* THEN the substance needs to be considered for "corrosive" classification.

The performance of the BfR physico-chemical rule-base for predicting the absence of skin effects has been assessed by the RIVM (Rorije and Hulzebos, 2005), whereas the structural rule-base for predicting the occurrence of skin effects has been assessed by the ECB (Gallegos Saliner *et al.*, 2007). The endpoint is the former EU (DSD) classification and the algorithms and domain of applicability are transparent. However, the exact chemical structures of the training set are not disclosed to users of the model, due to the data originating from the confidential notification procedure at the time of the development of the system. Though the rules are empirically derived, a mechanism of action can be deduced. Thus, in principle, the resulting predictions can be used as the basis for classification by comparison with CLP criteria. It should be determined, on a case-by-case basis, whether the predictions for a given substance provide a sufficient basis for classification, or whether additional information is needed in a *Weight-of-Evidence* approach.

OECD QSAR TOOLBOX

The freely downloadable OECD QSAR Toolbox software (<http://www.qsartoolbox.org/>) covers the skin corrosion/irritation endpoint with one experimental database and two profilers.

In more detail, the database of experimental data (called "Skin irritation" in the software) refers to the endpoint primary irritation index and collects the data available in:

1. The RIVM Skin Irritation database, which contains Primary Skin Irritation Indices from skin irritation tests from the following sources: ECVAM Workshop 6 on Corrosivity (Barratt (1995b); Botham *et al.* (1995)), and ECETOC Technical Report No.66 on Skin Irritation and Corrosion Reference Chemicals Data Bank (ECETOC, 1995).
2. Experimental results for Primary Skin Irritation Indices from LjMU. Additional experimental results gathered from OECD SIDS Dossiers published between 1992 and 2009 were added in 2010.

The OECD QSAR Toolbox allows for the identification of analogues based on mechanistic and endpoint specific profilers, and for the prediction of skin irritation/corrosion through the use of profilers (BfR rule-base), readacross, trend analysis and QSAR models. Information about inclusion and exclusion rules, details on the performance of the exclusion rules, and applicable chemical class-specific rules for the results of the Skin irritation/corrosion profiler can be found by searching the context menu in the the OECD QSAR Toolbox software.

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Appendix R.7.2–3 (Q)SARs and expert systems for serious eye damage and eye irritation

Content of Appendix R.7.2-3:

- Literature-based QSAR models
- Commercial models
- BfR decision support system
- OECD QSAR Toolbox

In principle, Annex XI to the REACH Regulation allows for an adaptation of the standard information requirements by using (Q)SARs, including the prediction of non-irritancy. However, for the endpoint serious eye damage/eye irritation, only very few of the currently available models are suitable for this purpose if used as stand-alone methods. Nevertheless, such models can still have merit when used as supporting information or in *Weight-of-Evidence* approaches and for positive prediction of serious eye damage/eye irritation.

LITERATURE-BASED QSAR MODELS

In the open scientific literature, (Q)SARs have been based on continuous (e.g. molar eye scores) or categorical (e.g. EU classifications) measures of eye irritation. Examples of mathematical (continuous) models have been published by Sugai *et al.* (1991) and Cronin *et al.* (1994), whereas examples of categorical models have been published by Sugai *et al.* (1990) and by Barratt (1997).

Regression models based on solvatochromic parameters can be used for predicting the degree of eye irritation, as illustrated by Abraham and coworkers (Abraham, 1993; Abraham *et al.*, 1998). The mechanistic basis of these models is that a substance is transferred from a pure organic liquid to an organic solvent phase consisting of the tear film and cell membranes on the surface of the eye. The more soluble the organic liquid in the initial phase, the greater the degree of irritation is. These models are worthy of further characterisation. However, for routine regulatory use, information on a number of so-called Abraham descriptors would also need to be made available.

Neural network approaches can also be used to model eye irritation (e.g. Patlewicz *et al.*, 2000). At present, however, many of these models lack transparency, especially in the algorithm. However if the training sets are provided as well as validation information they could possibly be used in a *Weight-of-Evidence* approach. Mechanistic reasoning should also be provided.

An approach called Membrane-Interaction QSAR analysis, developed by Kulkarni *et al.* (2001), provides a means of incorporating molecular dynamic simulations to generate membrane-solute interaction properties. The development and application of models based on molecular simulations requires the use of specialised expertise and software. They could be used to increase understanding of the mechanisms of eye irritation.

A classification approach called Embedded Cluster Modelling (ECM) provides a means of generating *elliptic models* in two or more dimensions (Worth and Cronin, 2000), so that irritants can be transparently identified as those substances located within the boundaries of the ellipse. The statistical significance of these “embedded clusters” can be verified by cluster significance analysis (CSA), as illustrated for an eye irritation dataset by Cronin (1996).

Different methods were applied to a dataset of 119 organic liquids classified as I (irritant) or NI (non-irritant) according to former EU classification criteria. The classification models (CMs) were developed by applying linear discriminant analysis (LDA), binary logistic regression (BLR), and classification tree (CT) analyses, using a single predictor variable (molecular weight), and assigning equal probabilities for the two classes (I/NI). (Worth and Cronin, 2003).

All of these models are simple to apply and are associated with a transparent algorithm. The statistics illustrate the inevitable trade-offs that result from the selection of different cut-off values. Thus, the BLR model does not identify many irritants, but it does so with a high degree of confidence. Conversely, the CT does not identify many of the non-irritants, but it has a low false negative rate. Thus, the combined use of the BLR and CT models could be useful for distinguishing between eye irritants and non-irritants.

An overview on the available literature-based models for serious eye damage/eye irritation is provided in [Table R.7.2-7](#).

Table R.7.2-7 Available literature-based models for serious eye damage/eye irritation.

Reference	Content
QSAR models	
Abraham et al. (2003) Draize rabbit eye test compatibility with eye irritation thresholds in humans: a quantitative structure-activity relationship analysis.	Draize rabbit eye test scores, as modified maximum average score (MMAS), for 68 pure bulk liquids were adjusted by the liquid-saturated vapor pressure P. These 68 adjusted scores, as log (MMAS/P), were shown to be completely equivalent to eye irritation thresholds (EIT), expressed as log (1/EIT), for 23 compounds in humans. Thus, for the first time the Draize eye test in rabbits for pure bulk liquids is shown to be perfectly compatible with eye irritation thresholds in humans.
Barratt (1995) The role of structure-activity relationships and expert systems in alternative strategies for the determination of skin sensitisation, skin corrosivity and eye irritation.	This paper describes the derivation of a set of structural alerts for skin sensitisation, which have been incorporated into the expert system DEREK, and of Quantitative structure-activity relationships (QSARs) derived for predicting the skin corrosivity (for organic acids and bases) and for the eye irritation potential (for neutral organic chemicals).
Gerner et al. (2005) Assessment of the Eye Irritating Properties of Chemicals by Applying Alternatives to the Draize Rabbit Eye Test: The Use of QSARs and <i>In Vitro</i> Tests for the Classification of Eye Irritation.	This paper evaluates and discusses the nature of eye lesions and their importance for classification and labelling of possible hazards to human eyes, with a view to promoting the development of specific <i>in vitro</i> assays which are able to discriminate between eye damage, moderate eye irritation, and minor irritation effects which are completely reversible within a few days. Structural alerts for the prediction of eye irritation/corrosion hazards to be classified and labelled according to international classification criteria, are presented, which should be validated in accordance with internationally agreed (OECD) principles for (Q)SAR system validation. Physicochemical limit values for prediction of the absence of any eye irritation potential relevant for human health can make available a definition of the applicability domains of alternative methods developed for the replacement of the Draize eye irritation test.
Solimeo et al. (2012) Predicting Chemical Ocular Toxicity Using a Combinatorial QSAR Approach.	This paper describes QSAR models for a set of small molecules with animal ocular toxicity data compiled by the National Toxicology Program Interagency Center for the Evaluation of Alternative Toxicological Methods.

Reviews and evaluation of existing models	
Gallegos Saliner et al. (2006) Review of Literature-Based Models for Skin and Eye Irritation and Corrosion.	This report reviews the state-of-the-art of <i>in silico</i> and <i>in vitro</i> methods for assessing dermal and ocular irritation and corrosion. In this review, emphasis is placed on literature-based QSAR models for skin and eye irritation and corrosion as well as computer-based expert systems.
Gallegos Saliner et al. (2008) Review of (Q)SAR Models for Skin and Eye Irritation and Corrosion.	This paper reviews the state-of-the-art of <i>in silico</i> methods for assessing dermal and ocular irritation and corrosion. It is based on an in-depth review performed by the European Chemicals Bureau of the European Commission: Joint Research Centre. The most widely used <i>in silico</i> approaches are classified into methods to assess (1) skin irritation, (2) skin corrosion and (3) eye irritation. In this review, emphasis is placed on literature-based (Q)SAR models.
Tsakovska et al. (2005) Evaluation of (Q)SARs for the prediction of Eye Irritation/Corrosion Potential - physicochemical exclusion rules.	In this study, an evaluation was performed of the physicochemical BfR-DSS rule-base (comprising 31 physicochemical exclusion rules) for predicting the absence of eye irritation/corrosion. According to the results of this study: a) the physicochemical exclusion rules for eye irritation/corrosion comply well with the OECD validation principles; b) predictions of no adverse effect (NOT R34/R35/R36/R41) can be made for 20 out of the 199 chemicals in the test set; c) 3 of the 45 irritants/corrosives are falsely predicted as non-irritant or non corrosive; d) the probability of a negative prediction being correct (Negative Predictive Value) is 0.87; and e) approximately 10% of Draize rabbit eye tests could be avoided by relying on the predictions of no adverse effect.
Tsakovska et al. (2007) Evaluation of SARs for the prediction of eye irritation/corrosion potential - structural inclusion rules in the BfR decision support system.	This work summarises the results of a study carried out by the ECB to assess the performance of the BfR structural rule-base. The assessment included: (a) evaluation of the structural alerts by using the training set of 1341 substances with experimental data for eye irritation and corrosion; and (b) external validation by using an independent test set of 199 substances. The test set of 199 substances contained 154 (77%) non-labelled substances and 45 (23%) labelled as eye irritants/corrosives, subdivided as follows: (i) 10 R36 substances (5%); (ii) 28 R41 substances (14%); and (iii) 7 substances (4%) labeled R34 or R35.

Further details on these models can be found in Chapter 4 of the JRC report "Alternative methods for regulatory toxicology - a state-of-the-art review" (Worth *et al.*, 2014).

COMMERCIAL MODELS

There is a number of software tools available that provide access to QSARs for serious eye damage/eye irritation.

The **TOPKAT** software (<http://accelrys.com/solutions/scientific-need/predictive-toxicology.html>) includes models for eye irritation based on structural fragments. These models were originally developed by Enslein *et al.* (1988). The TOPKAT algorithm is not very transparent. The model predicts a probability of a weak/mild/moderate and severe irritation. It states that probabilities <0.3 and >0.7 give sufficient certainty of the prediction. The model gives the sensitivity and specificity values of the specific classes such as acyclic, which are mostly around or above 90%. It also shows similar structures from the TOPKAT perspective including the experimental result. The TOPKAT predictions weak/mild/moderate and severe

irritation need to be translated to consider them for classification. The models indicate whether the prediction is in the applicability domain of the model.

There is a rulebase for irritation in **Derek Nexus** (Sanderson and Earnshaw, 1991; Combes and Rodford, 2004), which is developed and regularly updated by LHASA Ltd (<http://www.lhasalimited.org/products/derek-nexus.htm>). See for a general outline the skin irritation section on (Q)SARs. The Derek Nexus rule-base has five alerts that are specific to eye irritation, plus one for eye lachrymation. If Derek Nexus does not make a prediction of irritation or corrosivity, it cannot be concluded that there is no effect – it could mean that none of known alerts was found to be present in the substance of interest or it was outside the applicability domain of that specific alert. The Derek Nexus model is transparent in its algorithm, when the model is fired showing the structural alert and its limitations. The alert is underlined with literature references and sometimes with example substances, which is not sufficient to consider them internally validated. The example substances underline the mechanistic reasoning. The Derek Nexus model can be used for positive identification of skin irritation. The confidence levels have to be translated to consider them for classification. The Derek Nexus model cannot be used to predict non-serious eye damage/eye irritation as the model only contains alerts that detect the presence of serious eye damage/eye irritation.

The fragment-based **MultiCASE** approach (http://www.multicase.com/case-ultra-models#skin_eye_tox_bundle) has been used to model eye irritation (Klopman *et al.*, 1993; Enslein *et al.*, 1988; Rosenkranz *et al.*, 1998; Klopman, 1998). The publications on these models do not define the algorithms. In principle, the MultiCASE model can be used for positive and negative indication for eye irritation. The structural alert is provided as well as the internal validation. The MultiCASE model also indicates whether it is in the applicability domain of the model. The MultiCASE predictions of weak/mild/moderate and severe irritation need to be translated to consider them for classification. The prediction should be underlined with mechanistic reasoning using other models or expert judgement.

ACD/Labs Percepta Predictors (<http://www.acdlabs.com/products/percepta/predictors.php>), developed by ACD/Labs, includes a module for skin and eye irritation. It estimates the potential of a compound to cause eye or skin irritation in a standard rabbit Draize test. The predictions are reported as qualitative irritation categories (not irritating, slightly irritating, irritating, highly irritating, and corrosive). Probabilistic models are supplemented by an expert system that identifies Structural Alerts relevant in the irritational properties of compounds. Overall, 21 structural alerts were formulated for rabbit eye irritation, and 17 alerts for the rabbit skin irritation case.

PaDEL-DDP Predictor includes several models for skin and eye irritation and corrosion (<http://padel.nus.edu.sg/software/padelddpredictor/>). The models have been built from a training set of 1707 compounds using one and two dimensional descriptors. The final predictions rely on consensus models based on majority voting from base models predictions. The applicability domain is defined by the range of descriptors for compounds in the training set.

BFR DECISION SUPPORT SYSTEM

The decision support system (DSS) developed by the German Federal Institute for Risk Assessment (BfR) uses physico-chemical exclusion rules to predict the absence of serious eye damage/eye irritation potential in combination with structural inclusion rules (SARs) to predict the presence of such potential (Gerner *et al.*, 2005). These rules are used analogously to those described in the skin corrosion and irritation section above. The physico-chemical and structural rule-bases are designed to predict the former EU risk phrases for eye irritation (R36) and severe eye irritation/corrosion (R41). Independent assessments by the ECB support the performance of the physico-chemical rule-base for predicting the absence of eye effects

(Tsakovska *et al.*, 2005), as well as the performance of the structural rulebase for predicting the occurrence of eye effects (Tsakovska *et al.*, 2007).

OECD QSAR TOOLBOX

The freely downloadable OECD QSAR Toolbox software (<http://www.qsartoolbox.org/>) covers the serious eye damage/eye irritation endpoint with one experimental database and two profilers.

In more detail, the database of experimental data (called "Eye irritation ECETOC" in the software) refers to the endpoint Modified Maximum Average Score (MMAS) and collects experimental results on rabbit eye irritation described in ECETOC Technical Report No.48 on Eye Irritation Reference Chemicals Data Bank (ECETOC, 1992).

The OECD QSAR Toolbox allows for the identification of analogues based on mechanistic and endpoint specific profilers, and for the prediction of skin irritation/corrosion through the use of read across, trend analysis and QSAR models. Information about inclusion and exclusion rules, details on the performance of the exclusion rules, and applicable chemical class-specific rules for the results of the Eye irritation/corrosion profiler can be found by searching the context menu in the the OECD QSAR Toolbox software.

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R.7.3 Skin and respiratory sensitisation

R.7.3.1 Introduction

A number of diseases are recognised as being, or presumed to be, allergic in nature. These include asthma, rhinitis, conjunctivitis, allergic contact dermatitis (ACD), urticaria and food allergies (the latter is not discussed in this document). In this Section, the endpoints discussed are those traditionally associated with occupational and consumer exposure to substances. Photosensitisation is potentially important but its mechanism of action is poorly understood, and it is not discussed in this document.

R.7.3.1.1 Definition of skin and respiratory sensitisation

A skin sensitizer is an agent that will lead to an allergic response in susceptible individuals following skin contact. As a consequence of a secondary - usually organ-specific - subsequent re-exposure, adverse health effects on the skin (allergic contact dermatitis).

A respiratory sensitizer is an agent that will lead to hypersensitivity of the airways following inhalation exposure to that agent. Respiratory sensitisation (or hypersensitivity) is a term that is used to describe asthma and other related respiratory conditions (rhinitis, extrinsic allergic alveolitis), irrespective of the mechanism (immunological or non-immunological) by which they are caused. In contrast, skin allergy is based on an immunological mechanism. In this sense, it is important to distinguish an ACD e.g. from non-immunological contact urticaria (NICU) or toxic/irritant contact eczema which represent non-immunologically mediated responses of the skin.

When directly considering human data in this document, the clinical diagnostic terms asthma, rhinitis, extrinsic allergic alveolitis and allergic contact dermatitis have been retained.

These definitions are reflected in the criteria for the classification of skin and respiratory sensitizers, which provide a useful reference against which the hazardous properties of a substance can be judged. These criteria are given in the Regulation (EC) No 1272/2008 on the classification, labelling and packaging of substances and mixtures (CLP Regulation).

Classification and labelling under the CLP Regulation

Substances and mixtures causing skin sensitisation and/or respiratory sensitisation can be further characterised by their classification under the CLP Regulation. Currently (at the time of publication of this Guidance) the CLP (and UN GHS) criteria for classifying sensitizers are based on standard animal data, human data and data obtained from non-standard methods, e.g. read-across or non-standard test methods may be used in combination in a *Weight-of-Evidence* approach. Discussions at UN GHS level are ongoing to include *in vitro* based criteria for classification. Information on the sensitising potency of a substance is important for the classification and labelling of mixtures since, depending on this sensitising potency, different specific concentration limits need to be used for mixture classification. As described in section R.7.3.3 of this guidance, an exposure assessment and risk characterisation has to be made according to REACH for a skin sensitizer.

Detailed information on the classification and labelling of substances and mixtures can be found in the [Guidance on the Application of the CLP criteria](#) and in the CLP Regulation.

a) For skin sensitisation

Skin sensitisers are classified in Category 1 with the signal word “warning” and the Hazard statement H317 “*May cause an allergic skin reaction*”. Where data are sufficient, skin sensitisers can be divided into sub-categories. If data are not sufficient for sub-categorisation, Category 1 must be chosen.

- **Sub-category 1A:** Substances showing a high frequency of occurrence in humans and/or a high potency in animals can be presumed to have the potential to produce significant sensitisation in humans. Severity of reaction may also be considered.
- **Sub-category 1B:** Substances showing a low to moderate frequency of occurrence in humans and/or a low to moderate potency in animals can be presumed to have the potential to produce sensitisation in humans. Severity of reaction may also be considered.

b) For respiratory sensitisation

Respiratory sensitisers are classified in Category 1 with the signal word “danger” and the Hazard statement H334 “*May cause allergy or asthma symptoms or breathing difficulties if inhaled*”. Where data are sufficient, respiratory sensitisers can be divided into sub-categories. If data are not sufficient for sub-categorisation, Category 1 must be chosen.

- **Sub-category 1A:** Substances showing a high frequency of occurrence in humans; or a probability of occurrence of a high sensitisation rate in humans based on animal or other tests. Severity of reaction may also be considered.
- **Sub-category 1B:** Substances showing a low to moderate frequency of occurrence in humans; or a probability of occurrence of a low to moderate sensitisation rate in humans based on animal or other tests. Severity of reaction may also be considered.

R.7.3.1.2 Objective of the guidance on skin and respiratory sensitisation

The general objectives of this guidance are:

- to establish whether information from physical/chemical data, from non-testing methods (grouping, QSARs and expert systems), from *in chemico*, *in vitro* or *in vivo* studies⁴⁹ or human experience data, provides sufficient evidence that the substance has skin or respiratory sensitisation potential or the lack thereof; or
- to establish whether new information needs to be generated to meet the information requirements under the REACH Regulation by providing a testing and assessment strategy as presented in this document⁵⁰.

⁴⁹ These terms are defined as follows: an *in vitro* study is a study using cells, tissues or organs and conducted in glass or plastic vessels in a laboratory; an *in vivo* study is a study conducted in a living organism; an *in chemico* study is a study using abiotic (i.e. not conducted in animals or *in vitro*) measurements of the reactivity or other physico-chemical properties of a substance.

⁵⁰ The testing and assessment strategies are also referred to as Integrated Approaches to Testing and Assessment (IATAs).

Therefore, in the sections on skin sensitisation and respiratory sensitisation, firstly an overview of types of data is given that may provide information on sensitisation, followed by guidance on the process of judging the available data in terms of adequacy, completeness and remaining uncertainty. In Sections [R.7.3.6](#) and [R.7.3.11](#) guidance is given on application of the data to reach a conclusion on suitability for classification and labelling, including potency, if possible. However, when performing new tests, registrants must take account of the new requirement for potency determination for skin sensitising substances (Cat. 1A or not). Finally, in Sections [R.7.3.7](#) and [R.7.3.12](#), a testing and assessment strategy is presented for skin sensitisation and respiratory sensitisation, respectively.

SKIN SENSITISATION

R.7.3.2 Mechanisms of skin sensitisation

Contact allergens are reactive substances (usually organic substances or metal ions) of low molecular weight (<500 - 1000 Da) and have a lipophilicity that favours dermal penetration. Skin sensitisation is considered to be a delayed type hypersensitivity (Type IV according to Gell and Coombs) (Karlberg *et al.*, 2008; Martin *et al.*, 2011; Martin, 2014). Some of the mechanisms leading to skin sensitisation (allergic contact dermatitis in humans) are relatively well understood. In 2012 the OECD published an **Adverse Outcome Pathway** (AOP), which describes the biological mechanisms of skin sensitisation initiated by the covalent binding of substances to skin proteins (OECD, 2012). It should be noted that this AOP does not cover metals or allergens of biological origin, but only substances that form a covalent binding to skin proteins. The key events of this skin sensitisation pathway are: 1) covalent binding of the electrophilic substance to skin proteins; 2) release of pro-inflammatory cytokines and induction of cyto-protective pathways in keratinocytes; 3) activation and maturation of dendritic cells, and their migration to the local lymph nodes; 4) presentation of the chemical allergen by the dendritic cells (allergen processed by the dendritic cell and displayed in its surface as an epitope) to naïve T-cells, which leads to their differentiation and proliferation into allergen-specific memory T-cells. Even though not considered as being part of the key events from one to four leading to the adverse outcome, dermal bioavailability (penetration and, if applicable, metabolism) is a prerequisite for a substance to cause skin sensitisation, i.e. the substance needs to reach the viable epidermis in its reactive form.

The mechanisms of metals leading to induction of the innate immune system and, concomitantly, possibly leading to skin sensitisation are not completely understood (Thierse *et al.*, 2005; Martin *et al.*, 2006). Some types of metal ions may act as non-classical haptens, i.e. they do not require stable binding to processed proteins but may directly or indirectly cause structural changes in the major histocompatibility complex (MHC) molecule-peptide complex (by metal-protein complex formation) which then lead to recognition and activation of T-cells *via* T-cell receptors (Templeton, 2004; Gammerdinger *et al.*, 2003; Lu *et al.*, 2003). Less is known about other types of metal ions such as beryllium (Bowerman *et al.*, 2014). Therefore, skin sensitisation for metals should be evaluated on a case-by-case basis depending on the metal and amount of available information.

Traditionally the development of skin sensitisation has been divided in two phases, i.e. induction and elicitation. In the induction phase the naïve individual becomes sensitised to the allergenic agent, e.g. through the molecular events as described above, leading to the formation of allergen-specific memory T-cells. Those specific memory cells migrate into the dermis and epidermis for the repeated encounter with the specific allergen. In the elicitation phase the memory T-cells, created before in the induction phase, re-encounter the specific allergen which leads to the quick proliferation and activation of those allergen-specific T-cells. The activated cells start secreting specific cytokines, which in turn mobilise other inflammatory cells leading to the clinical outcome of allergic contact dermatitis.

R.7.3.3 Information requirements for skin sensitisation

The information on skin sensitisation that is required to be submitted for registration and evaluation purposes is specified in Annexes VI to XI to the REACH Regulation. According to Annex VI, the registrant should gather and evaluate all existing available information before considering further testing. This includes structural considerations, physico-chemical properties, (Q)SAR ((Quantitative) Structure-Activity Relationship), information from structurally similar substances, *in vitro/in chemico* data, animal studies, and human data. For

classified substances, information on exposure, use and risk management measures should also be collected and evaluated in order to ensure that potential risks are identified and adequate risk management measures are taken.

If these data are inadequate for hazard and risk assessment, including classification and labelling, further testing should be carried out in accordance with the **requirements of Annex VII (≥ 1 tpa) to the REACH Regulation**.

The standard information requirements at this tonnage level for skin sensitisation (see Sections 8.3, 8.3.1. and 8.3.2 in Column 1 of Annex VII) are as follows:

8.3. Column 1: Skin sensitisation

Information allowing:

- *a conclusion whether the substance is a skin sensitiser and whether it can be presumed to have the potential to produce significant sensitisation in humans (Cat. 1A), and*
- *risk assessment, where required.*

8.3.1 Column 1: Skin sensitisation, in vitro/in chemico

Information from in vitro/in chemico test method(s) recognised according to Article 13(3), addressing each of the following key events of skin sensitisation:

- (a) molecular interaction with skin proteins;*
- (b) inflammatory response in keratinocytes;*
- (c) activation of dendritic cells.*

8.3.2. Column 1: Skin sensitisation, in vivo

Column 2 of Annex VII lists specific rules according to which the required standard information indicated in column 1 may be omitted, replaced by other information, or adapted in another way. If the conditions are met under which column 2 of this Annex allows adaptations, the fact and the reasons for each adaptation should be clearly indicated in the registration dossier. Column 2 specific rules for adaptation are as follows:

8.3. Column 2:

The study(ies) under point 8.3.1. and 8.3.2. do not need to be conducted if:

- *the substance is classified as skin corrosion (Category 1), or*
- *the substance is a strong acid ($\text{pH} \leq 2,0$) or base ($\text{pH} \geq 11,5$), or*
- *the substance is spontaneously flammable in air or in contact with water or moisture at room temperature.*

8.3.1. Column 2:

The(se) test(s) do not need to be conducted if:

- *an in vivo study according to point 8.3.2 is available, or*
- *the available in vitro/in chemico test methods are not applicable for the substance or are not adequate for classification and risk assessment according to point 8.3.*

If information from test method(s) addressing one or two of the key events in column 1 already allows classification and risk assessment according to point 8.3, studies addressing the other key event(s) need not be conducted.

8.3.2. Column 2:

An in vivo study shall be conducted only if in vitro/in chemico test methods described under point 8.3.1. are not applicable, or the results obtained from those studies are not adequate for classification and risk assessment according to point 8.3.

The Murine Local Lymph Node Assay (LLNA) is the first-choice method for in vivo testing. Only in exceptional circumstances should another test be used. Justification for the use of another in vivo test shall be provided.

In vivo skin sensitisation studies that were carried out or initiated before 10 May 2017, and that meet the requirements set out in Article 13(3), first subparagraph, and Article 13(4) shall be considered appropriate to address this standard information requirement.

This means that when new data are generated the assessment must not only investigate whether a substance is a skin sensitiser but also the skin sensitisation potency ('whether the substance can be presumed to have the potential to produce significant sensitisation in humans (Cat. 1A)').

In case the substance is a skin sensitiser based on *in vitro/in chemico* testing and the results of *in vitro/in chemico* testing allow a sufficiently reliable conclusion that the substance has the potential to produce significant sensitisation in humans (Cat. 1A), no further testing is required if that classification (Cat. 1A) is applied.

In case the substance is a skin sensitiser based on *in vitro/in chemico* testing, and if the results of *in vitro/in chemico* testing allow a sufficiently reliable conclusion that the substance does not have the potential to produce significant sensitisation in humans, it can be presumed that the substance would be a moderate skin sensitiser (Cat. 1B). In this case, no further testing is needed. ECHA recommends that this reasoning and the consequent classification be followed by the registrant.

However, in case significant sensitisation (Cat. 1A) cannot be excluded with sufficient confidence based on *in vitro/in chemico* testing, additional information (*in silico/in vitro/in chemico*) would need to be generated to strengthen the weight of evidence. If still no reliable conclusion can be reached, as a last resort *in vivo* testing (LLNA) would need to be performed.

When all data sources have been considered, a decision whether a substance is presumed to produce "*significant sensitisation in humans*" can normally be made by using a *Weight-of-Evidence* approach. In rare cases where all relevant data sources and possibilities have been considered and that decision still cannot be made, Cat. 1 classification can be applied (See also footnote e to [Figure R.7.3-2](#)).

Following the change of legislation in 2016, potency assessment is now required. However, in case there is already existing *in vivo* information (study initiated or conducted before 10 May 2017) that does not allow assessment of skin sensitisation potency, this information can still be used to fulfil the information requirement and no additional testing is required. In such cases, any information on skin sensitisation potency coming from such studies should be used together with existing information from other sources or with additional non-animal test data to refine classification and risk assessment for skin sensitising substances.

General provisions for the generation of information on intrinsic properties of substances are contained in REACH Article 13 which states that, in particular for human toxicity, information

must be generated whenever possible by means other than vertebrate animal tests, through the use of alternative methods, for example *in vitro* methods or qualitative or quantitative structure-activity relationship models or from information from structurally related substances (grouping or read across), provided that the conditions specified in Annex XI are met.

In addition to the specific rules of adaptation (column 2 of Annexes VII to X), Annex XI 1.2 to 1.5 to the REACH Regulation lays out general rules of adaptation to the standard information requirements. The specific rules for adaptations are: use of non-animal test methods (e.g. *in vitro/in chemico*) combined with other approaches within a *Weight-of-Evidence* approach (section 1.2), use of (Q)SARs (section 1.3), use of *in vitro* methods (section 1.4, note these refer to stand-alone methods) or the use of read-across (section 1.5). In the case of Annex XI adaptation as well, this fact and the reasons for each adaptation should be clearly indicated in the registration dossier, i.e. in IUCLID.

Guidance on application of these rules is given in the testing and assessment strategy described in Section [R.7.3.7](#) of this Guidance.

The Murine Local Lymph Node Assay (LLNA), allowing assessment of potency, is the first-choice method for *in vivo* testing in case new *in vivo* testing is justified. Only in exceptional circumstances should another test be conducted. This means that in certain cases other *in vivo* methods may be used. In such cases convincing scientific justification for conducting another test must be provided in the registration dossier.

R.7.3.4 Information sources on skin sensitisation

R.7.3.4.1 Non-human data for skin sensitisation

Experimental data available in databases

Registrants need to collect all available relevant information on their substance. It is advised to start by looking at all experimental data for skin sensitisation that may already be available from REACH registration dossiers or from other sources (e.g. from the literature). ECHA's dissemination website is the primary source for REACH data. Data can also be found through the OECD QSAR Toolbox or the eChemPortal. Training sets from computational tools like TIMES, Ambit, Topkat, Vitic Nexus, and others are also a valuable source of experimental data. All these sources compile heterogeneous results originating from different standard and non-standard tests. It is very important to assess how the original data have been interpreted in these tools and a consultation with original sources is always recommended, if possible. More details on data sources are given in [Appendix R.7.3-1](#).

Non-testing data for skin sensitisation

The adaptation of standard information requirements can be used if relevant and reliable alternative data can be provided for the substance of interest. As specified in Annex XI to the REACH regulation, the use of non-testing methods needs to be justified and sufficiently documented. Read-across and (Q)SAR models are non-testing methods that can provide data for skin sensitisation.

Read-across

Read-across/chemical categories are described in Sections R.6.1 and R.6.2 of Chapter R.6 of the [Guidance on IR&CSA](#). The scientific basis for building grouping arguments and read-across cases were revisited in the second version of the OECD Guidance Document on grouping of chemicals (OECD, 2014) and in the OECD Guidance Document on the reporting of defined approaches and individual information sources to be used within integrated approaches to testing and assessment (IATA) for skin sensitisation (OECD, 2016b). More detailed advice on

the assessment of read across can be found in ECHA's Read-Across Assessment Framework - RAAF (see <http://echa.europa.eu/support/grouping-of-substances-and-read-across>). Developing and assessing read-across for skin sensitisation was discussed and exemplified in the work of Patlewicz *et al.* (2015).

(Q)SAR models

In the case of QSARs and expert systems, registrants need to prepare property predictions by completion of a QSAR Prediction Reporting Format (QPRF). The QPRF is a harmonised template for summarising and reporting substance-specific predictions generated by (Q)SAR models. For filling a data gap under REACH, it is also necessary to provide information on the prediction model employed following a QSAR Model Reporting Format (QMRF) document. The QMRF is a harmonised template for summarising and reporting key information on (Q)SAR model validity, including the results of any validation studies. The information is structured according to the OECD (Q)SAR validation principles (for further information see <http://www.oecd.org/env/ehs/risk-assessment/validationofqsarmodels.htm>). The JRC QSAR Model Database is an inventory of information on available QMRFs, freely accessible online (<https://eurl-ecvam.jrc.ec.europa.eu/databases/jrc-qsar-model-database>). More detailed guidance on QSAR models, their use and reporting formats, including the QMRF, is provided in Section R.6.1 of Chapter R.6 of the *Guidance on IR&CSA*. There is also an initiative started recently (QsarDB) that aims to develop a dynamic repository for QSAR models and datasets for giving access to them and to facilitating predictions from selected literature models that are transparent enough and reproducible.

There are some (Q)SAR models for skin sensitisation reported in the peer-reviewed literature. Available models include local and global (Q)SARs as well as expert systems. If not implemented in a software tool, their use might be restricted due to accessibility issues of a technical nature. Exploring the reaction chemistry of substances forms the basis of most read-across justifications and many of the available skin sensitisation (Q)SARs. According to the OECD AOP for covalent binding to proteins, the skin sensitisation potential of a substance is related in the first place to its ability to react with skin proteins to form covalently linked conjugates and recognition of these by the immune system. In the vast majority of cases, this is dependent on electrophilic reactivity of the substance or a derivative produced by metabolisms or abiotic degradation (Barratt *et al.*, 1997). There are various types of electrophile-nucleophile reactions in skin sensitisation, of which perhaps the most frequently encountered are: Michael-type reactions, S_N2 reactions, S_NAr reactions, acylation reactions and Schiff-base formation. These chemical reaction mechanisms can serve as a means of describing the domain of applicability (the scope) of a (Q)SAR model or form the basis for grouping substances into chemical categories. Recent work in this area has been described elsewhere (Aptula *et al.*, 2005; Aptula and Roberts, 2006; Roberts *et al.*, 2007, 2011; Schultz *et al.*, 2009; Natsch *et al.*, 2012; Enoch and Roberts, 2013).

Some (Q)SAR models that may be useful for predicting several REACH relevant endpoints, including skin sensitisation, have been included in software packages. A list of available tools was compiled within ANTARES, an EU LIFE project whose results are freely available online (<http://www.antares-life.eu/index.php?sec=modellist>). QSAR predictions regarding skin sensitisation (and a range of other toxicological and ecotoxicological endpoints) of nearly all discrete organic pre-registered substances under REACH are included in the freely available Danish QSAR database (<http://qsar.food.dtu.dk/>), See further description in [Appendix R.7.3-1](#)). The reliability of each prediction needs to be assessed for every substance depending on information for model applicability. The OECD Guidance on grouping of chemicals (OECD, 2014) also provides a summary of tools that might be useful in predicting endpoints of regulatory relevance, including skin sensitisation (see also: <http://www.oecd.org/chemicalsafety/risk-assessment/groupingofchemicalschemicalcategoriesandread-across.htm>).

A non-exhaustive list of available tools is also given in [Table R.7.3–1](#).

More details on data and (Q)SAR models in scientific publications and in *in-silico* tools are given in [Appendix R.7.3–1](#), while Section [R.7.3.5.1](#) discusses the evaluation of non-testing methods.

Table R.7.3–1 *In silico* tools for skin sensitisation prediction. Note that qualitative models might have been developed using thresholds different from those indicated in REACH and CLP. The user is advised to use as many different softwares as possible to gather different pieces of information.

Name of the software	Model/module	Model type	Endpoint and/or training set data type
QSAR Toolbox, free http://www.qsartoolbox.org/	Protein binding profilers: • OASIS v1.3 • OECD • Potency • Alerts for skin sensitisation by OASIS v1.3	Structural alerts	Protein binding
	“Endpoint” (i.e. Databases): • Skin sensitisation • Skin sensitisation ECETOC • ECHA Chem	Data repositories	Mainly LLNA and GMPT
	Data gap filling: • Read across • Trend analysis	Qualitative ⁵¹ or quantitative	Mainly LLNA and GMPT
ToxTree, free http://toxtree.sourceforge.net/skinsensitisation.html	Skin sensitisation reactivity domains	Structural alerts	Reactivity mode of action

⁵¹ E.g. OASIS scale in the QSAR Toolbox: Non-sensitiser (EC₃ ≥ 50%), Weak sensitiser (10% ≤ EC₃ < 50%), Strong sensitiser (EC₃ < 10%)

VEGA, free http://www.vega-qsar.eu/	Skin sensitisation CAESAR	Qualitative ⁵²	LLNA
CASE Ultra, commercial http://www.multicase.com/case-ultra	SkinEye Toxicity models	Qualitative ⁵³	Many (GMPT, LLNA, other)
Derek Nexus, commercial http://www.lhasalimited.org/products/derek-nexus.htm	Skin sensitisation	Qualitative ⁵⁴	GPMT and LLNA
TIMES, commercial http://oasis-lmc.org/products/models/human-health-endpoints/skin-sensitization.aspx	TIMES Skin Sensitisation	Qualitative ⁵⁵ (Metabolic simulator + "local" QSARs)	Mainly GPMT and LLNA
TOPKAT, commercial http://accelrys.com/products/collaborative-science/biovia-discovery-studio/qsar-admet-and-predictive-toxicology.html	Two models: • Non-sensitisers vs. Sensitisers • Weak/Moderate vs. Strong sensitisers	Qualitative ⁵⁶	GPMT
Danish (Q)SAR database http://qsar.food.dtu.dk/	Skin sensitisation (CASE Ultra, Leadscape and SciQSAR)	Qualitative ⁵⁷	GPMT and Allergic Contact Dermatitis in Humans)

⁵² Skin sensitisation model in VEGA: aggregated data from Geberick *et al.* (2005) (extreme, strong and weak sensitisers coded as positive, NC as negative)

⁵³ CASE Ultra model for LLNA: Weak (EC3 < 100%), Moderate (EC3<10%), Strong (EC3<1%).

⁵⁴ The outcome of Derek that can be regarded as "positive" alerts are: CERTAIN, PROBABLE, PLAUSIBLE or EQUIVOCAL (in descending order of likelihood), because the query molecule contains a toxicophore that matches one of the alerts that have been coded. EQUIVOCAL is recommended as being a "positive" result in Derek because the alert has still fired for the molecule, but there is an indication that there is(are) (an)other aspect(s) that Derek has considered in its calculation (such as physico-chemical properties).

⁵⁵ Aggregated data from different sources and tests converted into three categories (Strong, Weak and Non-sensitisers) according to the following scheme: (i) Data from LLNA: extreme, strong and moderate LLNA data converted into "Strong", weak to "Weak" and non-sensitiser to "Non-sensitiser"; (ii) Data from GPMT: strong and moderate GPMT data converted into "Strong", weak into "Weak" and non-sensitiser into "Non-sensitiser"; (iii) Data from BfR (Schlede *et al.*, 2003): category A converted into "Strong", category B into "Weak", category C into "Non-sensitiser".

⁵⁶ TOPKAT scale (GPMT, % of animals positive): Weak (1-30%), Moderate (30-70%), Strong (70-100%).

⁵⁷ Several factors were considered in the activity classification including: the type of assay used, i.e. human or guinea pig maximization test; use of adjuvant; dose used for challenge; and the sensitisation rate. Weak, moderate, strong and extreme sensitizers were included in the model training set as positive and non-sensitisers were included as negative (see further information in the QMRFs available in the website of the database).

Testing data for skin sensitisation

Internationally adopted test methods for skin sensitisation are described in the Annex to the EU Test Methods (TM) Regulation (Council Regulation (EC) No 440/2008) and in OECD Test Guidelines (TGs) (available at <http://www.oecd.org/env/ehs/testing/oecdguidelinesforthetestingofchemicals.htm>).

Please note that the latest version of an adopted test guideline should always be used when generating new data, independently from whether it is published by the EU or the OECD.

The testing and assessment strategy developed for skin sensitisation (see Section [R.7.3.7](#) of this Guidance) emphasises the need to evaluate **all** available information (including structural considerations and physico-chemical properties) before attempting any *in chemico*, *in vitro* or *in vivo* testing.

In chemico/in vitro data

Internationally adopted *in chemico/in vitro* test methods to assess whether a substance is a skin sensitizer or not are listed in [Table R.7.3–3](#). In addition this table lists test methods that have undergone validation and have been submitted to validation authorities for independent peer-review or are currently in the peer review process. The development of Test Guidelines for some of these methods is under consideration by the OECD. More information on the specific scope and limitations of these tests is provided in Section [R.7.3.5.1](#) under “Testing data on skin sensitisation”.

In case several EU/OECD adopted test methods are available for a key event, the registrant should select the most appropriate test method available for their substance based on the applicability of the test method.

Table R.7.3–2 Validation and adoption status of *in chemico/in vitro* methods for skin sensitisation ⁵⁸

AOP Key event measured ⁵⁹	Test method	Validation status, regulatory acceptance	EU Test Methods/ OECD test guideline	Outcome according to the test method/guideline	EURL ECVAM DB-ALM protocol Nr.
Skin sensitisation					
Key Event 1 Peptide/protein binding	DPRA	Validated and regulatory acceptance	B.59/TG 442C	SS or NS with complementary information	154

⁵⁸ Note: The test methods have each been validated independently, with a limited scope. This means that each has its limitations and cannot be used as a stand-alone test method; however the limitations may in many cases be compensated when used together with additional information e.g. by using information from similar substances within a *Weight-of-Evidence* approach. The test methods have not been validated for predicting potency and cannot be used currently on their own to sub-categorise or to predict potency. However, even though the *in chemico/in vitro* methods have not been validated for potency assessment, they provide quantitative or concentration-response information that can be used to assess whether the substance is presumed to have the potential to produce significant sensitisation in humans.

⁵⁹ Some of the methods described under a specific key event may cover mechanisms under other key events as well.

Key Event 2 Keratinocyte response	KeratinoSens™	Validated and regulatory acceptance	B.60/TG 442D	SS or NS with complementary information	155
	LuSens ⁶⁰	Under validation assessment	N.A/N.A	SS or NS with complementary information	184
	SENS-IS ⁶¹	Under validation assessment	N.A/N.A	SS or NS with complementary information	N.A
Key Event 3 Monocytic /Dendritic cell response	h-CLAT	Validated and regulatory acceptance	N.A/TG 442E	SS or NS with complementary information	158
	U-SENS™ ⁶⁰	Validated and under regulatory adoption	N.A/draft TG available	SS or NS with complementary information	183
	IL-8 Luc Assay ⁶²	Validated and under regulatory adoption	N.A/draft TG available	SS or NS with complementary information	N.A.
Key Event 4 ⁶³ T-cell response	N.A	N.A	N.A/N.A	N.A.	N.A.

NOTE: "Validated" means that the test method has gone through a validation process, e.g. by EURL ECVAM, ICCVAM or JaCVAM.

Abbreviations: SS = skin sensitiser; N.A. = not available; NS = non-sensitiser; DPRA = Direct Peptide Reactivity Assay; h-CLAT = human Cell Line Activation Test; IL = Interleukin; Luc = Luciferase; TG: Test Guideline.

⁶⁰ The LuSens and the U-SENS™ test methods have undergone industry-led validation studies (Ramirez *et al.*, 2016; Alépée *et al.*, 2015). The information generated in the validation studies has been submitted to EURL ECVAM and is currently under evaluation. For U-SENS™ ESAC peer review has been performed and EURL ECVAM recommendation is under preparation. A standard project submission form (SPSF) for a Test Guideline concerning U-SENS™ was submitted to the OECD in 2015 and a draft OECD TG is available. The project has been included in the OECD work programme.

⁶¹ The SENS-IS test method underwent an industry lead validation and has been submitted to EURL ECVAM (Cottrez *et al.*, 2016). An SPSF for the development of a Test Guideline was submitted to the OECD in 2015.

⁶² The IL-8 Luc Assay underwent a validation study coordinated by JaCVAM (Kimura *et al.*, 2015). The test method was peer-reviewed by JaCVAM (April 2016). A SPSF for the development of a Test Guideline was submitted to the OECD in 2014 and a draft OECD TG is available.

⁶³ It is important to note that there are currently no validated or adopted *in vitro* test methods to address Key Event 4 (human T-cell activation) of the AOP. However, this is a key step in ACD, reflecting the adaptive immune response, similarly to the LLNA, even though the LLNA does not measure antigen specific T-cell responses *per se* but only cell proliferation (Dietz *et al.*, 2010; Richter *et al.*, 2013).

The test methods indicated in [Table R.7.3–3](#) are either *in chemico* assay(s) (DPRA), or cell-based assays (Keratinosens™, LuSens, h-CLAT, U-SENS™, IL-8 Luc Assay, SENS-IS). These test methods were developed to address specific events of the skin sensitisation AOP (OECD, 2012). The AOP for skin sensitisation describes the current understanding of key events linked to skin sensitisation. As each of the test methods only addresses a specific key event of skin sensitisation, currently they should not be used in isolation to identify a skin sensitiser or a non-sensitiser. In the future, stand-alone methods may become available (e.g. SENS-IS may have the potential to serve as a stand-alone method, however this will be clarified once the scientific validity of the test method has been established). More information on how these test methods can be used in the REACH context can be found in [Section R.7.3.7.2](#) of this Guidance.

It is important to note that currently several non-animal test methods are under development or evaluation, and therefore their regulatory use and predictive value has not been assessed yet. It is advised to monitor the status of current developments through the scientific literature. Test methods under evaluation by EURL ECVAM or other international validation bodies can be monitored through the EURL ECVAM Test Method Submission webpage (<https://eurl-ecvam.jrc.ec.europa.eu/test-submission/>). The registrant is also advised to follow any updates to the ECHA webpage concerning Testing methods and alternatives (see: <http://echa.europa.eu/support/oecd-eu-test-guidelines>).

Animal data

- Guideline-compliant tests

For new *in vivo* testing of skin sensitisation potential, the murine local lymph node assay (LLNA), which is currently the best *in vivo* method to assess skin sensitisation potency, is the REACH Annex VII-endorsed *in vivo* method. This assay has been validated internationally and has been shown to have clear animal welfare benefits and scientific advantages compared with the guinea pig tests described below. The LLNA is designed to detect the potential of substances to induce sensitisation as a function of lymphocyte proliferative responses induced in regional lymph nodes (induction phase). This method is described in EU B.42/OECD TG 429, which contain the standard LLNA method and the rLLNA (in the rLLNA only one dose is used, which does not allow the assessment of potency). In addition, there are different variants of the LLNA adopted by the EU and OECD, i.e. EU B.50/OECD TG 442A (Local Lymph Node Assay: DA) and EU B.51/OECD TG 442B (Local Lymph Node Assay: BrdU-ELISA). The main differences compared to the OECD TG 429 is that these test methods do not use radioactive labelling and that there are currently no CLP criteria available for predicting skin sensitisation potency with these methods. However, the dose-response relationship information obtained may provide some information on skin sensitisation potency that can be used within a *Weight-of-Evidence* approach. It is recommended that when new *in vivo* data need to be generated the “standard” LLNA according to EU B.42/OECD TG 429 be used, if possible.

Two further animal test methods for skin sensitisation are described in EU B.6/OECD TG 406: the guinea pig maximisation test (GPMT) and the Buehler test. The GPMT is an adjuvant-type test in which the acquisition of sensitisation is potentiated by the use of Freund’s Complete Adjuvant (FCA) and in which both intradermal and topical exposure are used during the induction phase. The Buehler test is a non-adjuvant method involving for the induction phase topical application only. Both test methods assess the elicitation phase, i.e. the adverse outcome of skin sensitisation.

Both the GPMT and the Buehler tests are able to detect substances with moderate to strong sensitisation potential, including those with relatively weak sensitisation potential. In such methods the activity is measured as a function of challenge-induced dermal hypersensitivity reactions elicited in test animals compared with controls. Since the LLNA is the preferred method for new *in vivo* testing, the use of the standard guinea pig tests to obtain new data on

the skin sensitisation potential of a substance will be acceptable only in exceptional circumstances and will require scientific justification. However, existing data of good quality that were generated before 10 May 2017, or for which the study was initiated before 10 May 2017, and derived from such tests are acceptable; and if these tests provide clear results that are adequate for classification, even when a conclusion on potency (Cat. 1A or not) cannot be drawn, they will preclude the need for further *in vivo* testing. However, existing information from other sources or non-animal test data should be used to refine the classification, if available. The CLP Regulation includes criteria for when data generated by the LLNA test (EU B.46/OECD TG 429) or GMPT/Buehler test (EU B.6/OECD TG 406) will allow potency assessment for skin sensitisers (i.e. classification in category 1A or 1B; see Section [R.7.3.5.1](#) under "Testing data on skin sensitisation"). However, due to the individually chosen test design in the guinea pig tests, it is often not possible to conclude whether the test substance is a strong/extreme (i.e. Cat. 1A) skin sensitiser.

ECETOC Monograph 29 (2000) contains a useful discussion of these tests.

- [Non-guideline compliant tests and refinements to the standard assays](#)

Existing data may be available from tests that do not have an OECD guideline, for example:

- i. other guinea pig skin sensitisation test methods (such as the Draize test, optimisation test, split adjuvant test, open epicutaneous test);
- ii. additional tests (such as the mouse ear swelling test, local lymph node cell count method (Basketter *et al.*, 2012)).

Information may also be available from other endpoints, for example, repeated dose dermal studies that show effects indicative of an allergic response, such as persistent erythema and/or oedema. In this case, care must be taken to distinguish allergenic effects from irritancy or non-immunological effects (such as non-immunological contact urticaria).

Data obtained from non-guideline compliant tests or from refinements of a standard assay need to be assessed according to Annex XI, section 1.1.2. In case these data do not alone fulfil the criteria of Annex XI, section 1.1.2, a *Weight of Evidence* approach according to Annex XI, section 1.2 needs to be applied. In addition, it should be noted that non-guideline test data as described above are not referred to in the CLP Regulation with specific classification criteria. Information from such studies can only be used as supporting evidence which would normally require expert judgement in a *Weight of Evidence* approach and will generally not be considered adequate for classification on their own.

R.7.3.4.2 Human data on skin sensitisation

Human data on cutaneous (allergic contact dermatitis and urticarial) reactions may come from a variety of sources:

- consumer experience and comments, preferably followed up by medical examinations (e.g. diagnostic patch tests);
- diagnostic clinical studies (e.g. patch tests, repeated open application tests);
- records of workers' experience, accidents, and exposure studies including medical surveillance;
- case reports in the general scientific and medical literature;

- consumer tests (monitoring by questionnaire and/or medical surveillance);
- epidemiological studies;
- existing human experimental studies such as the human repeat insult patch test (HRIPT) (Stotts, 1980; McNamee *et al.*, 2008) and the human maximisation test (HMT) (Kligman, 1966), although it should be noted that *new* experimental testing in humans for hazard identification, including HRIPT and HMT, is not acceptable.

R.7.3.5 Evaluation of available information on skin sensitisation

For hazard identification and potency assessment, it is important that the data provided are reliable and relevant. Conclusion on the usefulness of data may rely on one data point or on a *Weight-of-Evidence* approach, as described in Section [R.7.3.7.2](#) (under “How to perform and report *Weight-of-Evidence* analysis based on non-animal approaches”) and in REACH Annex XI Section 1.2. Such a *Weight-of-Evidence* approach also includes an evaluation of the available data as a whole, i.e. both over or across endpoints, starting with careful evaluation of sensitisation data. However, information obtained from skin corrosion/irritation and/or dermal toxicity test(s) may provide additional useful information for the *Weight-of-Evidence* approach. For example, skin corrosion data may enable a specific adaptation (Annex VII, column 2, section 8.3 of the REACH Regulation) and may help in distinguishing irritation effects from sensitisation.

When a non-animal testing approach, as described in the Testing and Assessment Strategy in Section [R.7.3.7](#) is followed in order to meet the information requirement, weight should be given to the validated and/or adopted *in chemico/in vitro* methods described below.

The *Weight-of-Evidence* approach provides a basis to decide whether further information is needed on endpoints for which existing data appear inadequate or data are not available, or whether the requirements are fulfilled.

In the following sections some additional remarks are made on the adequacy of the various types of data that may be available.

R.7.3.5.1 Non-human data on skin sensitisation

Non-testing data on skin sensitisation

Read-across

The use of read-across requires the use of data from one or more source substance(s) for assessing the hazardous properties of a target substance. The read-across assessment framework (RAAF) document, which is published on the ECHA website (<http://echa.europa.eu/en/support/grouping-of-substances-and-read-across>), describes ECHA’s assessment of the suitability of a read-across. Among other things, the RAAF emphasises the importance of transformation processes (metabolic or abiotic) for the potential activation of substances (in the case of sensitisation: pre- and pro-haptenation). This is only one of the factors that determine the selection of scenario for assessment; some other general questions for assessing the suitability of selected analogues are:

- is the same endpoint considered?
- are there any additional functional groups or additional substituents that might influence the reactivity and sensitising behaviour (applicability domain considerations)?

- are the physico-chemical parameters similar (e.g. LogP, applicability domain considerations)?
- are there impurities that influence the sensitisation profile?
- is the likely chemical mechanism the same?

In the case of skin sensitisation, the most robust means of comparing two or more substances is through an evaluation of their likely chemical reactivity. Work in this area has investigated means of encoding reactivity for the different chemical reaction type domains in the form of rules (Aptula and Roberts, 2006; Aptula *et al.*, 2006; Schultz *et al.*, 2009; Roberts and Aptula, 2014)⁶⁴. If the chemical reactivity is not known, or cannot be determined through experimentation, then a pragmatic means of identifying similar substances can be through a substructural/analogue search. In this context the RAAF requires on the one hand to specify why the commonalities between two or more analogue structures suggest similar biological action; on the other hand justification needs to be provided as to why structural dissimilarities are not expected to result in dissimilar biological action or quantitative differences in potency.

(Q)SAR models

When evaluating the reliability of (Q)SAR predictions, the assessment depends on both the substances of interest and the (Q)SAR model(s) used to make a prediction. General advice on (Q)SARs including an evaluation of OECD principles for QSAR validation is given in Section R.6.1.3 of Chapter R.6 of the [Guidance on IR&CSA](#). Clearly there is a wide range of different (Q)SARs and expert systems available for the estimation of skin sensitisation hazard. The approaches are quite varied and each has been developed based on different sets of *in vivo* data (principally GPMT and LLNA). Whilst efforts have been made to characterise a number of the literature-based models in terms of the OECD principles for QSAR validation (see Roberts *et al.*, 2007 as an example), further work is still required for some of the commercial systems (ECETOC, 2003). In addition, in many cases these models have been demonstrated to be reasonable for predicting skin sensitisers correctly but they are limited in predicting non-sensitisers correctly (Roberts *et al.*, 2007; ECETOC, 2003). For this reason, careful interpretation of model predictions needs to be considered in light of other pieces of information, e.g. analogue read-across (other similar substances with respect to their mechanistic domain). A good practice is to use the results from these models as building blocks for *Weight of Evidence* (e.g. the prediction from a reactivity-based model addressing the molecular initiating event of the AOP could be used the same way as the DPRA, if the prediction is considered reliable).

Further work is needed to encode more knowledge/rules for non-reactive substances as well as for those substances likely to undergo chemical or metabolic transformation.

Consideration of which model(s) to apply will be dependent on the specific substance(s) of interest, the underlying training dataset and the applicability domain of the model(s), (i.e. only predictions within the applicability domain of the models should normally be considered. Models with training sets containing analogues close to the target substance should be preferred). These issues are described more fully in Section R.6.1 of Chapter R.6 of the [Guidance on IR&CSA](#) and in the [Practical Guide 5](#) on "How to use and report (Q)SARs". An example is illustrated here: if the substance falls into a chemistry reactivity domain that is well characterised, then a local (Q)SAR model developed for this domain (such as those previously

⁶⁴ This approach might involve the systematic generation of *in vitro* reactivity data for these different mechanistic domains.

described) may give rise to the most robust prediction of skin sensitisation. Where the mechanism is not understood or not known *a priori* one or more of the expert systems such as TOPKAT, Derek Nexus or the others already described will be best candidates to provide an estimate. These systems may not be fully transparent but they often provide a reasonable amount of supporting information to enable the robustness of a prediction to be evaluated.

The prediction needs to be evaluated by taking into account the likely chemical reactivity and the presence of similar substances within the training set of the model. This type of information is needed to assess whether the prediction derived is meaningful and relevant. For global models available in the literature, the training set(s) and the algorithm(s) are usually available to allow such comparisons to be made.

The QMRF and QPRF were developed to provide templates for including specific model and prediction information. More details are provided in Section R.6.1 of Chapter R.6 of the [Guidance on IR&CSA](#).

Other information

Other information such as results in other assays, e.g. the Ames test (a common feature of genotoxic substances is that they can bind covalently to DNA and cause direct DNA damage) or aquatic toxicity tests, may provide supporting information about the electrophilicity of the substance of interest and hence its likely sensitisation ability. It is notable that *in vitro* genotoxicity assays do not always address molecular (DNA) binding. Also, abiotic transformation of the substance in an aquatic toxicity test may lead to differences in the availability of the actually active substance. Some of this work explores correlations between aquatic toxicants and skin sensitisers (Aptula *et al.*, 2006) and between experimentally identified mutagens and sensitisers (Wolfreys and Basketter 2004; Patlewicz *et al.*, 2014). More recently, the use of mutagenicity data was proposed as part of an integrated approach to testing and assessment (IATA) for skin sensitisation (Patlewicz *et al.*, 2014) (please see Section [R.7.3.7.2](#), under “How to deal with the lack of or limited metabolic capacity of non-animal test methods”).

Testing data on skin sensitisation

In chemico/in vitro data

There are several validated test methods for the assessment of skin sensitisation potential *in chemico/in vitro* and, for some of them, EU/OECD- adopted test guidelines are available (see Section [R.7.3.4.1](#)). These test methods have not been developed as stand-alone test methods, but rather as test methods to be used together with other pieces of information in a *Weight-of-Evidence* approach⁶⁵, e.g. by using several *in chemico/in vitro* methods together, as described in section 8.3.1 of Annex VII to the REACH Regulation.

Annex VII to the REACH Regulation specifies that when new data need to be generated to fulfil the standard information requirement for skin sensitisation, as a first step *in chemico/in vitro* studies assessing three key events of skin sensitisation should be performed, unless data from

⁶⁵ For fulfilling the information requirement of Annex VII, 8.3.1, it is necessary to consider the information obtained from the three key events (unless data from fewer key events already allow classification and risk assessment, as specified in Annex VII, section 8.3), in a *Weight-of-Evidence* approach, even though no formal *Weight-of-Evidence* in the meaning of Annex XI, section 1.2 needs to be submitted.

fewer key events already allows classification and risk assessment, as specified in Annex VII, section 8.3, column 2.

In case a conclusion cannot be made on whether the substance is a skin sensitiser or not and whether it can be presumed to have the potential to produce significant sensitisation in humans (Cat. 1A or not), it may be possible to conduct additional *in silico/in vitro/in chemico* test(s) to strengthen the evidence. In case a conclusion still cannot be made, an *in vivo* study, preferably the LLNA according to EU B.42 / OECD TG 429, needs to be performed.

However the REACH Regulation gives several options for adapting this standard information requirement, such as the specific rules for adaptation in column 2 of Annex VII, sections 8.3, 8.3.1 and 8.3.2 or the general rules for adaptations in Annex XI. As a consequence, data from the tests described below may be accepted to fulfil Annex VII requirement when used **in combination** with each other if the conditions of Annex VII, section 8.3 are met, as the methods described below are not stand-alone methods. In case the key event(s) described in Annex VII, section 8.3.1. are addressed by the use of methods other than *in vitro/in chemico* methods, e.g. with *in silico* methods or in combination with other pieces of evidence like read-across, a *Weight-of-Evidence* approach according to Annex XI, section 1.2 needs to be generated (see Section [R.7.3.7](#)). As *in chemico/in vitro* test methods have not been developed to be used as stand-alone methods their results must be used in combination in the context of a *Weight of Evidence*⁶⁶. In practice, the acceptability depends on whether the specific use of the methods for a given substance is within their applicability domain.

The test methods described below are not currently suitable **on their own** for predicting skin sensitising potency. Indicators of potency such as the level of peptide depletion and concentration-responses can be obtained from the existing *in chemico* and *in vitro* tests, respectively. While there is no standardised prediction model (based on a single test) or data interpretation procedure (based on multiple tests/sources of information) to integrate these potency indicators into an adequate potency assessment (Cat. 1A or not), some approaches have been proposed in the scientific literature, which can be applied on case-by-case basis. However, in case additional information is needed to conclude on skin sensitisation potency, supporting data on potency, e.g. from structurally similar substances obtained *via* the use of the OECD QSAR Toolbox, may be helpful. In case no firm conclusion on the skin sensitisation potency (Cat. 1A or not) can be drawn while there is some evidence, e.g. from peptide reactivity, that the substance may be a strong sensitiser, a precautionary Cat. 1A classification may be considered. A review of different approaches is provided in [Appendix R.7.3-4](#). At this point in time (at the time of publication of this Guidance) no firm guidance can be provided on how potency estimation should be performed (this needs to be done on a case-by-case basis). Therefore, the registrant is advised to carefully follow the recent developments in this area e.g. *via* ECHA's webpage on "[Testing methods and alternatives](#)".

It is important to note that, currently, several non-animal test methods are under development or evaluation, however their regulatory use and value have not been assessed yet. It is advised to monitor the status of current developments through the scientific literature (e.g. provide reference to the most recent scientific reviews in the area). Test methods under evaluation by EURL ECVAM or other international validation bodies can be monitored through the EURL ECVAM Test Method Submission webpage (<https://eurl-ecvam.jrc.ec.europa.eu/test-submission/>).

⁶⁶ For fulfilling the information requirement of Annex VII, 8.3.1, it is necessary to consider the information obtained from the three key events (unless data from fewer key events already allow classification and risk assessment, as specified in Annex VII, section 8.3), in a *Weight-of-Evidence* approach, even though no formal *Weight-of-Evidence* in the meaning of Annex XI, section 1.2 needs to be submitted.

- Direct Peptide Reactivity Assay (DPRA) – B.59 / OECD TG 442C

The DPRA aims to provide information on the molecular initiation event of skin sensitisation i.e. protein binding of low molecular weight substances using synthetic heptapeptides containing cysteine and lysine amino acids. In the assay, peptide reactivity is addressed by measuring the depletion of the synthetic heptapeptides by HPLC using UV detection. However, when considering this limitation, it should be kept in mind that the relative percentage of substances reacting preferably with amino acids other than cysteine and lysine is at present unclear and that the cysteine and lysine peptides represent softer to harder model nucleophiles which would cover different reaction mechanisms. More information can be obtained from the EURL ECVAM Recommendation (available at: <https://eurl-ecvam.jrc.ec.europa.eu/eurl-ecvam-recommendations/eurl-ecvam-recommendation-on-the-direct-peptide-reactivity-assay-dpra>).

The specific limitations of the test method according to the current test guideline are:

- It is only applicable to test substances that are soluble in an appropriate solvent at a final concentration of 100 mM. Substances that are not soluble at this final concentration can still be tested at lower soluble concentrations. In such a case, positive results could still be used to identify a test substance as a sensitiser whereas negative results obtained with concentrations < 100 mM should be considered inconclusive;
- Co-elution (i.e. the substance and the peptide elute at the same time) may hamper the determination of peptide reactivity, therefore appropriate co-elution controls need to be included in the test design;
- It is not applicable to the testing of metals and metal compounds (known to react with proteins *via* mechanisms other than covalent binding), and to complex mixtures of unknown composition, substances of unknown or variable composition, complex reaction products or biological materials (i.e. UVCB substances) due to the defined molar ratio of the test substance and peptide. It is only applicable to multi-constituent substances where a reasonably well defined molar ratio of the test substance and peptide can be established;
- The test system has no metabolic capacity, therefore pro-haptens (i.e. substances requiring metabolic activation to exert their sensitising activity) may produce false negative results. Pre-haptens (i.e. substances activated by abiotic transformation, e.g. auto-oxidation or hydrolysis) may produce false negative results, especially in the case of slow oxidizers. However, identification of slow oxidizers would also fail by using *in vivo* methods (Casati *et al.*, 2016);
- Test substances with exclusive reactivity towards amino-acids other than cysteine or lysine (e.g. nucleophilic sites of histidine) may lead to false negative results;
- Potential over-predictions may be due to substances that do not covalently bind to the peptide but do promote its oxidation (e.g. cysteine dimerisation).
- Even though results of this method cannot be used directly to predict skin sensitisation potency, the amount of depleted peptide and/or the chemical reactivity class obtained (i.e. low, moderate and high reactivity) may be useful to inform potency assessment when used together with other information sources. Examples of how this information has been used for potency prediction can be found in the case studies presented in the OECD Guidance document (OECD, 2016c).

- ARE-Nrf2 Luciferase Test Method (KeratinoSens™) – B.60 / OECD TG 442D

The Keap1-Nrf2-ARE pathway is considered to be a major regulator of cyto-protective responses to electrophile and oxidative stress by controlling the expression of detoxification, antioxidant and stress response enzymes and proteins. In the assay, induction of the luciferase gene is measured as an indicator of the activity of the pathway. As the majority of substances causing skin sensitisation are electrophiles reacting with nucleophilic centres in skin proteins, this pathway is relevant for skin sensitisation. However, the Keap1-Nrf2-ARE signalling pathway is not only related to keratinocytes but is also detectable in other cell types, and in addition it may also be affected by non-electrophilic modulators (e.g. corrosive/irritating substances) and may hence produce false positive responses (Richardson *et al.*, 2015). More information can be obtained from the EURL ECVAM Recommendation (available at: <https://eurl-ecvam.jrc.ec.europa.eu/eurl-ecvam-recommendations/recommendation-keratinosens-skin-sensitisation>).

The specific scope and limitations of the test method according to the current test guideline are:

- It is applicable to test substances that are soluble or that form a stable dispersion either in water or DMSO, or another appropriate solvent if its choice is scientifically justified. Test substances that do not fulfil these conditions at the highest final required concentration of 2000 µM may still be tested at lower concentrations. In such a case, positive results could be used to identify a test substance as sensitiser whereas negative results obtained with concentrations < 1000 µM should be considered inconclusive;
- The test system using the human keratinocyte cell line HaCaT has a limited metabolic capacity, therefore pro-haptens (i.e. substances requiring metabolic activation to exert their sensitising activity) may produce false negative results. Pre-haptens (i.e. substances activated by abiotic transformation, e.g. auto-oxidation or hydrolysis) may also produce false negative results, especially in case of slow oxidizers. However, identification of slow oxidizers would also fail by using *in vivo* methods (Casati *et al.*, 2016);
- Test substances with exclusive reactivity towards nucleophiles other than cysteine's sulfhydryl group (e.g. lysine residues) can produce false negative results in the assay;
- Test substances that do not act as sensitisers but are nevertheless chemical stressors may produce false positive results;
- Highly cytotoxic substances cannot always be reliably assessed;
- Test substances that interfere with the luciferase enzyme can affect its activity by either increasing or inhibiting the luminescence.
- Even though results of this method cannot be used directly to predict skin sensitisation potency, the concentration-response information may be useful to inform potency assessment when used together with other information sources. Examples of how this information has been used for potency prediction can be found in the case studies presented in the OECD Guidance document (OECD, 2016c).

- Human Cell Line Activation Test (h-CLAT) - OECD TG 442E

The h-CLAT assay aims to provide information on dendritic cell (DC) activation by using a human monocytic leukemia cell line (THP-1) as an alternative model to DCs. The DC activation

is measured by analysing the expression (upregulation) of specific cell surface markers known to be linked to DC maturation, i.e. CD86 and CD54, by using flow cytometry. Monocytic human THP1 cells used in this assay may give different signals of the same cellular molecules after stimulation with a specific substance compared to human dendritic cells (Lehtonen *et al.*, 2007). More information can be obtained from the EURL ECVAM Recommendation (available at: <https://eurl-ecvam.jrc.ec.europa.eu/eurl-ecvam-recommendations/eurl-ecvam-recommendation-on-the-human-cell-line-activation-test-h-clat-for-skin-sensitisation-testing>).

The specific scope and limitations of the test method according to the current draft test guideline are:

- It is applicable to test substances that are soluble or form a stable dispersion in an appropriate solvent;
- Test substances with $\text{Log Kow} \leq 3.5$ can be tested whereas substances with $\text{Log Kow} > 3.5$ tend to produce a higher rate of false negative results. For such substances with $\text{Log Kow} > 3.5$ positive results could be used to support the identification of a test substance as a sensitizer. Negative results for substances with $\text{Log Kow} > 3.5$ should not be considered. .
- The test system has a limited metabolic capacity, therefore pro-haptens (i.e. substances requiring metabolic activation to exert their sensitising activity) may produce false negative results. Pre-haptens (i.e. substances activated by abiotic transformation e.g. auto-oxidation or hydrolysis) may also produce false negative results, especially in case of slow oxidizers. However, identification of slow oxidizers would also fail by using *in vivo* methods (Casati *et al.*, 2016);
- Highly cytotoxic substances cannot always be reliably assessed;
- Since it uses a fluorescein isothiocyanate (FITC)-labelled antibody and propidium iodide (PI), strong fluorescent test substances emitting at the same wavelength as FITC may interfere with the flow cytometry light-signal acquisition. In such a case, other fluorochrome-tagged antibodies or other cytotoxicity markers, respectively, can be used as long as it can be shown that they provide results similar to those obtained with the FITC-tagged antibodies or PI, e.g. by testing the proficiency substances in Annex II to the test guideline.
- Even though results of this methods cannot be used directly to predict skin sensitisation potency, the concentration-response information may be useful to inform potency assessment when used together with other information sources. Examples of how this information has been used for potency prediction can be found in the case studies presented in the OECD Guidance document (OECD, 2016c).

Concerning the *in chemico/in vitro* test methods, any modification made to the adopted test guidelines is not recommended and should only be done in exceptional circumstances and needs to be properly documented and scientifically justified and shown to yield comparable results using the proficiency substances listed in the EU/OECD test guideline. The reporting template in Annex II of the OECD Guidance Document On The Reporting Of Defined Approaches To Be Used Within Integrated Approaches to Testing And Assessment (OECD, 2016a) can be used for this purpose and also to document *in chemico/in vitro* methods for which no adopted test guideline is available (see [Appendix R.7.3-2](#)). Proper documentation and justification is needed when the information submitted has been generated by using test methods that have not been formally validated and do not have adopted test guidelines.

Animal data

Well reported studies using internationally acceptable protocols, particularly if conducted in accordance with the principles of GLP, can be used for hazard identification. Other studies (see Section [R.7.3.4.1](#) and below), not fully equivalent to OECD test protocols, can, in some circumstances, provide useful information. Particular attention should be paid to the quality of these tests and the use of appropriate positive and negative controls. The specificity and sensitivity of all animal tests should be monitored through the inclusion of appropriate positive and negative controls. In this context, positive controls are the 6-monthly sensitivity checks with an appropriate positive control substance, and negative controls are the vehicle-treated control animals included as part of each test.

- [Guideline-compliant tests](#)

Murine Local Lymph Node Assay

For the conduct and interpretation of the LLNA the following points should be considered:

- i. the vehicle in which the test material and controls have been applied;
- ii. the concentrations of test material that have been used;
- iii. any evidence for local or systemic toxicity, or skin inflammation resulting from application of the test material;
- iv. whether the data are consistent with a biological dose-response;
- v. the submitting laboratory should be able to demonstrate its competency to conduct the LLNA.

EU B.42/OECD TG 429 provides guidance on the recommended vehicles, number of animals per group, concentrations of test substance to be applied and substances to be used as positive control. A preliminary study or evaluation of existing acute toxicity/dermal irritation data is normally conducted to determine the highest concentration of test substance that is soluble in the vehicle but does not cause unacceptable local or systemic toxicity. The submission of historical control data will demonstrate the ability of the test laboratory to produce consistent responses. Based on the incorporation of radioactive labelling (tritiated (³H)-methyl thymidine), substances that result in a stimulation index (SI) ≥ 3 at one or more test concentrations are considered to be positive for skin sensitisation. Both positive and negative responses in the LLNA conducted as described in EU B.42/OECD TG 429 meet the data requirements for classification of a substance as a skin sensitiser including potency estimations: no further testing is required.

Alternative vehicles to those listed in EU B.42/OECD TG 429 may be used in the LLNA if sufficient scientific justification is provided. There can be some variability due to the choice of vehicle (vehicle effect) that may enhance or suppress the response in the LLNA (see Section [R.7.3.6.1](#)).

The LLNA: DA test method described in EU B.50/OECD TG 442A measures ATP content by luminescence in the proliferating cells and hence does not require the use of radioactive labelling of cells. Substances that result in $SI \geq 1.8$ at one or more testing concentration(s) are considered to be positive for skin sensitisation. In case of borderline positive results ($1.8 \leq SI \leq 2.5$), linked to the sensitivity of the detection method, additional information may be considered such as the dose-response relationship, evidence of systemic toxicity or excessive irritation, and, where appropriate, statistical significance together with SI values to confirm that such results are indeed positives. Currently, there are no CLP criteria available for

predicting skin sensitisation potency with this test method, even though dose-response information obtained from the assay may provide information on skin sensitisation potency that can be used in a *Weight-of-Evidence* approach. It is recommended that when new *in vivo* data need to be generated the "standard" LLNA according to EU B.42/OECD TG 429 be used, if possible.

The LLNA: BrdU-ELISA test method described in EU B.51/OECD TG 442B uses the non-radiolabelled marker 5-bromo-2-deoxyuridine (BrdU) to measure lymphocyte proliferation. Substances that result in $SI \geq 1.6$ at one or more testing concentration(s) are considered to be positive for skin sensitisation. In case of borderline positive results ($1.6 \leq SI \leq 1.9$), linked to the sensitivity of the detection method, additional information may be considered such as the dose-response relationship, evidence of systemic toxicity or excessive irritation, and, where appropriate, statistical significance together with SI values to confirm that such results are indeed positives. Currently, there are no CLP criteria available for predicting skin sensitisation potency with this test method, even though dose-response information obtained from the assay may provide information on skin sensitisation potency that can be used in a *Weight-of-Evidence* approach. It is recommended that when new *in vivo* data need to be generated the "standard" LLNA according to EU B.42/OECD TG 429 be used, if possible. The EU B.50/OECD TG 442A (LLNA: DA) and EU B.51/OECD TG 442B (LLNA: BrdU-ELISA) recommend the use of the same vehicles as in the standard LLNA EU B.42/OECD TG 429.

Limitations of all the above LLNA variants include the following:

- False negative predictions can be obtained with certain metals (e.g. nickel, Schmidt and Goebler, 2015) and false positive predictions may be obtained with certain surfactant type substances (Kreiling *et al.*, 2008; Garcia *et al.*, 2010; Ball *et al.*, 2011) or siloxanes (Petry *et al.*, 2012).
- Low solubility of the substance may interfere with the accuracy of the predictions.
- The choice of vehicle may affect the prediction for certain substances. For instance DMSO as a polar solvent may enhance dermal bioavailability of some test substances and propylene glycol may suppress the proliferative effects of some test substances (e.g. DNCB) (Anderson *et al.*, 2011). Therefore, it is important to properly select the vehicle used in the study.

The updated OECD TG 429 of 2010 includes the reduced LLNA (rLLNA), in which only one concentration is tested and less animals are used. It is recommended to use this refinement method only in case a confirmation of a negative result obtained with another testing method is required. Since only one dose is used in the study design, the rLLNA cannot currently be used for estimating the skin sensitisation potency of a substance (Ezendam *et al.*, 2013), even though a proposal has recently been published for predicting potency from a single dose (Roberts, 2015). The TGs for the LLNA variants, i.e. DA and BrdU-ELISA test methods, do not include the use of the rLLNA study design.

Guinea pig studies

New guinea pig studies should only be conducted in exceptional circumstances. In such cases a justification for using a test method other than the LLNA must be provided in the IUCLID dossier (Annex VII, section 8.3, column 2 to the REACH Regulation).

The guinea pig test method described in EU B.6/OECD TG 406, the GPMT (Magnusson *et al.*, 1969; Schleder and Eppler, 1995) and the Buehler test do also provide suitable information for hazard identification. Recommendations on conducting and analysing these methods are provided by Steiling *et al.* (2001). Particular attention should be paid to the quality of these tests with consideration given to the following points:

- i. numbers of test and control guinea pigs;
- ii. number or percentage of test and control animals displaying skin reactions;
- iii. whether skin irritation was observed at the induction phase;
- iv. whether the maximal non-irritating concentration was used in the challenge phase;
- v. the choice of an appropriate vehicle (ideally, one that solubilises or gives a stable suspension or emulsion of the test material, is free of allergenic potential, is non-irritating, enhances delivery across the stratum corneum, and is relevant to the usage conditions of the test material, although it is recognised that it will not always be possible to meet all these conditions);
- vi. whether there are signs of systemic toxicity (a sighting study should be performed to determine an appropriate induction dose that causes irritation but not systemic toxicity);
- vii. staining of the skin by the test material that may obscure any skin reactions (other procedures, such as chemical depilation of the reaction site, histopathological examination or the measurement of skin fold thickness may be carried out in such cases);
- viii. results of rechallenge treatments if performed;
- ix. checking of strain sensitivity at regular intervals by using an appropriate control substance (as specified in OECD guidelines and EU Test Methods). Currently (at the time of publication of this Guidance), the recommended interval is 6 months.

The investigation of doubtful reactions in guinea pig tests, particularly those associated with evidence of skin irritation following a first challenge, may benefit from re-challenge of the test animals. In cases where reactions may have been masked by staining of the skin, other reliable procedures may be used to assist with interpretation; where such methods are used, the submitting laboratory should provide evidence of their value.

A justification for performing a new guinea pig test instead of an LLNA could be for example that the test substance contains nickel, as it is known that nickel is not correctly predicted in the LLNA.

There are criteria available for predicting skin sensitisation potency based on guinea pig tests. However, due to the individually chosen test design, it is often not possible to conclude whether the test substance is a strong/extreme (i.e. Cat. 1A) skin sensitiser. Nevertheless, in case such information is available, it may still be valuable in a *Weight-of-Evidence* assessment that may lead to a determination of whether the skin sensitising substance can be presumed to have the potential to produce significant sensitisation (Cat. 1A or not) in humans.

- [Non-guideline compliant tests and refinements to the standard assays](#)

The submitted dossier should include scientific justification for conducting any new test that is a modification or deviation from guideline methods. In such cases, it would be advisable to seek appropriate expert advice on the suitability of the assay before testing is begun.

Historically, guinea pig studies that are not fully equivalent to OECD test protocols have been conducted and can provide useful hazard information. These studies include, but are not limited to, the following: Draize test, optimisation test, split adjuvant test, open epicutaneous test and the cumulative contact enhancement test. In the case of positive results the

substance may be considered as a potential skin sensitiser. If, taking into account the above quality criteria, especially the positive and negative control data, there is a clear negative result, i.e. no animals displaying any signs of sensitisation reactions, then no further animal testing is required. Where there is a low level of response, the quality of the study is questionable, or where unacceptably low concentrations of the test material have been used for induction and/or challenge, further testing may be required. In addition, existing information may already be available e.g. from the mouse ear swelling test (MEST), which is a modification of the LLNA. The MEST has been evaluated in inter-laboratory studies, and it was concluded that the MEST could be used for identifying strong skin sensitisers (Dunn *et al.*, 1990; EC, 2004).

R.7.3.5.2 Human data on skin sensitisation

When reliable and relevant human data are available, they can be useful for hazard identification and even preferable over animal data. However, a lack of positive findings in humans does not necessarily overrule positive and good quality animal data.

Well conducted human studies can provide very valuable information on skin sensitisation. However, in some instances (due to lack of information on exposure, a small number of subjects, concomitant exposure to other substances, local or regional differences in patient referral, etc.) there may be a significant level of uncertainty associated with human data. Moreover, diagnostic tests are carried out to see if an individual is sensitised to a specific agent, and not to determine whether the agent can cause sensitisation. Evidence of skin sensitising activity derived from diagnostic testing may reflect the induction of skin sensitisation to the substance tested or cross-reaction with a chemically very similar substance. In both situations, the normal conclusion would be that this provides positive evidence of the skin sensitising activity of the substance used in the diagnostic test.

For evaluation purposes, existing human experience data for skin sensitisation should contain sufficient information about:

- the test protocol used (study design, controls);
- the substance or preparation studied (should be the main, and ideally, the only substance or preparation present which may possess the hazard under investigation);
- the extent of exposure (dose per square centimeter or concentration, frequency and duration);
- the frequency of effects (versus number of persons exposed);
- the persistence or absence of health effects (objective description and evaluation);
- the presence of interfering factors (e.g. pre-existing dermal health effects, medication, presence of other skin sensitisers);
- the relevance with respect to the group size, statistics, documentation;
- the *healthy worker* effect⁶⁷.

⁶⁷ Phenomenon observed initially in studies of occupational diseases: **workers** usually exhibit lower overall death rates than the general population because severely ill and disabled people are excluded from employment.

Human experimental studies on skin sensitisation are not normally conducted and should be avoided. Where human data are available, quality criteria and ethical considerations as presented in ECETOC monograph no 32 (ECETOC, 2002) should be taken into account.

Ultimately, where a very large number of individuals (e.g. 10^5) have frequent (daily) skin exposure for at least two years and there is an active system in place to pick up complaints and adverse reaction reports (including via dermatology clinics), and where no or only a very few isolated cases of allergic contact dermatitis are observed, then the substance is unlikely to be a significant skin sensitiser. However, information from other sources should also be considered in making a judgement on the substance's ability to induce skin sensitisation.

It is emphasised that testing with human volunteers is strongly discouraged, but when there are good quality data already available they should be used as appropriate in well justified cases.

R.7.3.6 Conclusions on skin sensitisation

R.7.3.6.1 Remaining uncertainty on skin sensitisation

Data that comply with REACH standard information requirements can be generated from well designed and well conducted non-animal and animal studies. However, it should be noted that no toxicological test is perfect and each test method has to balance between the sensitivity (rate of true positives) and specificity (rate of true negatives) of the prediction of the effect in the human population. The use of adjuvant in the GPMT may lower the threshold for irritation and so lead to false positive reactions, which can therefore complicate interpretation (running a pre-test with FCA treated animals can provide helpful information). In international trials, the LLNA has been shown to predict >80% of skin sensitisers when compared to human data but, like the guinea pig tests, it is dependent on the vehicle used. Variability due to the choice of vehicle (vehicle effect) may enhance or suppress the response in the LLNA by one order of magnitude (Anderson *et al.*, 2011; Hoffmann, 2015; Wright *et al.* 2010), therefore there may be some inherent uncertainty associated to the vehicle selection. It has been claimed that the LLNA may create false positives for (irritating) surfactants (non-specific lymphocyte proliferation) (Garcia *et al.*, 2010; Kreiling *et al.*, 2008). However, Basketter and Kimber (2011) state that, if the study is performed according to the dose selection criterion as specified in the OECD TG 429, no false positives results should be obtained based only on the irritating properties of the substance. Careful consideration should be given to circumstances where exposure may be sub-optimal due to difficulties in achieving a good solution and/or a solution of sufficient concentration. In some circumstances inconsistent results from guinea pig studies, or between guinea pig and LLNA studies, might increase the uncertainty of making a correct interpretation. Finally, for existing human data consideration must be given to whether inter-individual variability is such that it is not scientifically sound to generalise from a limited population.

The non-animal test methods (*in chemico/in vitro*) currently available have no or limited metabolic capacity (Fabian *et al.*, 2013). Therefore substances requiring metabolic activation before becoming sensitisers may not be correctly identified by such test methods. Also, some substances requiring abiotic transformation (e.g. auto-oxidation or hydrolysis) before becoming active may not be detected, however this issue is also applicable also to the animal test methods. More information on these limitations can be found in Section [R.7.3.7](#) of this Guidance. QSAR models also most often do not account for metabolism/abiotic transformation (e.g. auto-oxidation or hydrolysis) by themselves, or only do this implicitly by using as model training set both substances that do not require metabolic activation and substances that do require such metabolic activation. There are strategies, however, which can facilitate the

consideration of metabolic information (e.g. see Section [R.7.3.7.2](#) under “How to deal with the lack of or limited metabolic capacity of non-animal test methods”).

R.7.3.6.2 Concluding on suitability for Classification and Labelling

In order to conclude on an appropriate classification and labelling position with regard to skin sensitisation, the available data should be considered using the criteria according to Annex I to the CLP Regulation (EC) No 1272/2008. The CLP Regulation specifies that skin sensitisers should be allocated into sub-categories (i.e. 1A or 1B) whenever possible. In case the data are not sufficient for sub-categorisation, the substance must be classified in the general Category 1 (for further information, see Section 3.4 of the [Guidance on the Application of the CLP criteria](#)).

R.7.3.6.3 Concluding on suitability for chemical safety assessment: potency and dose-response assessment

Measurement of potency

According to section 8.3 of Annex VII to the REACH Regulation, in addition to the assessment of whether the substance is a skin sensitizer or a non-sensitizer, the potency of skin sensitizing substances must be assessed in order to be able to conclude whether it can be presumed to produce significant sensitisation in humans (Cat. 1A or not). If a decision on classification (Cat. 1A or not) and risk assessment can be made, no further testing is required. *In vivo* study(ies) carried out or initiated before 10 May 2017 and of good quality (i.e. guideline compliant and performed under GLP) are considered to fulfil the REACH information requirement even if they don't allow an assessment of skin sensitisation potency (Cat. 1A or not). However, in such cases existing information from other sources, e.g. read-across and/or QSARs, should be used to refine classification and risk assessment. Furthermore, generation of new non-animal test data can also be used to refine classification and risk assessment. Appropriate dose-response data can provide important information on the potency of the material being tested. This can facilitate the development of more accurate risk assessments. This section refers to potency in the induction phase of sensitisation.

Neither the standard LLNA nor the GPMT/Buehler test is specifically designed to evaluate the skin sensitising potency of test substances. Instead they are used to identify the sensitisation potential for classification purposes. However, these tests can all be used to estimate potency to varying degrees.

The relative potency of substances may be indicated by the percentage of positive animals in the guinea pig studies in relation to the intradermal or topical induction concentration(s) tested. Likewise, in the LLNA, the EC3 value (the dose estimated to cause a 3-fold increase in local lymph node proliferative activity) is used as a measure of potency (see the CLP Regulation, tables 3.4.3 and 3.4.4, and the [Guidance on the Application of the CLP criteria](#), Table 3.4.2.f). Often, linear interpolation of a critical effect dose from the EC3 is proposed (ECETOC, 2000), but more advanced statistical approaches basing conclusions on the characteristics of the dose-response curve and variability of the results is also used (Basketter *et al.*, 1999; van Och *et al.*, 2000). The dose-response data generated by the LLNA makes this test more informative than guinea pig assays for the assessment of skin sensitising potency.

EC3 data correlate quite well with HRIPT thresholds derived from historical testing data (Griem, 2003; Schneider and Akkan, 2004; Basketter *et al.*, 2005b; ICCVAM, 2011; Basketter and McFadden, 2012). However, the human data were not derived using a single well-defined protocol and thereby some uncertainty is associated with these comparisons. Furthermore, the thresholds derived cannot be applied directly to the general population. They must be subjected to a rigorous risk assessment process, including the application of several safety assessment factors (Api *et al.*, 2008; Basketter and Safford, 2015a). However a retrospective analysis performed by ICCVAM (2011) of LLNA data compared to human and/or guinea pig

data revealed that for 27 strong sensitising substances analysed, approximately half of them were underclassified based on an EC3 cut-off value of <2%. In the CLP Regulation there are criteria for determining potency based on animal data (both LLNA and GPMT/Buehler tests) and human data.

In the case of the GPMT and Buehler test, due to the dose selection criteria specified in the OECD TG 406, it is usually not possible to make a firm conclusion that a substance is a Category 1B sensitiser since classification in Category 1A cannot be excluded. Therefore, in case classification in Category 1A cannot be excluded, the general Category 1 classification must be chosen.

Several approaches for potency prediction by using non-animal approaches have been proposed in the scientific literature, and some of these could be useful, on a case-by-case basis, to support identification of strong sensitisers and setting of SCLs. A review of these approaches is given in [Appendix R.7.3-4](#). However, concerning classification and setting SCLs according to the CLP Regulation, currently (at the time of publication of this Guidance), no CLP criteria are available to classify based on *in vitro* data only and no widely accepted approach and data interpretation procedure based on non-animal data (*in chemico*, *in vitro*) is available. Combining the information obtained from *in chemico/in vitro* methods with information available from similar substances in a *Weight-of-Evidence* approach may still help in drawing a conclusion on skin sensitisation potency (Cat. 1A or not).

A lack of potency information and subsequent possibility to sub-categorise and to set SCLs, may result in a lower level of protection of humans. This is an important consideration, especially if the substance is used in a mixture and appropriate concentration limits are not used, leading to incorrect labelling of the mixture. Therefore, in order to fulfil REACH information requirements for the substance, as laid down in Annex VII, section 8.3, data from non-animal test methods (*in chemico/in vitro/in silico*) must allow conclusions on whether the substance is a skin sensitiser or a non-sensitiser and whether it can be presumed to have the potential to produce significant sensitisation in humans (Cat. 1A or not). A review of different approaches for assessing skin sensitisation potency is provided in [Appendix R.7.3-4](#). However, as work is still ongoing to address the lack of potency characterisation based on non-animal approaches, the registrant is advised to follow-up the recent and future developments in the field, e.g. *via* the ECHA website on testing methods and alternatives.

Derivation of a DNEL

Even though EC3 values obtained from the LLNA (B.42/OECD 429) (Basketter *et al.*, 2007) can be used for DNEL derivation, the first step should always be to perform a qualitative approach to assess and control the risks that may arise from exposure to a substance causing skin sensitisation. It should be noted that, currently, a quantitative assessment cannot be performed by using guinea pig, rLLNA, LLNA-DA, LLNA-BrdU-ELISA data or non-animal testing approaches. Guidance on how to use potency information for a qualitative assessment (see also Section E.3.4.2 of *Part E* of the [Guidance on IR&CSA](#)) and how to derive a DNEL as a second step in the safety assessment of sensitisers is given in Section R.8.6 and Appendix R.8-10 of *Chapter R.8* of the [Guidance on IR&CSA](#).

Quantitative risk assessment (QRA) approaches have been proposed for the identification of safe consumer exposure levels for skin sensitising substances. A QRA approach should use all the information available, i.e. human and animal data. Such an approach has been used e.g. by the fragrance industry (Api *et al.*, 2008; 2015). However, this approach has received criticism, especially when evaluated by the Scientific Committee on Consumer Safety (SCCS), as the safe levels identified by using QRA were not supported by existing data (SCCS, 2015a). Also the QRA methodology assessed by the SCCS at that time did not take e.g. aggregated and occupational exposures into account. The SCCS recommends further development of the

approach (SCCS, 2015b). In the framework of the IDEA project (<http://www.ideaproject.info>), the original QRA methodology (Api *et al.*, 2008) has been revised (Api *et al.*, 2015). These revisions include a revision of the safety assessment factors and the introduction of a probabilistic approach for aggregate exposure assessment. It should be noted that, despite its name, a QRA approach is not a precision tool.

Deriving the safe use levels for skin sensitisation can be problematic and may be associated with considerable uncertainty. Uncertainty assessment approaches have been published, e.g. by ECHA (see Chapter R.19 of the [Guidance on IR&CSA](#)) and IPCS/WHO (2014), and a draft guidance document is also available from EFSA (2015).

R.7.3.6.4 Additional considerations

Chemical allergy is commonly designated as being associated with skin sensitisation (ACD) and/or with sensitisation of the respiratory tract (asthma rhinitis and extrinsic allergic alveolitis). In view of this it is sometimes assumed that allergic sensitisation of the respiratory tract will result only from inhalation exposure to the causative substance, and that skin sensitisation necessarily results only from dermal exposure. This is misleading, and it is important for the purposes of risk management to acknowledge that sensitisation may be acquired by other routes of exposure. Since adaptive immune responses are essentially systemic in nature, sensitisation of skin surfaces may develop from encounter with contact allergens via routes of exposure other than dermal contact. Similarly, there is evidence from both experimental and human studies which indicate that effective sensitisation of the respiratory tract can result from dermal contact with a chemical respiratory allergen (Redlich, 2010; Kimber *et al.*, 2014c). Thus, in this case, it appears that the immune response necessary for the acquisition of sensitisation of the respiratory tract can be obtained *via* skin contact with chemical respiratory allergens (Arts and Kuper 2007; Kimber *et al.*, 2002). Such considerations have important implications for risk management. Thus, for instance, there is a growing view that effective prevention of respiratory sensitisation requires protection of both skin and respiratory tracts. This includes the cautious use of known contact allergens in products to which consumers are (or may be) exposed via inhalation, such as sprays. The generic advice for appropriate strategies to minimise the risk of sensitisation to chemical allergens is to provide protection of all relevant routes of exposure.

R.7.3.6.5 Information not adequate

A *Weight-of-Evidence* approach, comparing available adequate information with the tonnage-triggered REACH information requirements, may result in the conclusion that the requirements are not fulfilled. In order to proceed in further information gathering the testing and assessment strategy given in the next Section [R.7.3.7](#) can be adopted.

R.7.3.7 Testing and assessment strategy for skin sensitisation

R.7.3.7.1 Objective / General principles

The following testing and assessment strategy is recommended for developing adequate and scientifically sound data for the assessment and classification of the skin sensitisation properties of a substance. For existing substances with insufficient data, this strategy can also be used to decide which additional data, besides those already available, are needed. The objective is to collect all available information (including data from test methods and non-testing approaches) in order to assess the risk for skin sensitisation and/or to identify information gaps to be covered by generation of new information. The key principle of this strategy is that all available information is evaluated before another study is initiated. The strategy seeks to ensure that the data requirements are met in the most efficient and humane manner so that animal usage and costs are minimised.

The animal welfare considerations and the information requirements of the REACH Regulation have stimulated research on integrated strategies for skin sensitisation in the past few years. Some of these works give data on individual non-testing or non-animal testing methods e.g. *in silico* (Teubner *et al.*, 2013) or *in vitro* information (Martin *et al.*, 2010; Maxwell *et al.*, 2014; Reisinger *et al.*, 2015; Urbisch *et al.*, 2015), while others also make use of different combinations (Basketter *et al.*, 2013; Rorije *et al.*, 2013; Jaworska *et al.*, 2013). Besides the approaches mentioned above, a number of other data integration approaches, included as case studies, are documented in the OECD Guidance Document Annex I: Case Studies To The Guidance Document On The Reporting Of Defined Approaches and Individual Information Sources To Be Used Within Integrated Approaches To Testing And Assessment (IATA) For Skin Sensitisation (OECD, 2016c).

The testing and assessment strategy presented here comprises three parts (see [Figure R.7.3-1](#) and [Figure R.7.3-2](#)): Part 1 (elements 1 to 5) is about retrieving existing information, Part 2 (element 6) represents *Weight-of-Evidence* analysis and expert judgement, in case a conclusion cannot be reached based on a single element listed in Part 1. Part 3 (elements 7 to 9) is about generation of new information by testing, if needed. The elements presented in [Figure R.7.3-2](#) can be rearranged as appropriate, depending on the information needs to conclude on the substance's potential to cause skin sensitisation. This may be particularly helpful in cases where a conclusion can be drawn from certain elements without having to consider all of them.

The specific rules for adaptation of standard information requirements for skin sensitisation are described in column 2 of Annex VII to the REACH Regulation, whereas the general rules for adaptation from standard information requirements are given in Annex XI.

The new elements in the strategy are the recently EU/OECD- adopted and/or internationally validated *in chemico/in vitro* test methods for skin sensitisation (in particular the three test methods specified in element 5b and elements 7a to 7c in [Figure R.7.3-2](#)). These methods represent the key events that have been incorporated into the REACH Regulation as a standard information requirement as a first step when new information needs to be generated. These methods are based on the mechanistic understanding of the biological key events of skin sensitisation, initiated by the covalent binding of the substance to skin proteins. These key events have been codified in an Adverse Outcome Pathway (AOP) for skin sensitisation approved by the OECD (OECD, 2012). Three of these key events, i.e. peptide/protein reactivity, keratinocyte response and dendritic cell response, correspond to elements 5b (existing data), and to elements 7a, 7b and 7c (generation of new data) of [Figure R.7.3-2](#) below.

The **strategy** aims to help the registrant to find out how these *in chemico/in vitro* test methods for skin sensitisation can be used according to Annex VII, section 8.3.1, or in a *Weight-of-Evidence* approach according to the Annex XI, sections 1.2 – 1.5, to the REACH Regulation to enable hazard identification and appropriate classification decision, and risk assessment (where required) for a substance. Also other types of data, such as (Q)SAR, read-across and human data should be used in combination with the *in chemico/in vitro* test results. The key strengths and limitations of the *in chemico/in vitro* tests and other types of data are addressed below.

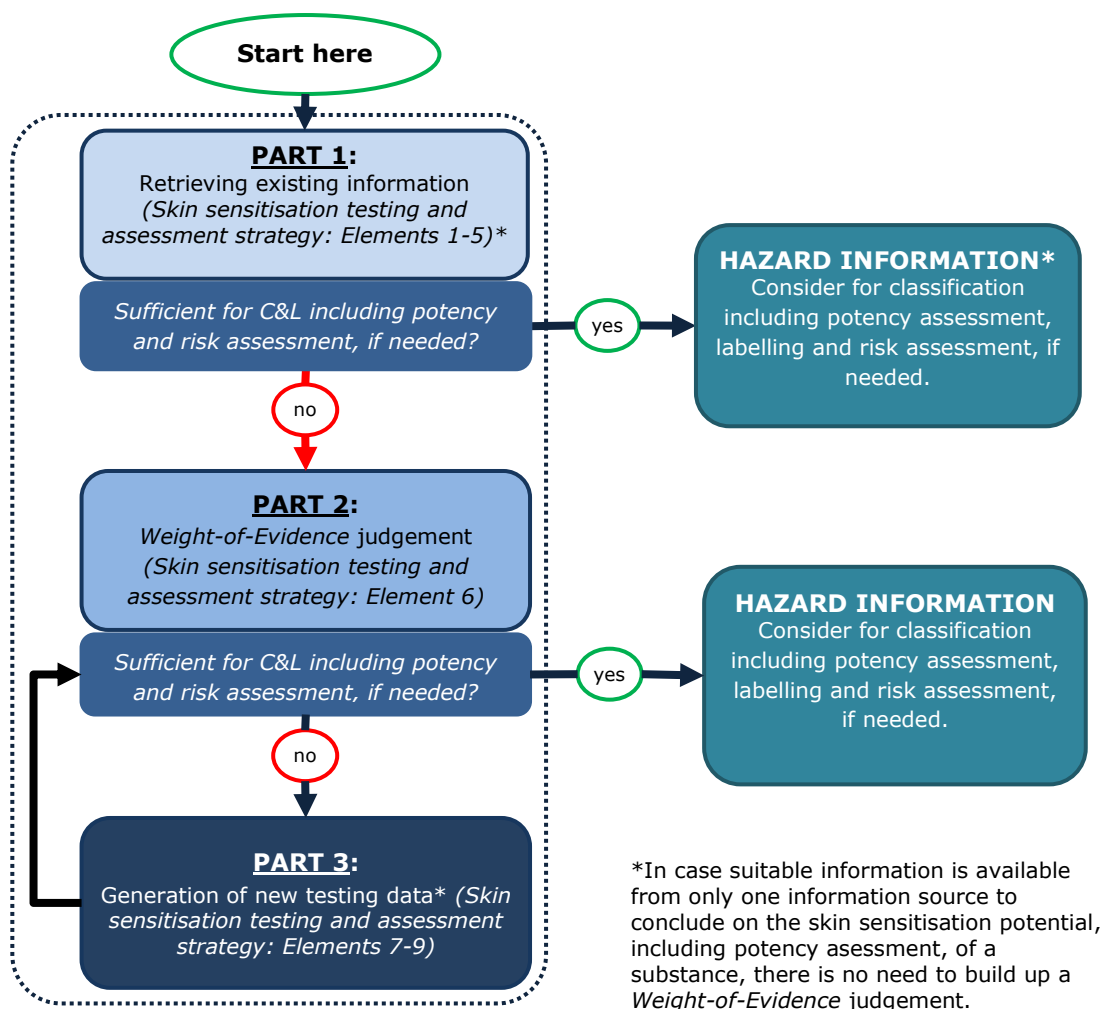


Figure R.7.3–1 Overview of the testing and assessment strategy for skin sensitisation

R.7.3.7.2 Application of the Testing and Assessment Strategy

Due to the recent developments in the field of non-animal test methods for skin sensitisation, and in line with Article 13(1) and section 8.3 of Annex VII to the REACH Regulation, registrants must investigate whether the information requirement for skin sensitisation can be fulfilled by using (existing) results from the non-animal test methods e.g. in a *Weight-of-Evidence* approach. It may be necessary to perform the *in vivo* test in case the non-animal test(s) results remain inconclusive, are documented as not applicable to the substance, or are not suitable for classification and risk assessment, as specified in Annex VII, section 8.3.

It is important to note that it is the responsibility of the registrant to ensure that the chosen test method (e.g. *in vitro*, *in chemico* or *in silico*) is suitable for testing the substance and obtaining adequate information. So before performing a specific non-animal test the registrant should consider whether there are substance-specific limitations that may hinder the performance of the test (e.g. low solubility or log Kow, UVCB nature of the substance for instance the DPRA is not applicable to UVCBs). There may also be some limitations of the test system like the absence of or limited metabolic capacity and hence pro-haptens may not be correctly detected and may give false negative results. Also substances requiring abiotic transformation, e.g. auto-oxidation or hydrolysis, (pre-haptens) may not be correctly

identified. In case the substance does not fall into the applicability domain of the non-animal test methods, an *in vivo* test (i.e. an LLNA) would need to be performed.

According to Step 1 of Annex VI to the REACH Regulation, all existing available test data should be gathered before any new testing is initiated. In Part 1 (elements 1 to 5c) of this strategy, existing and available information from the literature and databases is gathered and considered. The order of the different elements of Part 1 is only indicative and they may be arranged as appropriate. This may especially be helpful in cases where a reliable conclusion can be drawn from certain element(s) without having to consider all of them. For instance, if there are adequate human data (element 2) available that indicate that the substance should be classified as skin sensitiser according to the CLP Regulation, including a determination of whether the substance can be presumed to have the potential to produce significant sensitisation in humans (Cat. 1A or not), further testing is not required. At the end of Part 1, and if no final conclusion can be derived directly from one or several of the available pieces of information, all the information collected should be analysed using a *Weight-of-Evidence* approach (element 6).

In the information generation part (elements 7 to 9), new information on the skin sensitisation potential of the substance is produced by means of non-animal test methods or, as a last resort *in vivo* testing according to Annex VII, section 8.3.2 (element 9). The properties of the substance and existing information determine the need to generate new information, i.e. new data may not need to be generated for all elements under the information generation part as the order of the elements is only indicative.

While it is recommended that this approach be followed, other approaches may be more appropriate and efficient on a case-by-case basis.

Due to the complexity of the skin sensitisation endpoint, a combination of alternative test methods (e.g. *in silico*, *in chemico* and *in vitro*) in a *Weight-of-Evidence* approach needs to be considered to increase confidence in the final assessment of skin sensitisation, e.g. a combination of read-across and non-animal test methods can be useful in concluding on the assessment of skin sensitisation. The *in vitro* and *in chemico* test methods described in Sections [R.7.3.4.1](#) and [R.7.3.5.1](#) and in [Figure R.7.3-2](#) below (as elements 5 and 7) have not been developed as stand-alone methods, especially when negative results are obtained. The results obtained with *in silico* methods that aim at predicting the final endpoint (e.g. LLNA outcome, including EC3-value) could be used according to the REACH Regulation, if they fulfil the Annex XI, section 1.3 requirements. However, additional evidence such as read-across from analogues or results method(s) may be needed to confirm the reliability of the (Q)SAR prediction, which would otherwise be difficult to assess and accept.

In case no information on skin sensitisation is available for a substance it is recommended to start the assessment by using the OECD QSAR Toolbox (see Section [R.7.3.4.1](#)). The Toolbox can be used for many purposes. First, it facilitates the identification of existing *in chemico*, *in vitro* and *in vivo* data already available for the substance of interest. Second, it identifies skin sensitisation specific alerts and protein-binding alerts using profilers. Third, it can be used to predict and characterise metabolic or abiotic transformation (e.g. auto-oxidation or hydrolysis) products of the substance. Fourth, it facilitates the identification of analogues with experimental data for read-across, trend analysis and QSAR model building. In addition, the existing *in vivo* data for the substance and/or analogue substance(s) may provide useful information on the skin sensitisation potency, e.g. via EC3 values obtained from the existing LLNA studies. Note that the predictions can address the *in vivo* endpoints as well as *in vitro* ones (although for the moment there are not many *in vitro* data included in the Toolbox and the identification of analogues with data can be difficult).

In case all the available existing data, the use of the OECD QSAR Toolbox and/or other *in silico* tools do not enable to conclude on the skin sensitisation hazard including the sensitising

potency of a substance, it is necessary to investigate a sufficient number of key events (e.g. elements 7a, b and c in [Figure R.7.3-2](#)) as described in the AOP for skin sensitisation, by providing information from non-animal test methods or by other sources of information. The current understanding is that covering different key events by *in chemico/in vitro/in silico* or other data provides the best predictivity for the endpoint. Therefore, in case coverage of one or two key event(s) is omitted, the registrant would need to justify the approach taken based on the current knowledge. Based on the current knowledge, information obtained from peptide reactivity, whether obtained from *in chemico* or *in silico* methods, seems to show the highest predictive power and may provide more weight to the overall assessment of skin sensitisation (Natsch *et al.*, 2013, Urbisch *et al.*, 2015). However, it is important to consider the specific limitations of the approach with respect to the substance under investigation.

The OECD Guidance Document On The Reporting Of Defined Approaches To Be Used Within Integrated Approaches to Testing And Assessment (OECD, 2016a) aims to contribute to a harmonised approach to the reporting of defined approaches used as elements within IATAs (see Annexes I and II of the OECD Guidance Document, and [Appendix R.7.3-2](#) and [Appendix R.7.3-3](#) of this Guidance). In the case of an AOP-informed IATA, the different pieces of information would target key events along the defined toxicity pathway and the results used to inform a regulatory decision, as pointed out in [Figure R.7.3-2](#). The registrant is advised to use the template described in [Appendix R.7.3-3](#) to report a defined data interpretation procedure if this is used as part of the testing and assessment strategy described in [Figure R.7.3-2](#). In practical terms, this means that an individual endpoint study record should be created for each information source, e.g. *in vitro*, *in chemico* study data, in the IUCLID dossier. Then the *Weight-of-Evidence* approach wrapping all of the information sources together, e.g. by using the templates as described in the OECD Guidance Documents and [Appendix R.7.3-3](#) can be attached into the skin sensitisation endpoint summary.

Several approaches on how to combine and interpret data from non-animal testing approaches have been reported in the scientific literature, and independent assessments are under discussion. The use of positive predictions obtained from *in chemico/in vitro* test methods tends to be more straightforward than in case negative or conflicting predictions are obtained. Due to the specific limitations of each of the *in chemico/in vitro* test methods, in case a negative prediction is obtained, it is important to justify in the dossier how a potential false prediction can be ruled out. Supporting information on whether the substance is or is not a pro-hapten and whether metabolism is expected to occur *in vivo* can be obtained, e.g. from *in silico* methods or from test data for other endpoints (see Section [R.7.3.7.2](#), under “*How to deal with the lack of or limited metabolic capacity of non-animal test methods?*”). Also whether the substance is a pre-hapten and requires abiotic transformation (e.g. auto-oxidation or hydrolysis) to exert its skin sensitisation potential should be considered (note: the issue of not identifying pre-haptens correctly is not solely related to *in chemico/in vitro* methods, but can also occur with *in vivo* test methods).

Figure R.7.3–2 Testing and assessment strategy for evaluating the skin sensitisation potential of substances (footnotes a to f are detailed below the figure)

Element	Information	Conclusion
Existing data on physico-chemical properties		
1	Is the substance a strong acid (pH ≤ 2.0) or base (pH ≥ 11.5), corrosive to the skin or (spontaneously) flammable in air or in contact with water or moisture at room temperature?	YES: No <i>in vivo</i> testing required (Column 2 adaptation of Annexes VII, section 8.3) Note: extreme pH values/corrosive properties do not preclude performing <i>in chemico/in vitro/in vivo</i> test(s) at suitable concentrations and, therefore, it is possible to assess skin sensitisation hazard in sub-corrosive concentrations, if considered necessary.
Existing human data		
2	Are there adequate existing human data ^a , which provide evidence that the substance is a skin sensitiser?	YES: Consider classifying according to CLP criteria (Cat. 1A or not^e). If not conclusive on its own, use this information for <i>Weight-of-Evidence</i> analysis under point 6.
Existing animal data from sensitisation studies		
3	Are there data from existing studies <i>on skin sensitisation</i> in laboratory animals (LLNA, GPMT, or Buehler test, EU B.42, B.50, B.51 and B.6/OECD TGs 429, 442A, 442B and 406), which provide sound conclusive evidence that the substance is a sensitiser, or non-sensitiser?	YES: Consider classifying according to CLP criteria (Cat. 1A or not^e) or consider no classification. If not conclusive on its own, use this information for <i>Weight-of-Evidence</i> analysis under point 6. Note: <i>in vivo</i> study(ies) that were carried out or initiated before 10 May 2017, are guideline compliant and performed according to GLP can be used to fulfil the skin sensitisation requirements even if a conclusion on skin sensitisation potency cannot be reached. However, existing information from other sources or non-animal test data should be used to refine the classification.

Existing (Q)SAR data and read-across		
4	Do “read-across” from structurally and mechanistically related substances and/or do suitable (Q)SAR predictions reliably indicate skin sensitisation potential or the absence thereof of the substance? ^b	YES: Consider classifying according to CLP criteria (Cat. 1A or not^e) or consider no classification. If not conclusive on its own, use this information for <i>Weight-of-Evidence</i> analysis under point 6.
Existing in chemico and in vitro data		
5a	Is there evidence/hypothesis of dermal bioavailability based on physico-chemical, <i>in silico</i> , <i>in vitro</i> or <i>in vivo</i> data?	YES/NO: Use this information for <i>Weight-of-Evidence</i> analysis
5b	Has the substance demonstrated peptide/protein binding properties in an EU/OECD adopted <i>in chemico</i> test (e.g. OECD TG 442c)? (<i>Key event 1 of the AOP</i>), and/or Has the substance demonstrated activation of the Nrf2-Keap1-ARE toxicity pathway in an EU/OECD adopted <i>in vitro</i> test (e.g. OECD TG 442d)? (<i>Key event 2 of the AOP</i>), and/or Has the substance demonstrated induction of the cell surface markers (CD54 and/or CD86) on monocytic cells in a validated <i>in vitro</i> test, e.g. h-CLAT? (<i>Key event 3 of the AOP</i>). Data from <i>in chemico/in vitro</i> test methods that have been validated and are considered scientifically valid but are not yet adopted by the EU and/or OECD may also be used if the provisions defined in Annex XI to the REACH Regulation are met.	YES/NO: Consider classifying as Skin sensitiser (Cat. 1A or not^e) or consider no classification⁶⁸. If not conclusive, use this information for <i>Weight-of-Evidence</i> analysis under point 6.

⁶⁸ Currently (at the time of publication of this Guidance), there are no CLP (or UN GHS) criteria available on how to classify based on *in vitro* data only and how to derive potency. Discussions at the UN GHS level are ongoing. A review of different approaches assessing skin sensitisation potency is described in [Appendix R.7.3–4](#). *In vitro* data cannot be directly converted into potency outcomes similar to those derived from LLNA data. However, the *in vitro* data obtained may provide some indication of skin sensitising potency and, with support of e.g. read-across data, they may allow to derive the skin sensitising potency (Cat. 1A or not) of a substance.

5c	Are there data from (a) non-validated <i>in vitro</i> test(s), which provide evidence that the substance may be a skin sensitiser?	YES/NO: Consider classifying as Skin sensitiser (Cat. 1A or not^e)⁶⁸. If not conclusive, use this information for <i>Weight-of-Evidence</i> analysis under point 6.
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Weight-of-Evidence analysis

6	The “elements” described above may be arranged as appropriate. Taking all existing and relevant data (elements 1-5) into account, is there sufficient information to meet the information requirement of Section 8.3 of Annex VII and to make a decision on whether classification and labelling are warranted? For specific guidance on <i>Weight of Evidence</i> see below.	YES: Classify as Skin Sensitiser Cat. 1A or not^e or consider no classification⁶⁸. Classification as Skin Sensitiser Cat. 1 is only acceptable when based on existing <i>in vivo</i> data (see element 3). NO: Consider the next elements of the strategy.
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Generation of new non-animal data ^c

Note: Section 8.3.1 of Annex VII to the REACH Regulation specifies that when new information needs to be generated testing must start with *in chemico/in vitro* methods. Before performing *in chemico/in vitro* testing, it is important to consider whether the test method(s) to be used are suitable for the substance i.e. whether the substance fits in the applicability domain of a specific test method. In case the *in chemico/in vitro* tests are not suitable for the substance, an LLNA would need to be performed, as specified in Annex VII, section 8.3.2.

7a	Does the substance demonstrate peptide/protein binding properties in an EU/OECD adopted <i>in chemico</i> test (e.g. B. 59/OECD TG 442c)? (<i>Key event 1 of the AOP</i>) <i>In chemico</i> test methods that have been validated and are considered scientifically valid but are not yet adopted by the EU and/or OECD may also be used if the provisions defined in Annex XI to the REACH Regulation are met.	YES/NO: Consider classifying as Skin sensitiser (Cat. 1A or not^e)⁶⁸. If not conclusive, use this information for <i>Weight-of-Evidence</i> analysis under point 8.
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